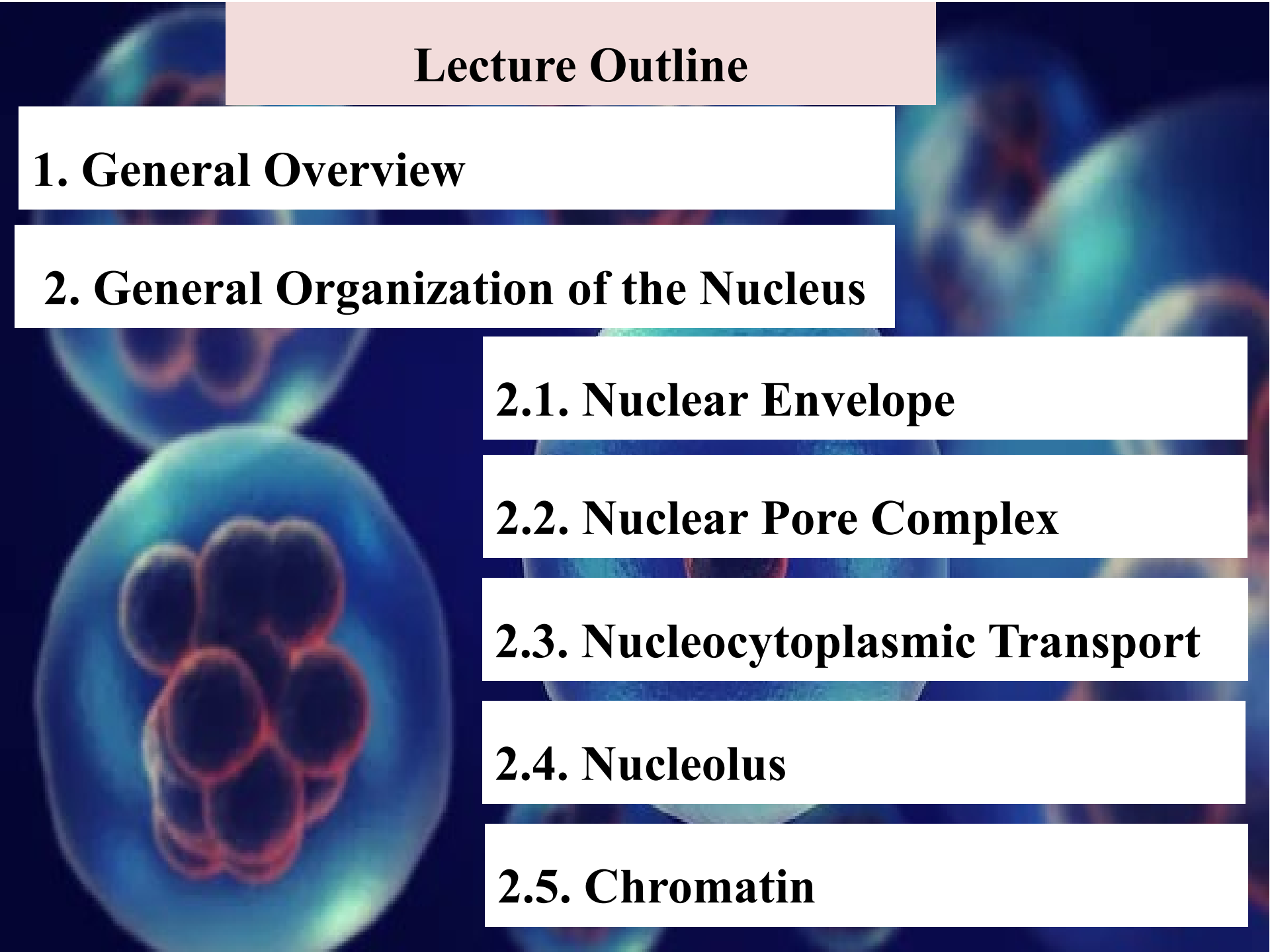


بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

I. The Cell Nucleus

**The Interphase Nucleus: Structure,
Organization, and Function**

The background of the slide features a microscopic image of a cell. A large, spherical nucleus is visible, containing a dense, dark nucleolus. The cytoplasm is filled with various organelles and structures, appearing in shades of blue and green. The overall image is slightly blurred, giving it a scientific, high-magnification feel.

Lecture Outline

1. General Overview

2. General Organization of the Nucleus

2.1. Nuclear Envelope

2.2. Nuclear Pore Complex

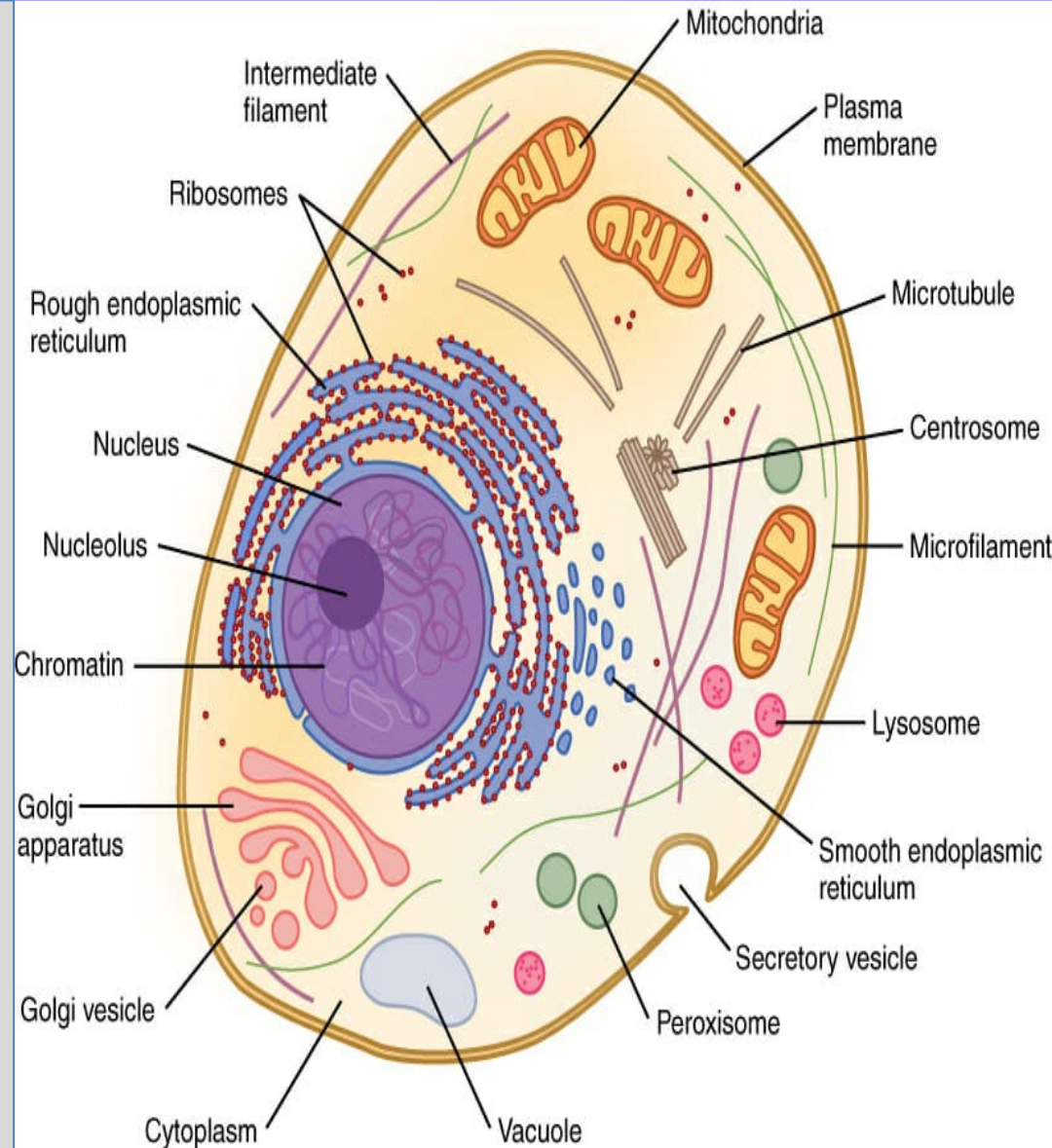
2.3. Nucleocytoplasmic Transport

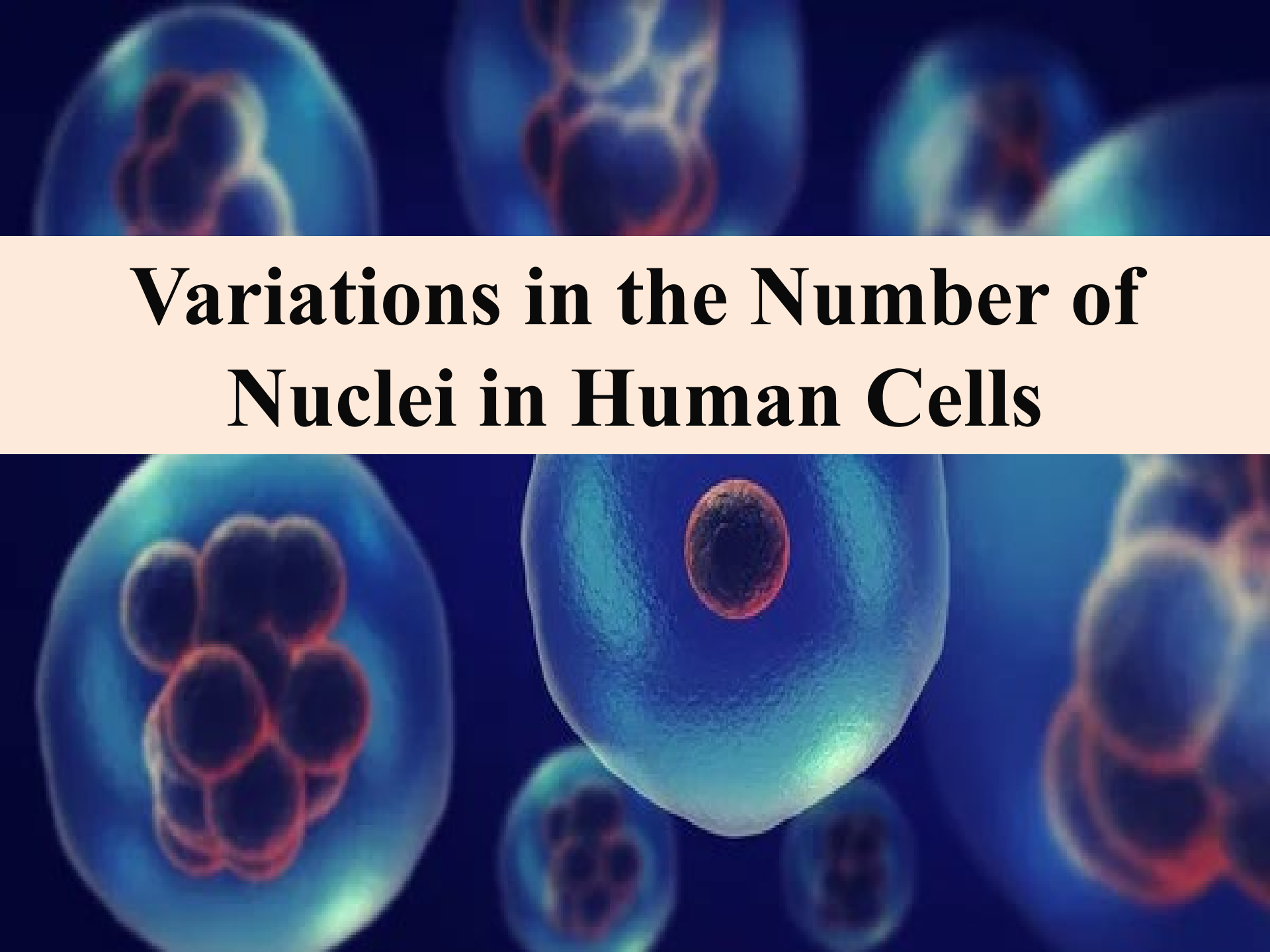
2.4. Nucleolus

2.5. Chromatin

1. General Overview

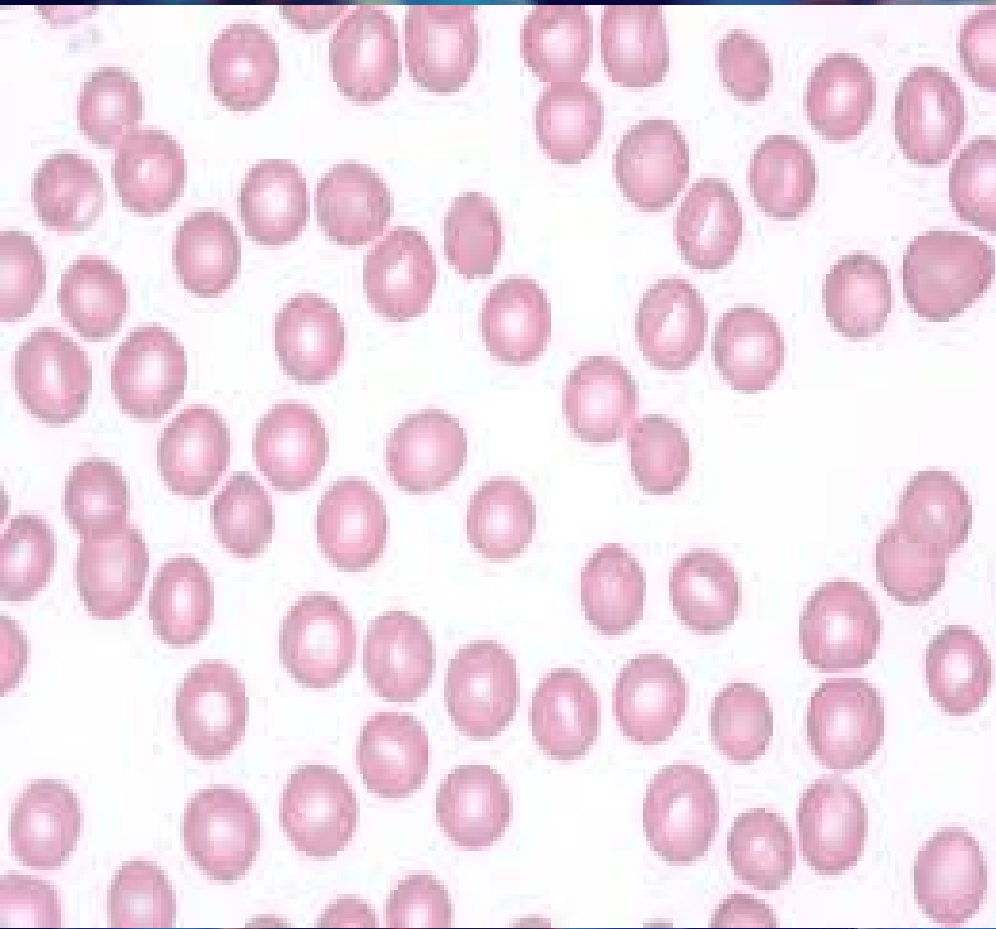
- **Shape:** Usually **spherical** or oval.
- **Diameter:** Between **5 and 10 μ m**
- **Main Role:** Stores the **genetic material (DNA)**
- **Key Feature:** Surrounded by a **nuclear envelope** with pores,
- **Nucleoplasm:** gelatinous fluid found inside the **nucleus**



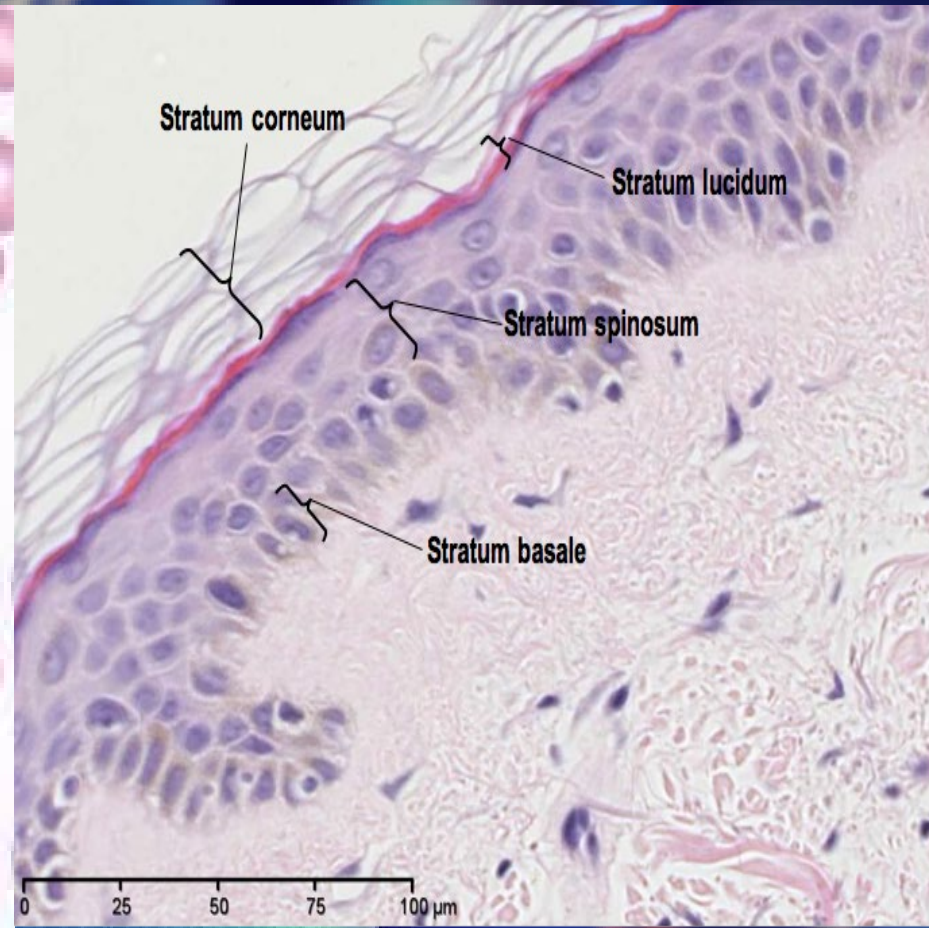
The background of the slide is a composite of several microscopic images of human cells. The cells are stained, with nuclei appearing in shades of red and pink, and the cytoplasm and cell membranes in various shades of blue and cyan. Some cells show a single nucleus, while others are multinucleated, containing two, three, or more nuclei. The cells are of varying sizes and are distributed across the slide, with some in sharp focus and others blurred in the background.

Variations in the Number of Nuclei in Human Cells

1. Anucleated Cells (0 Nuclei)



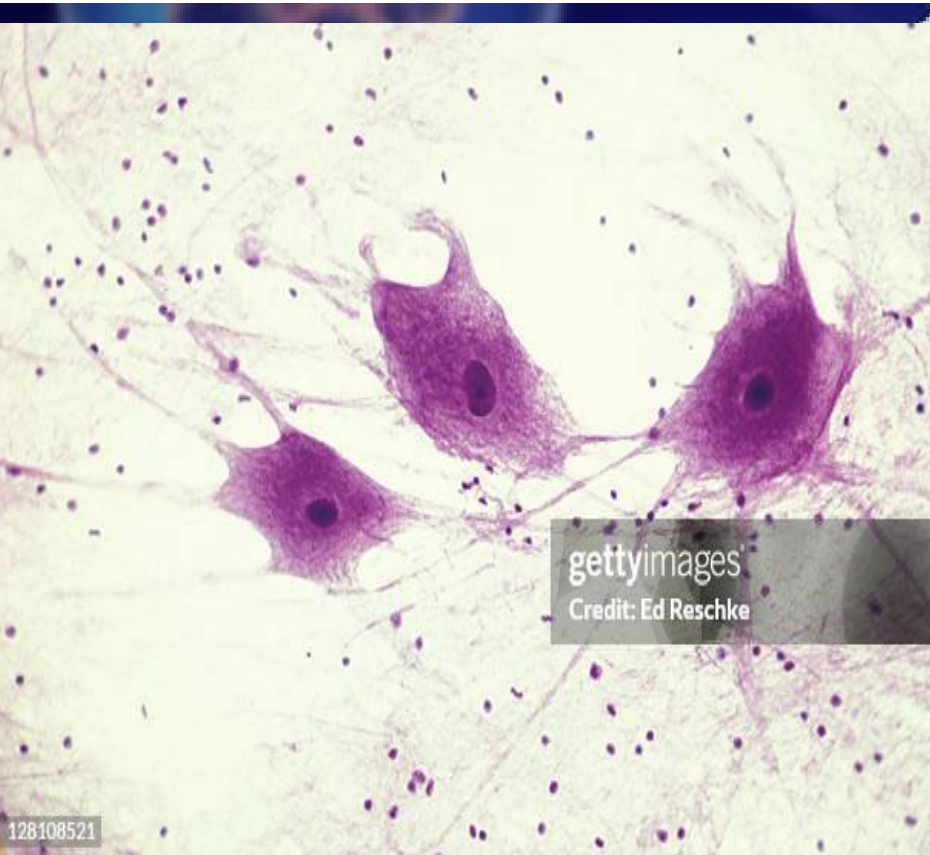
**Mature Red Blood Cells
(Erythrocytes)**



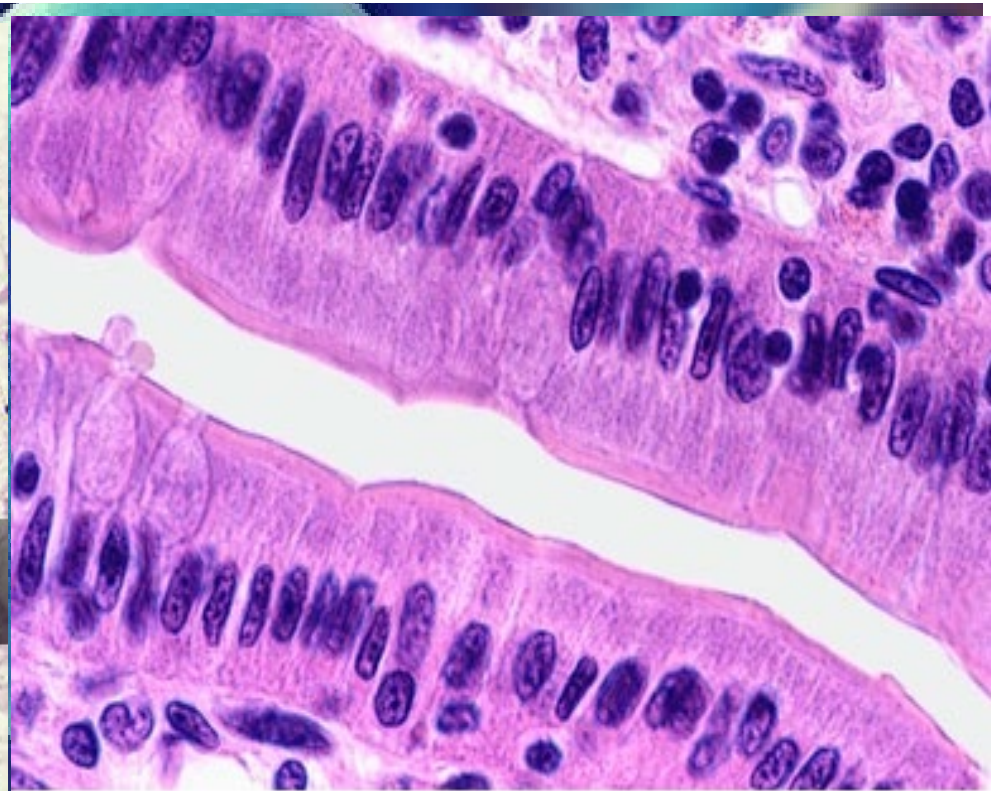
**Corneocytes (keratinocytes
anucleated)**

2. Mononucleated Cells (1 Nucleus)

- Examples:** Most somatic cells, including neurons (nerve cells), epithelial cells, and many immune cells like lymphocytes and monocytes.

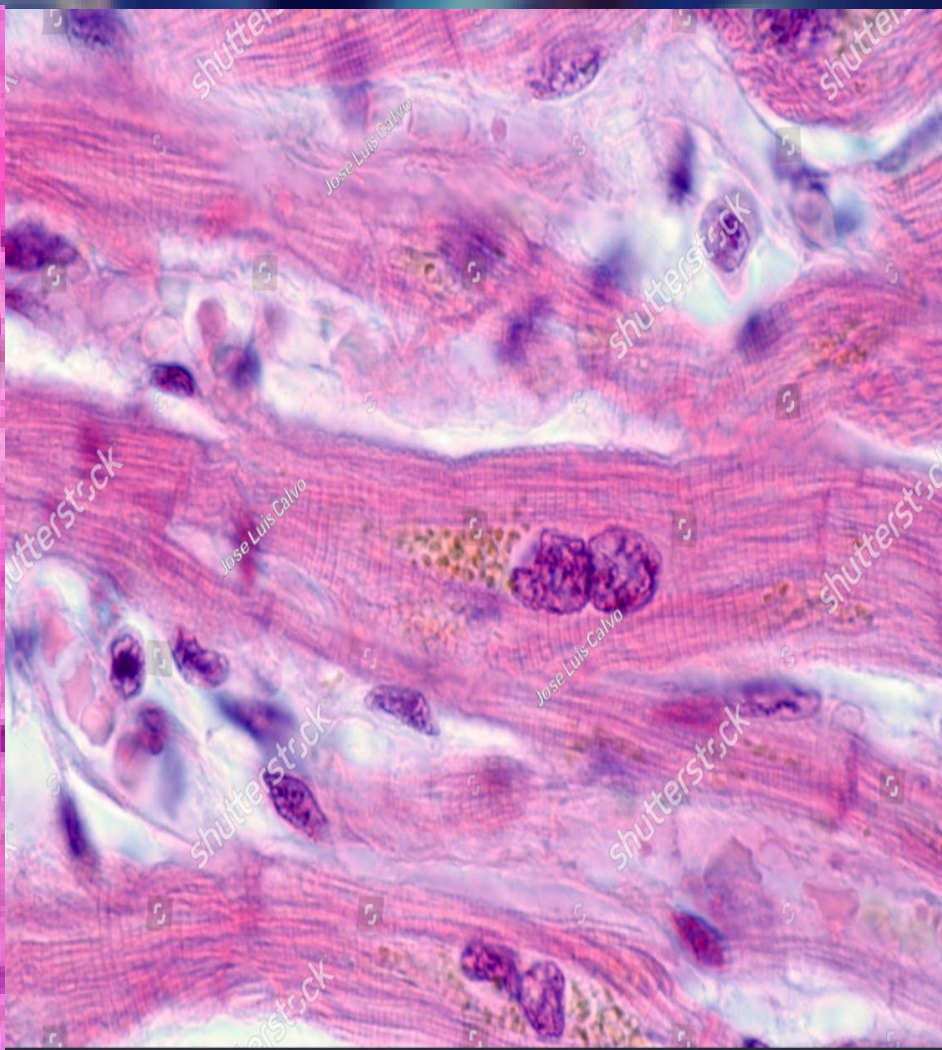
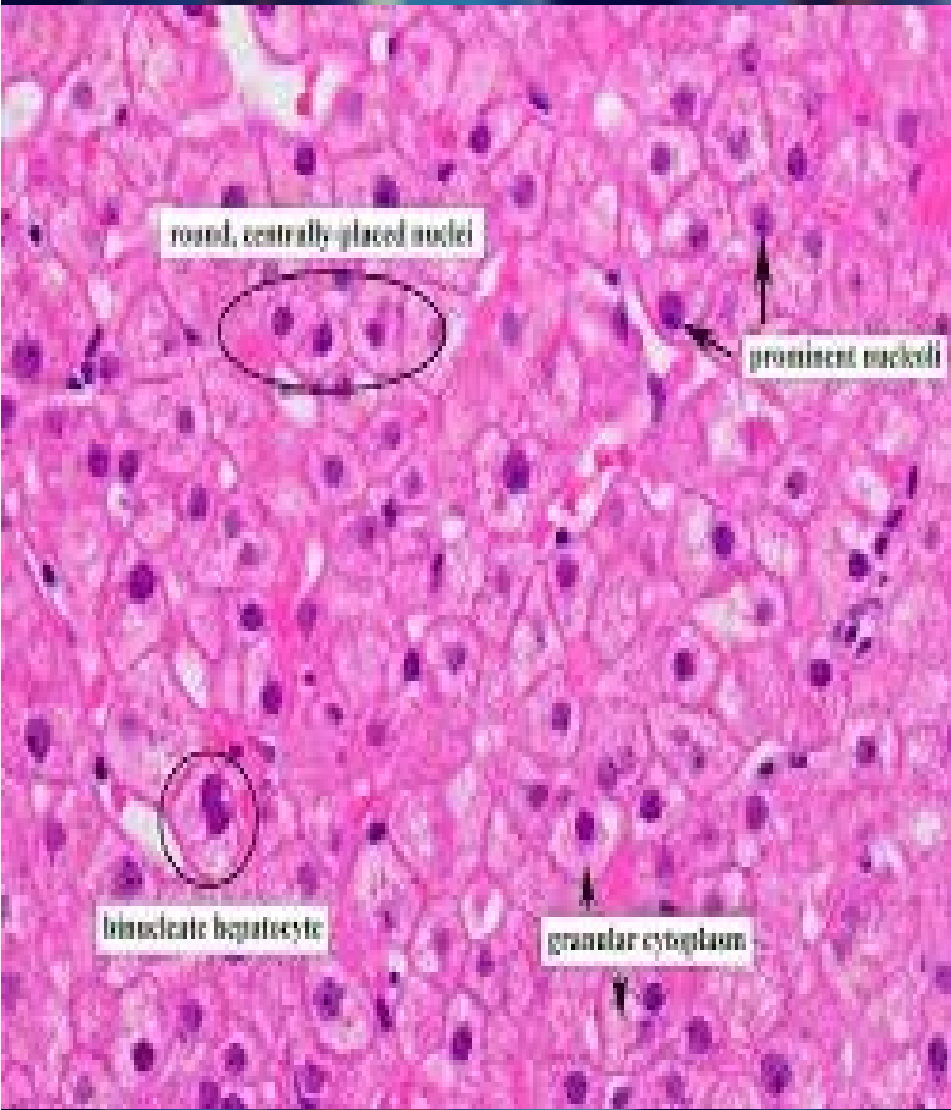


Nerve cells



Enterocytes

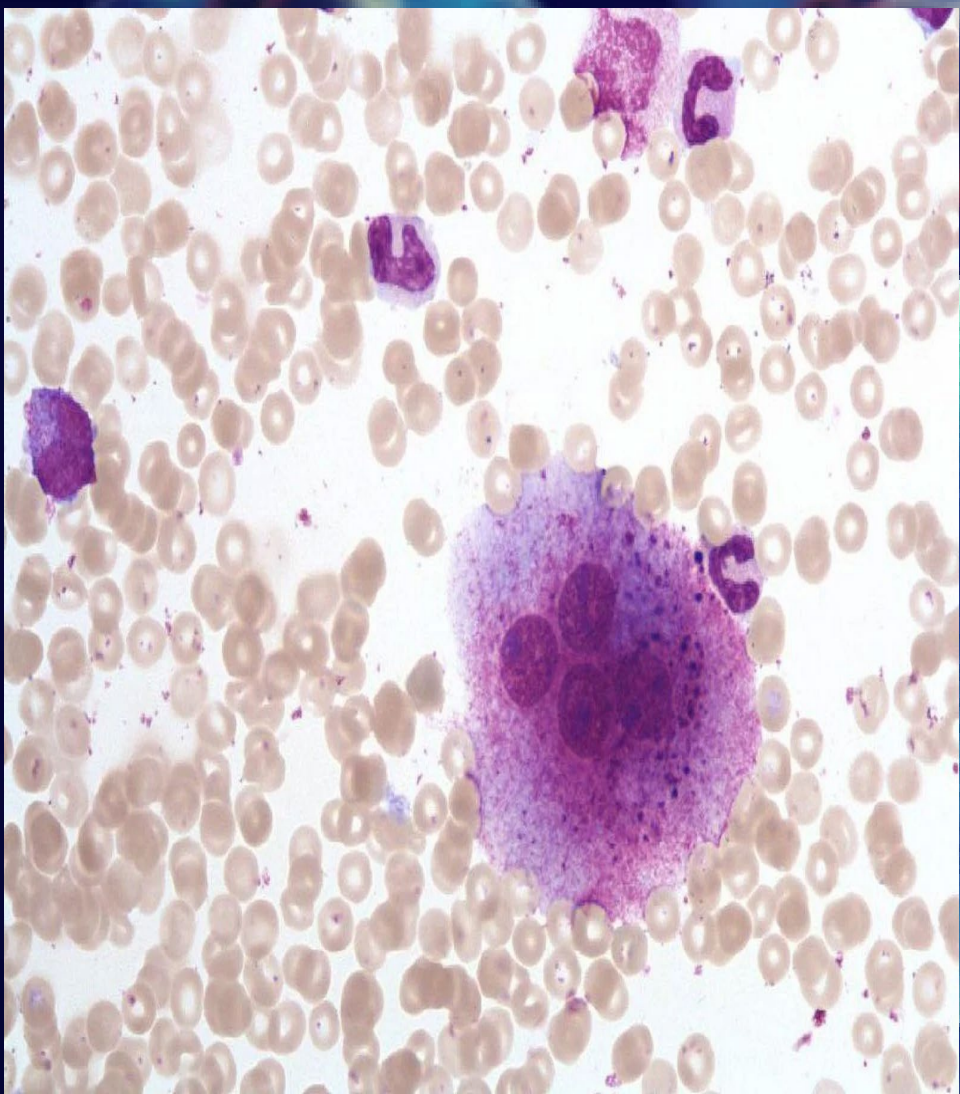
3. Binucleated Cells (2 Nuclei)



Hepatocytes (Liver Cells)

Cardiac Myocytes (Heart Muscle)

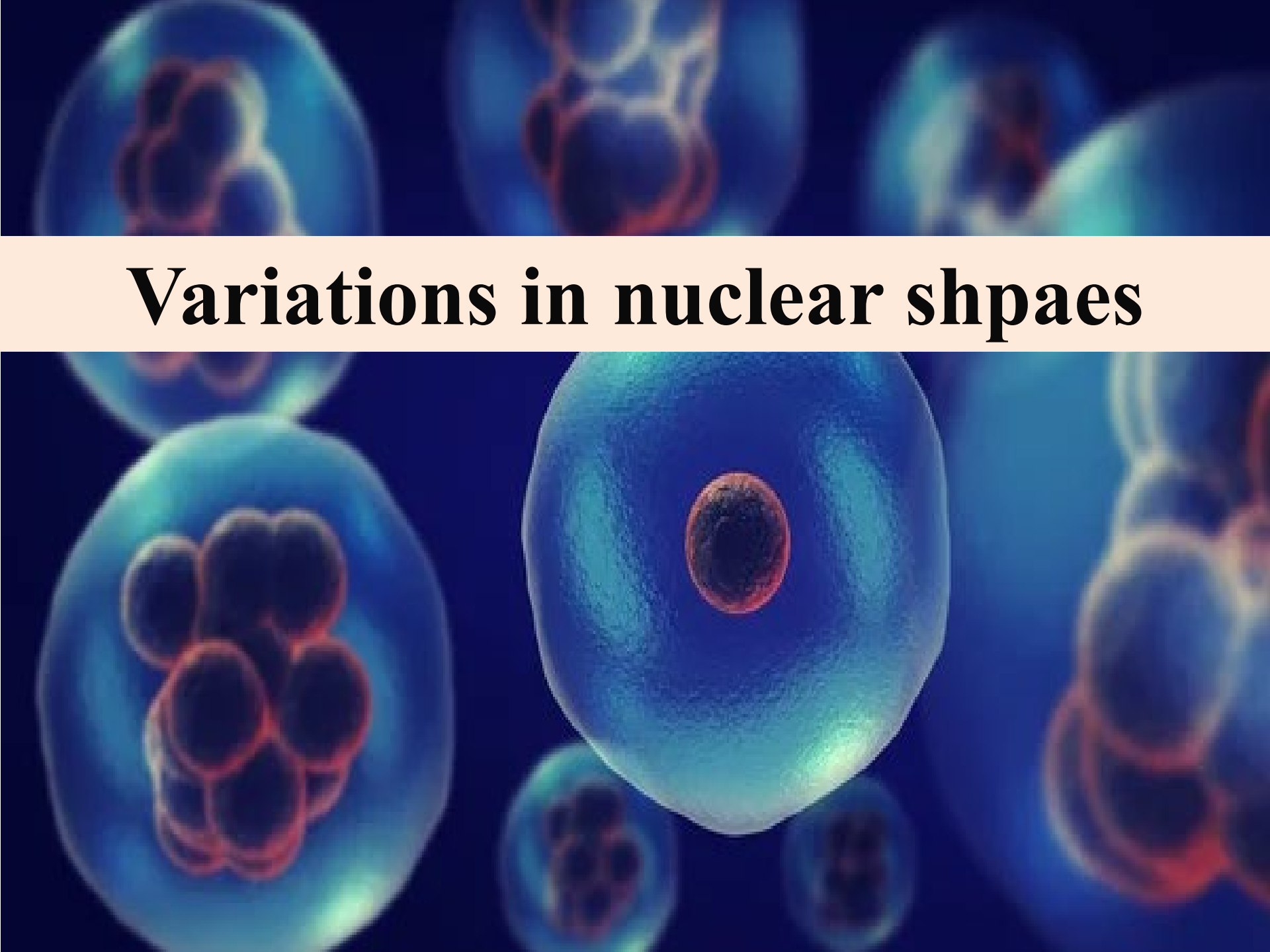
4. Multinucleated Cells (Many Nuclei)



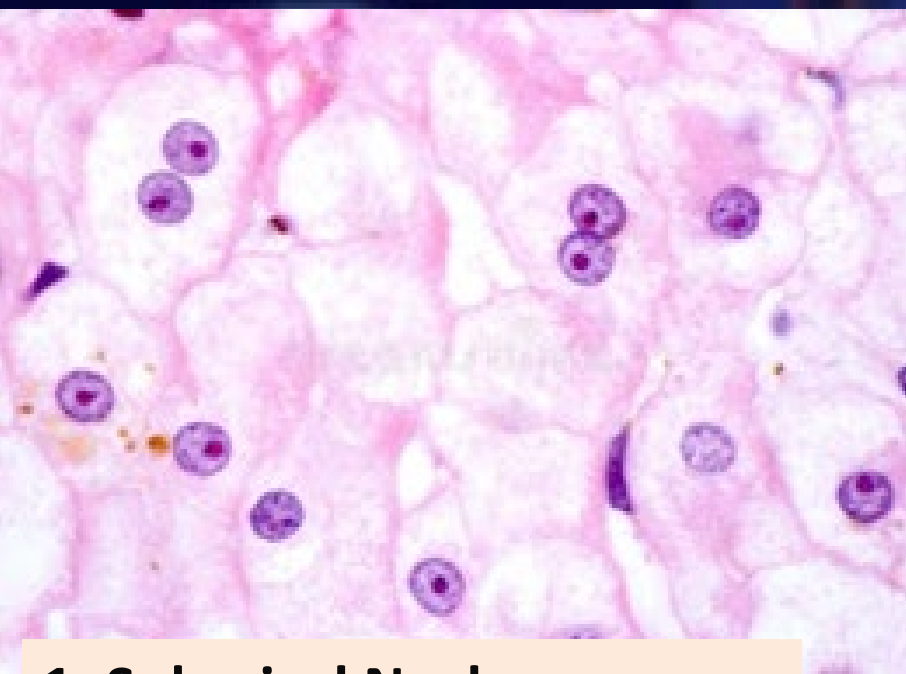
Osteoclasts



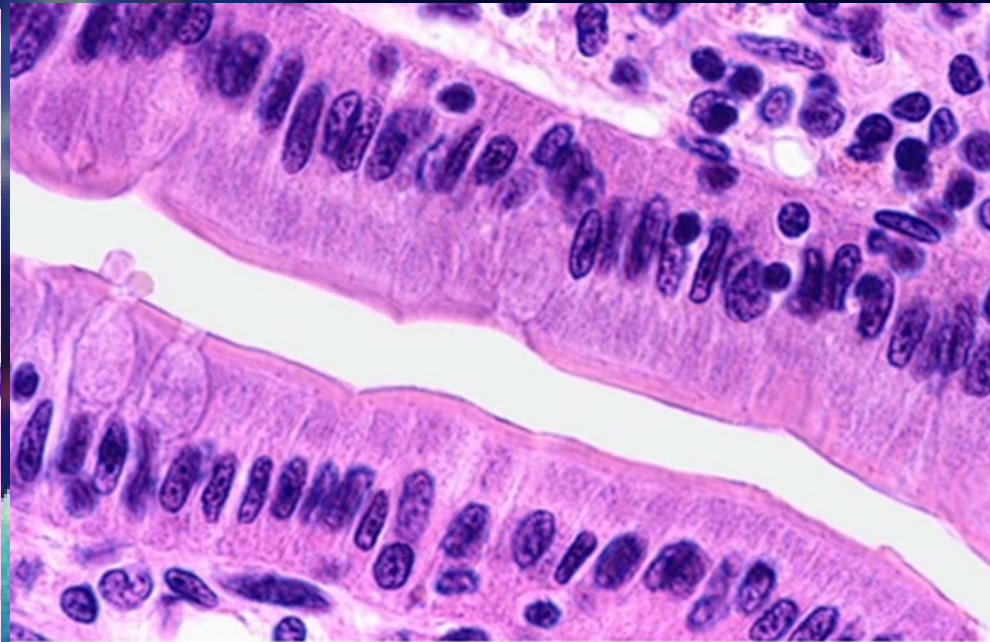
Skeletal Muscle Fibers

The background of the slide is a microscopic image of cells. The cells have a light blue, translucent cytoplasm and a darker red, more opaque nucleus. The nuclei are of various shapes and sizes, some appearing as single, large, oval structures, while others are smaller and more irregular. The overall image has a soft, out-of-focus quality, typical of a light micrograph.

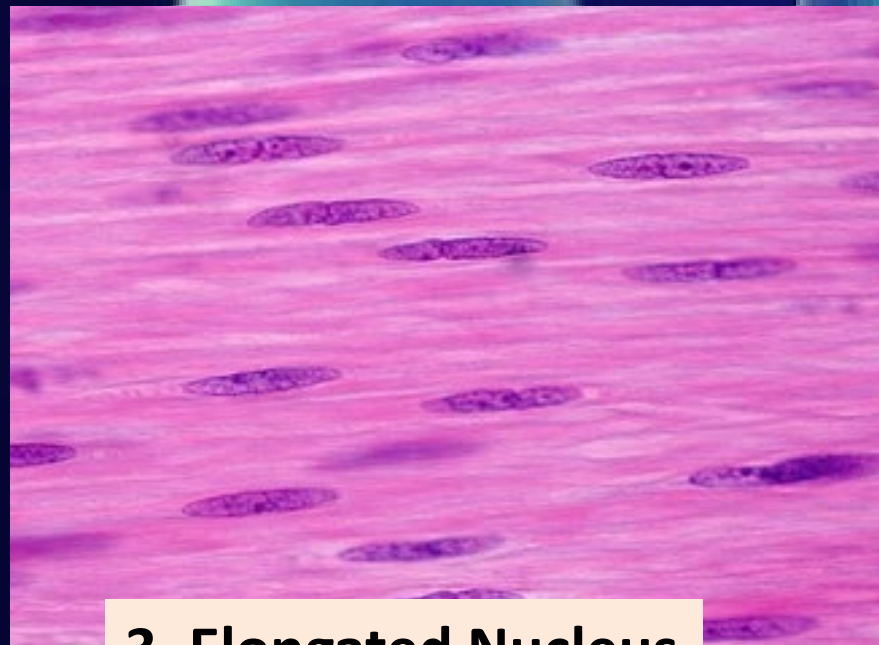
Variations in nuclear shapes



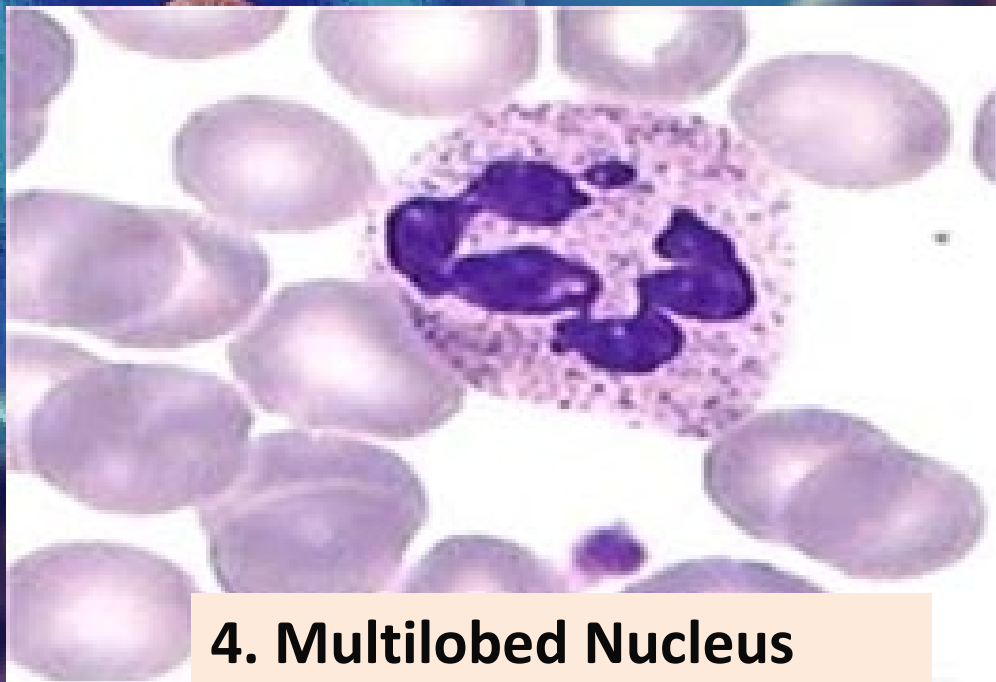
1. Spherical Nucleus



2. Ovoid Nucleus

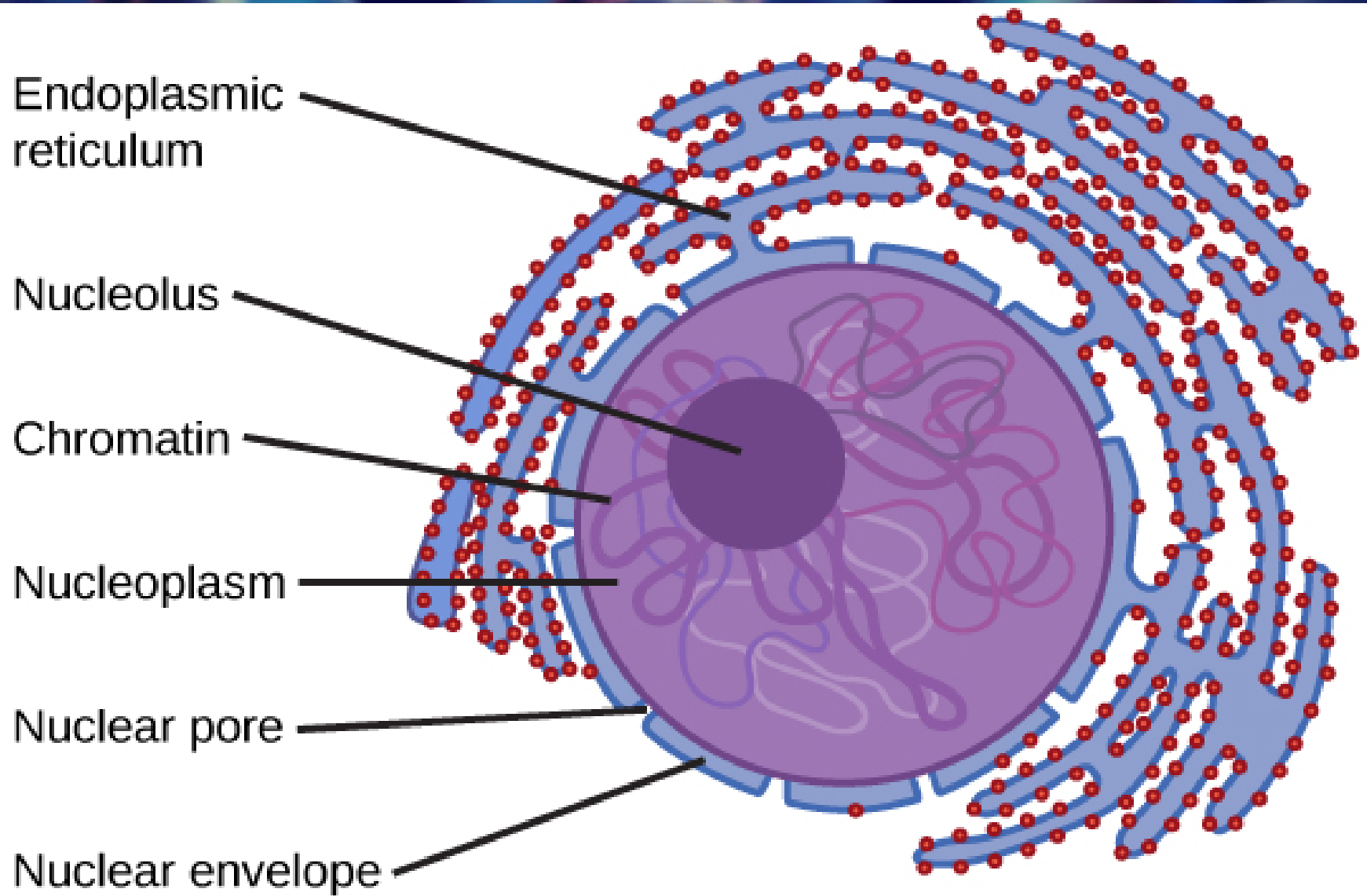


3. Elongated Nucleus

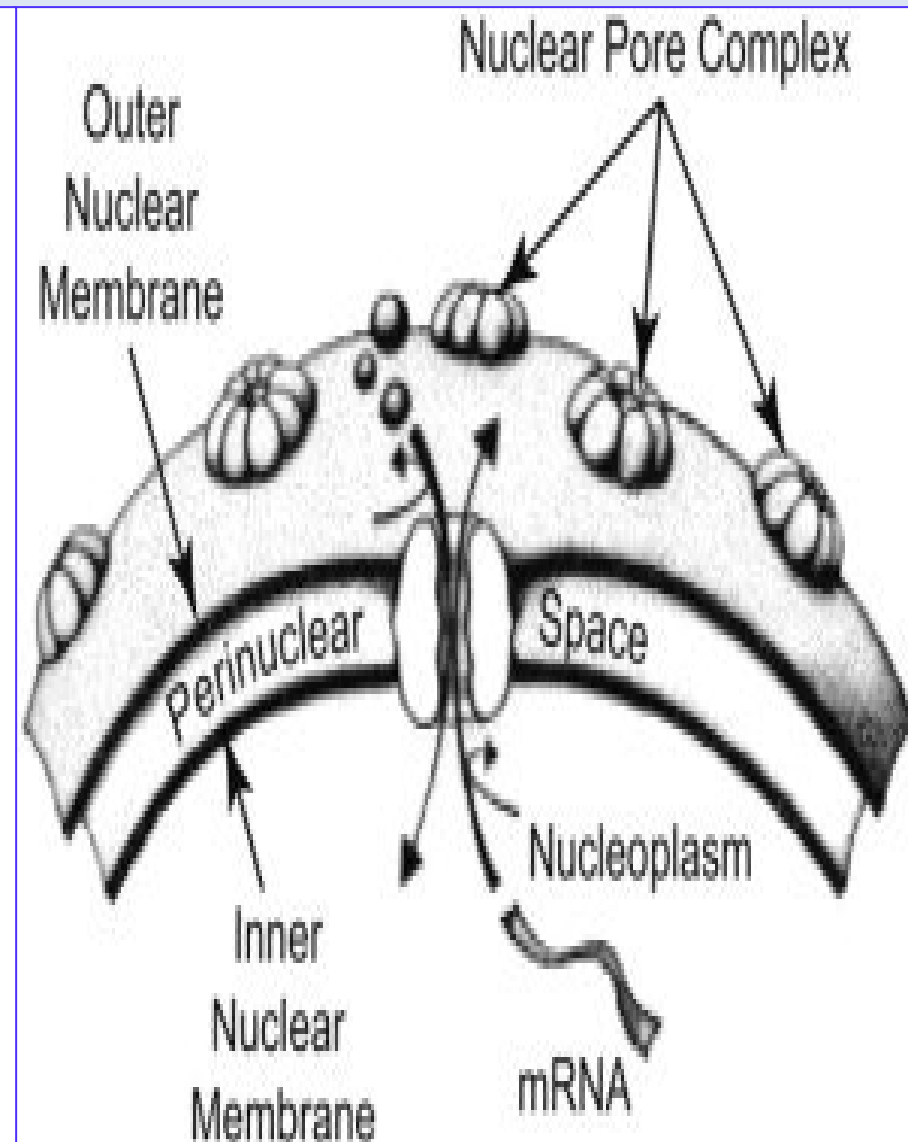
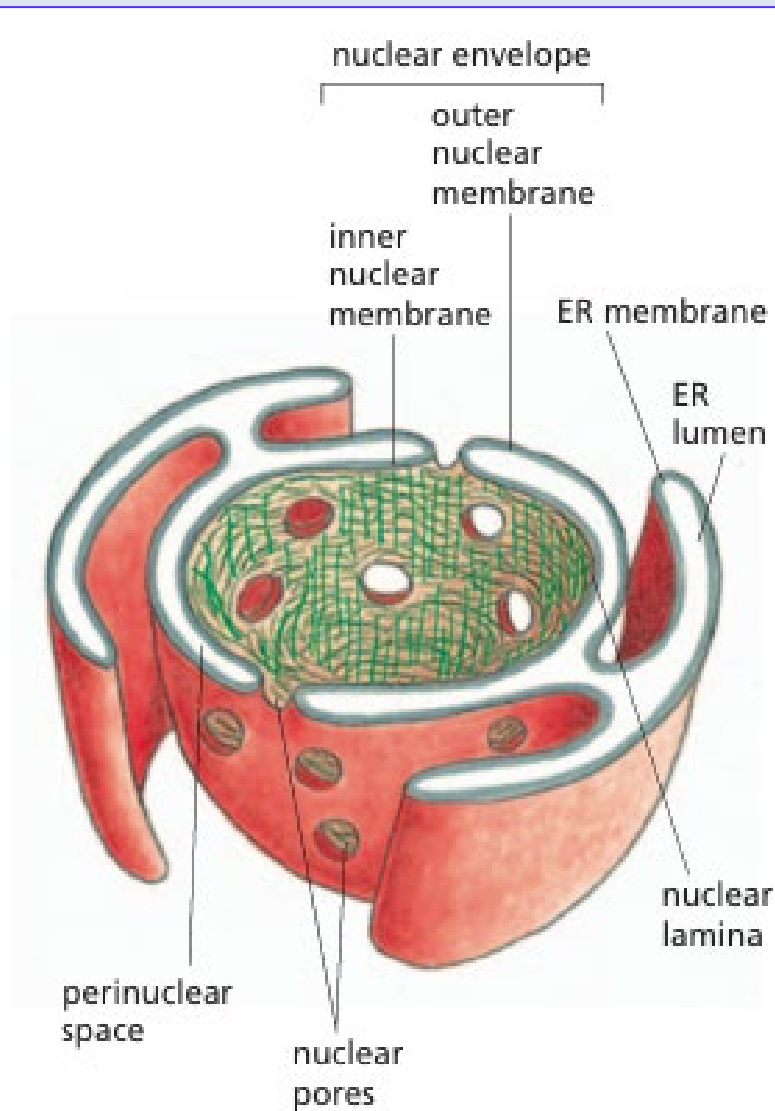


4. Multilobed Nucleus

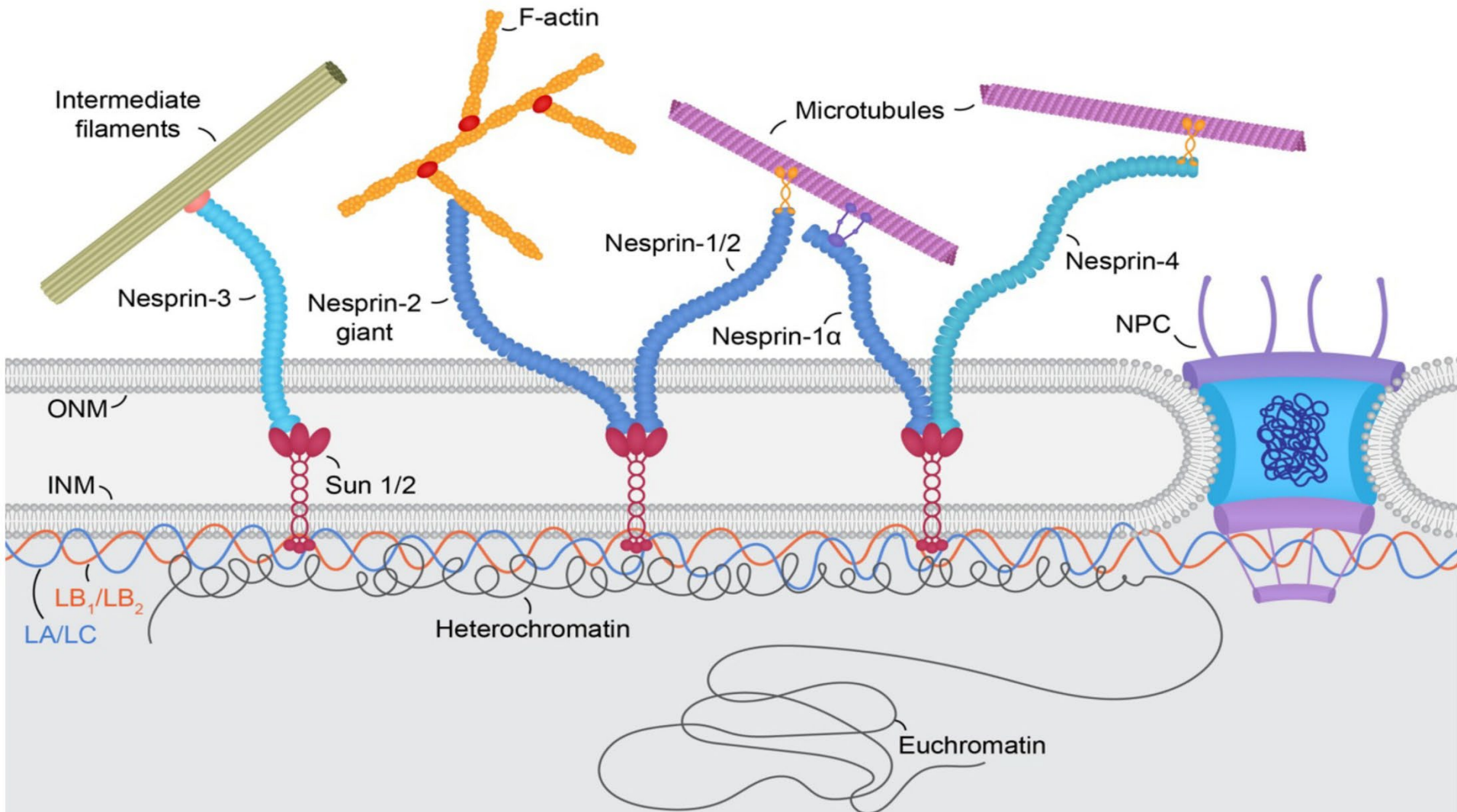
2. General Organization of the Nucleus



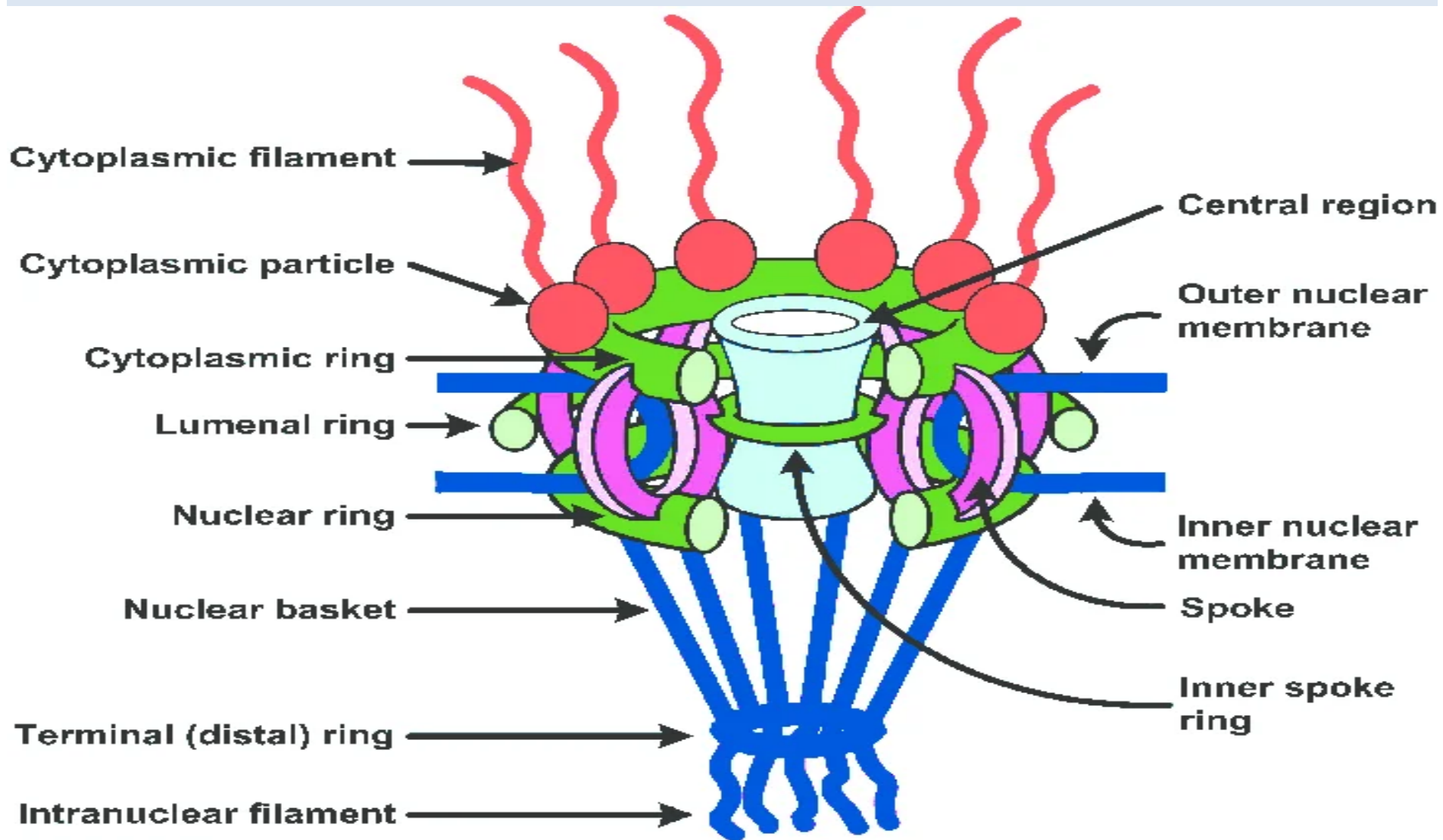
2.1. Structure and organization of the Nuclear Envelope



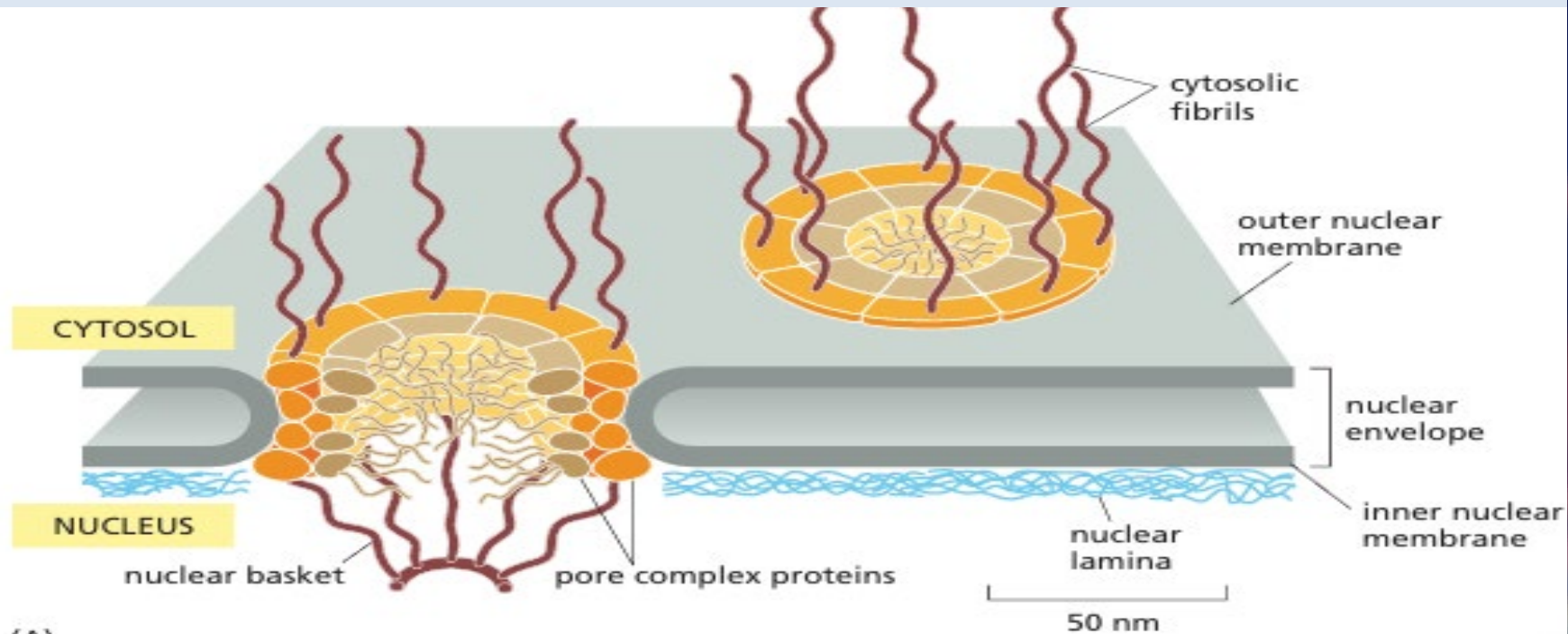
2.1. Structure and Molecular organization of the Nuclear Envelope



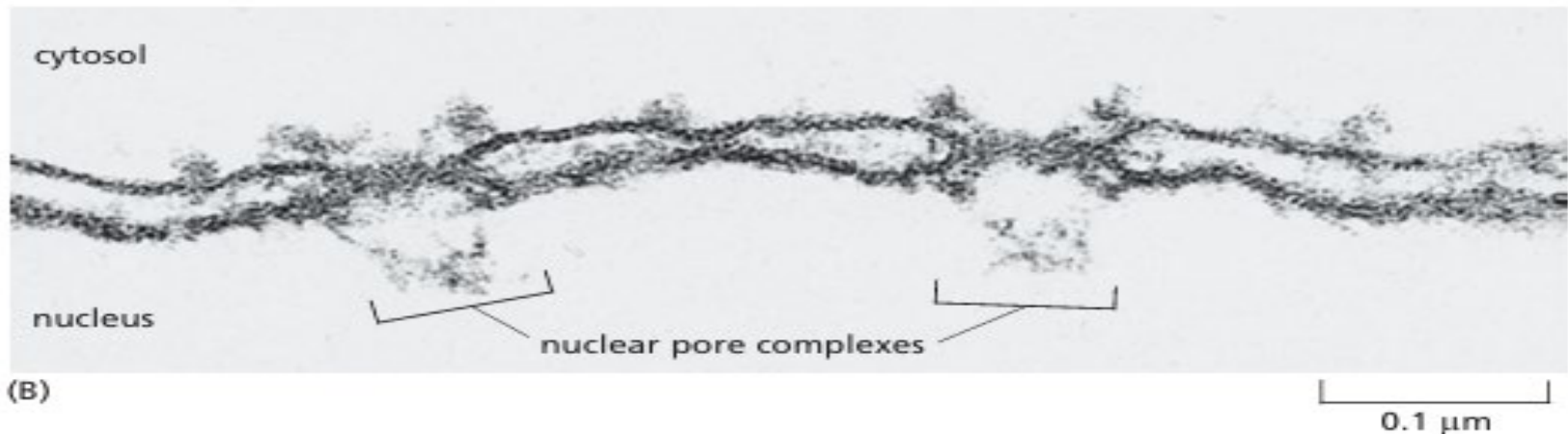
2.2.Molecular Organization of Nuclear Pore Complex



2.2.Molecular Organization of Nuclear Pore Complex

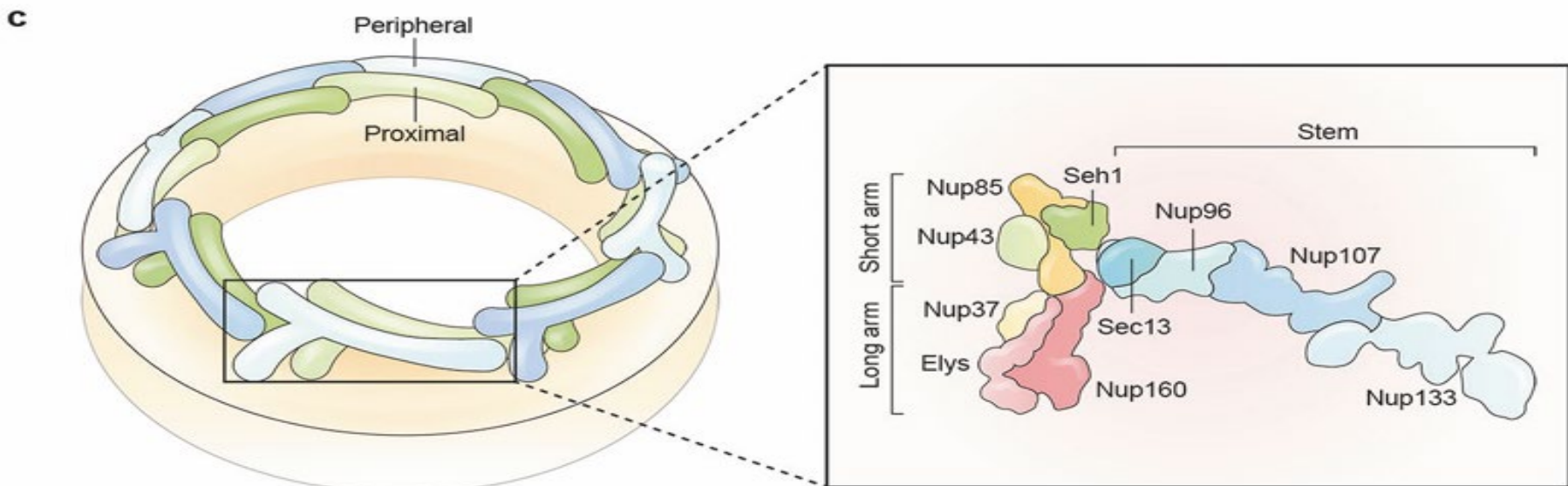
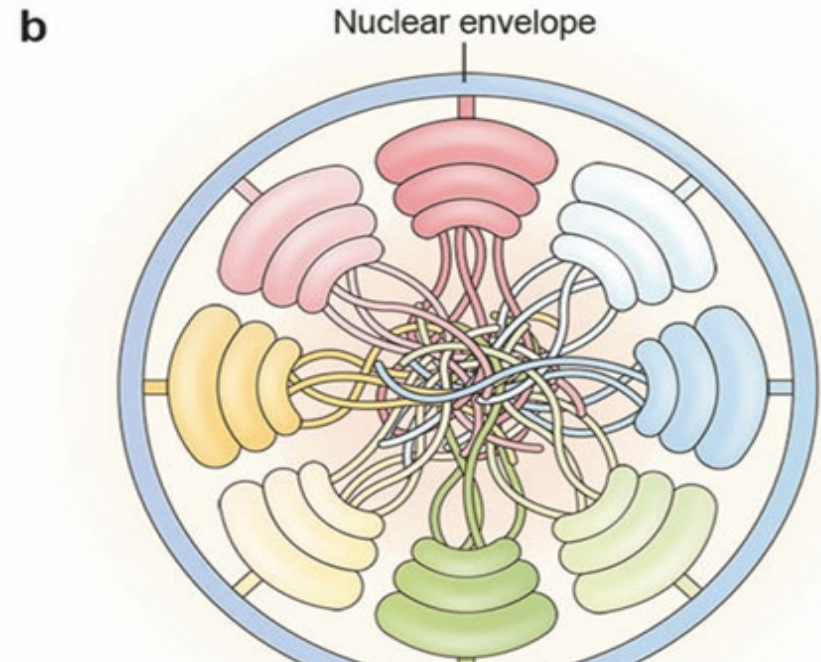
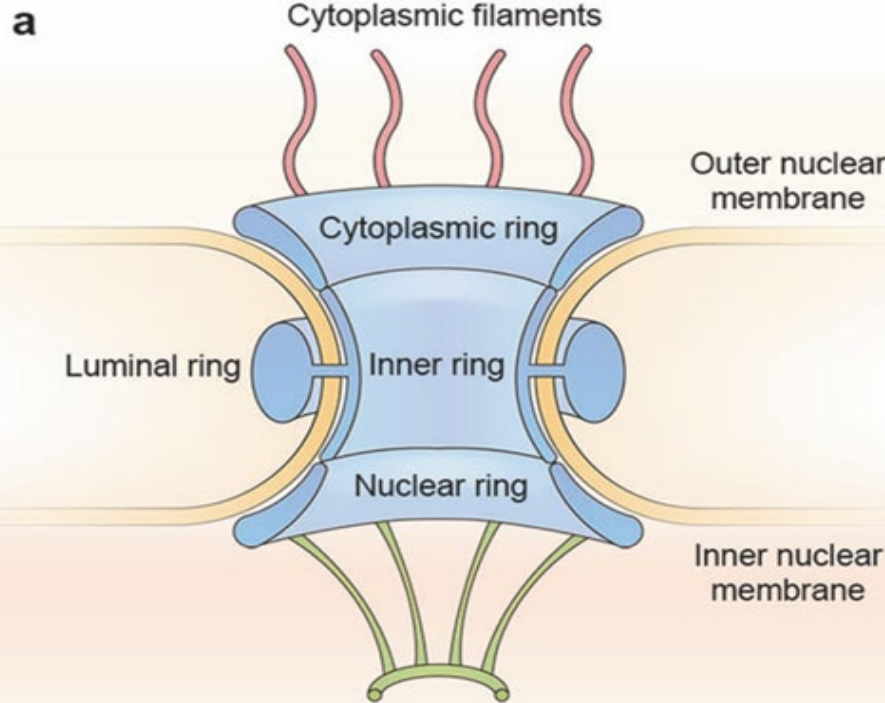


(A)

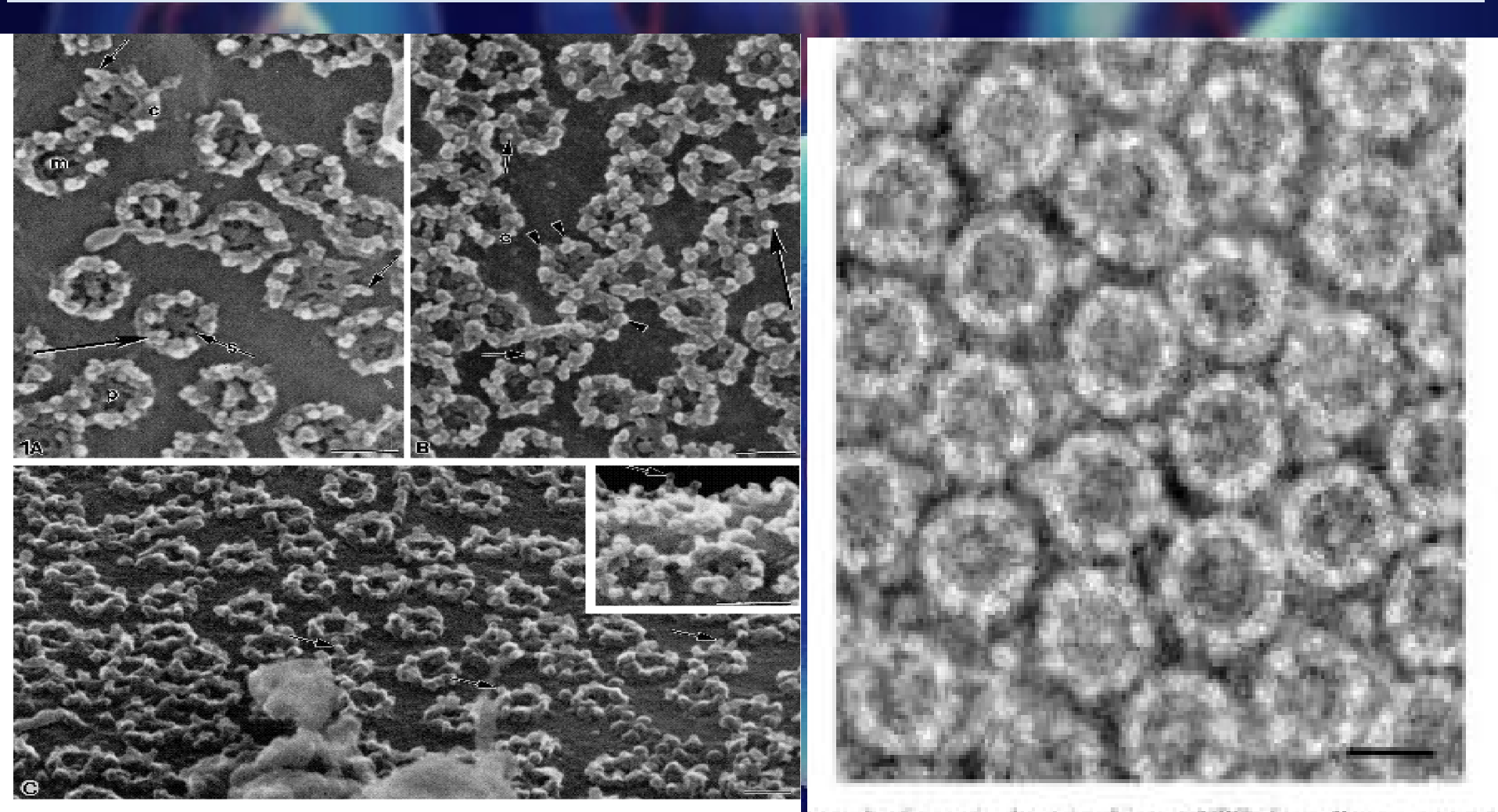


(B)

2.2.Molecular Organization of Nuclear Pore Complex

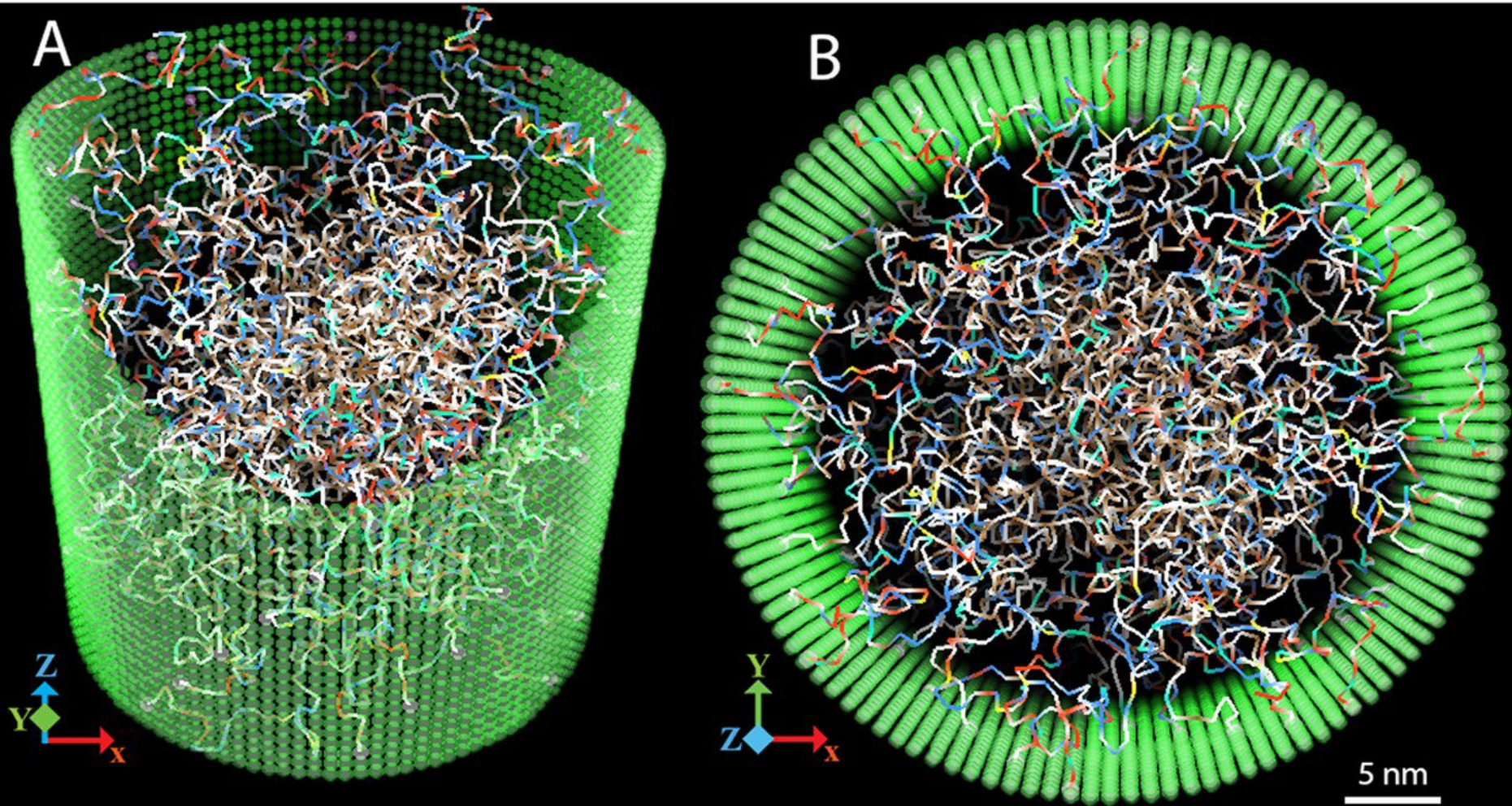


2.2.Molecular Organization of Nuclear Pore Complex



Scanning and transmission electron micrograph showing the distribution of nuclear pore complexes.

2.2.Molecular Organization of Nuclear Pore Complex

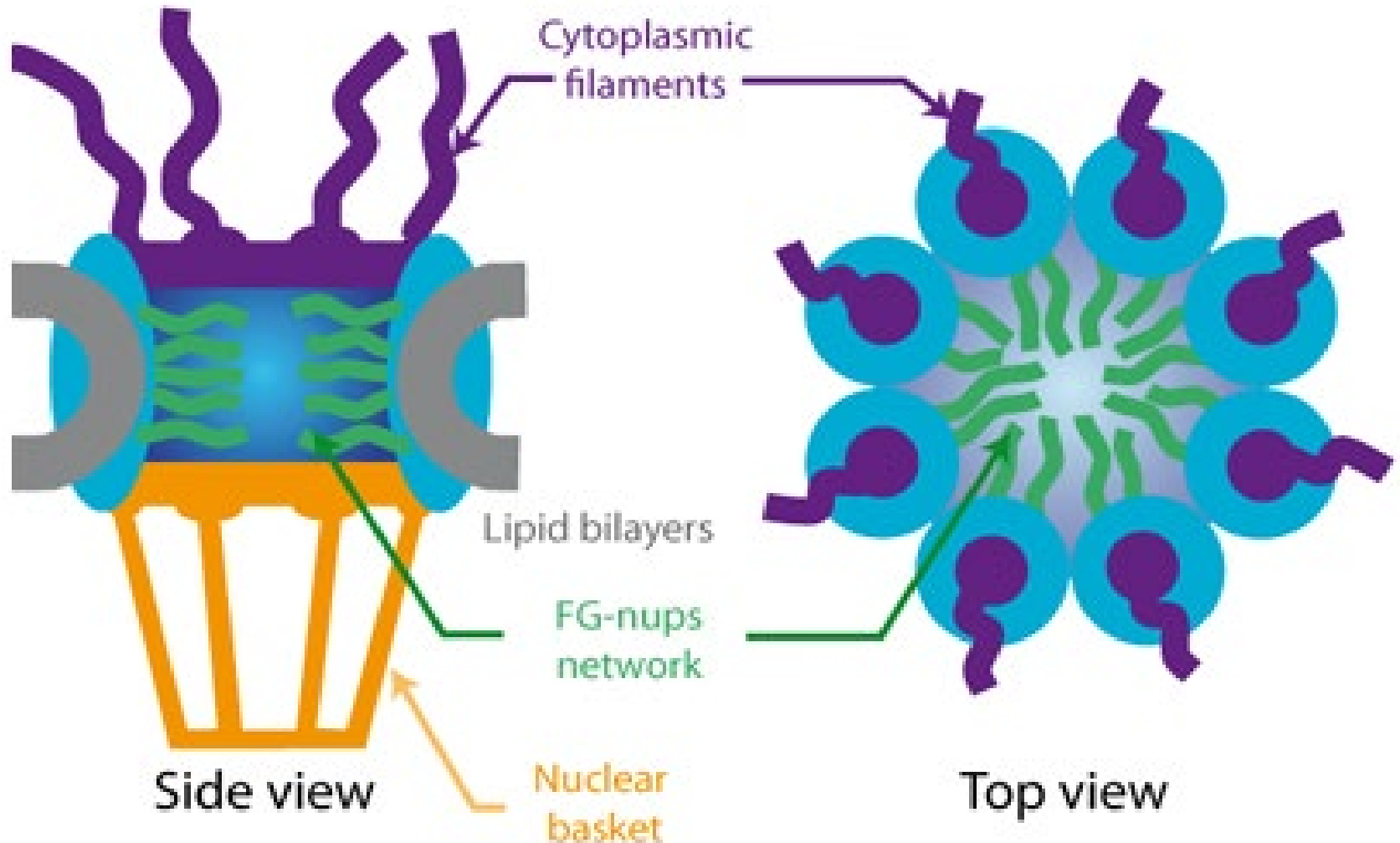


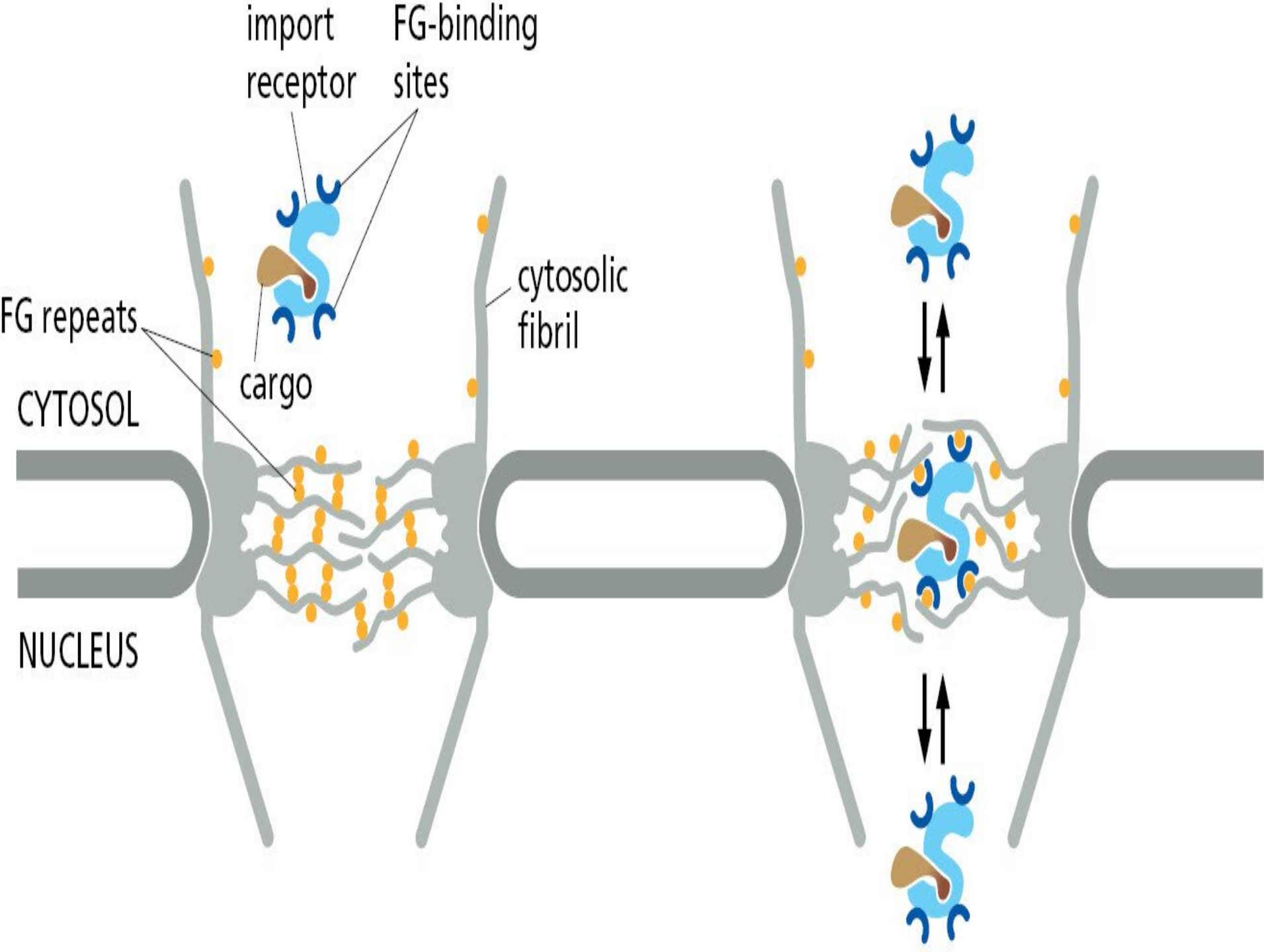
Central channel of the Nuclear Pore Complex with FG-repeats.

A: Side view

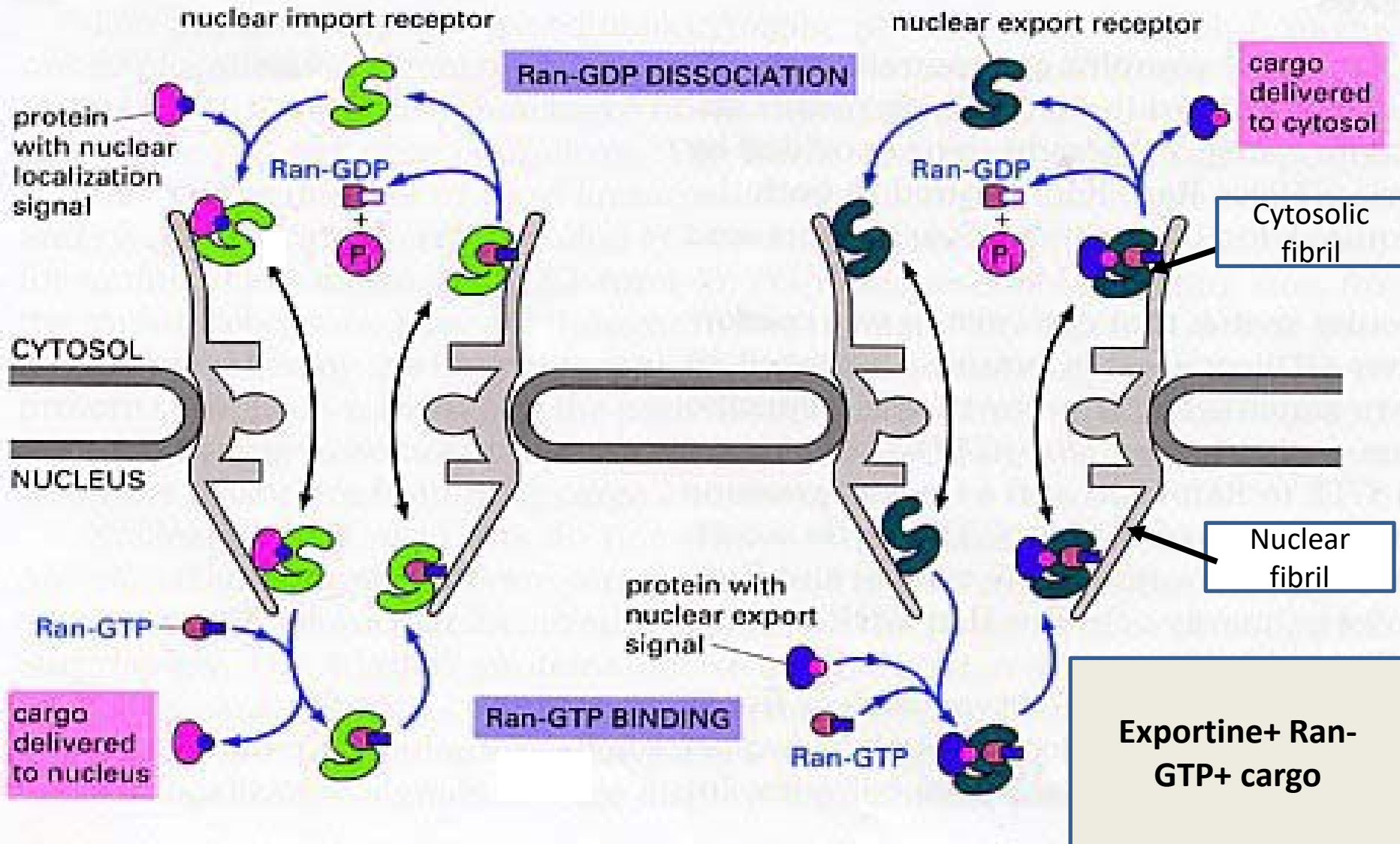
B: Top view

2.3. Mechanisms of exchange between the nucleus and the cytosol



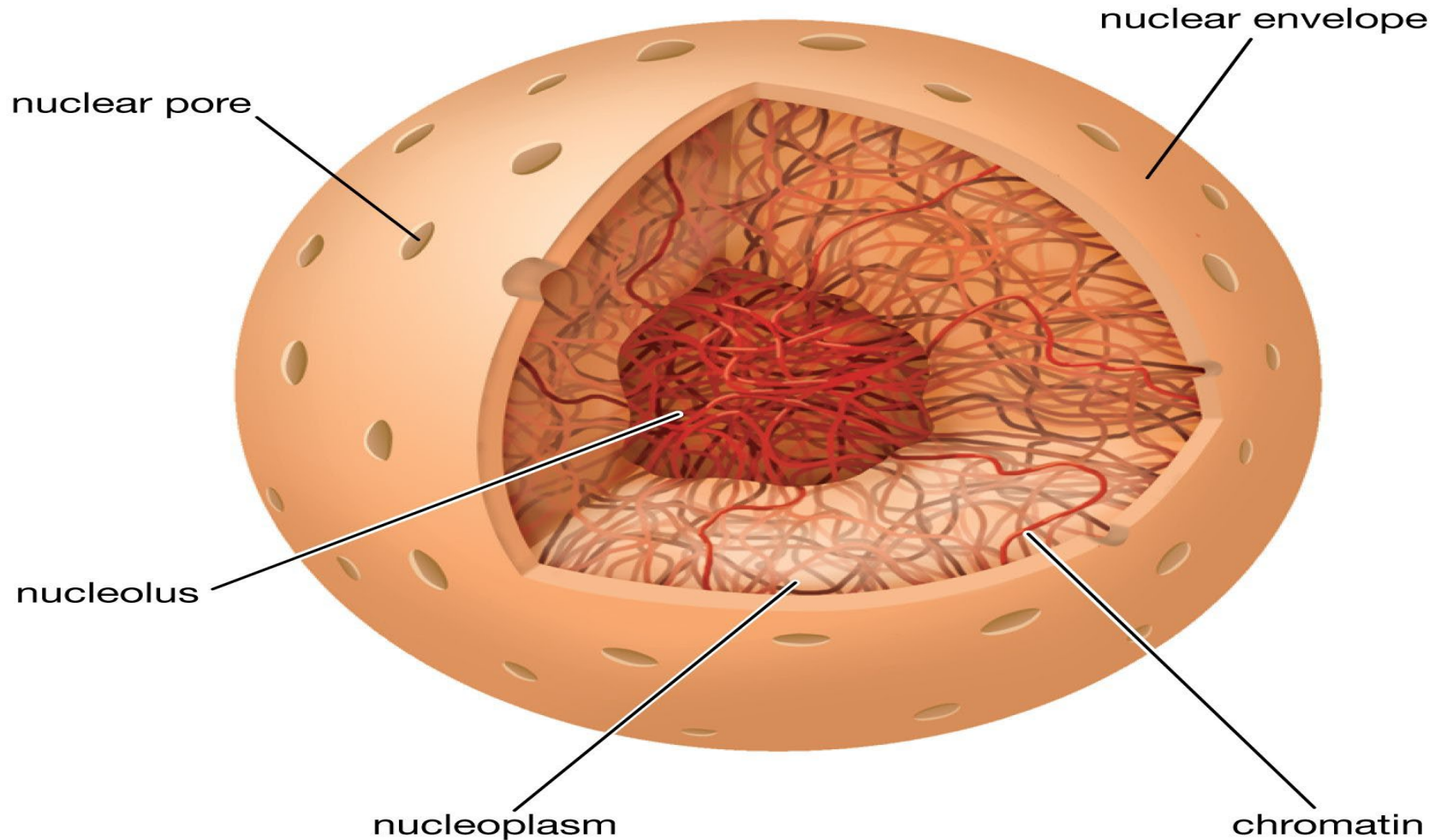


2.3. Mechanisms of exchange between the nucleus and the cytosol

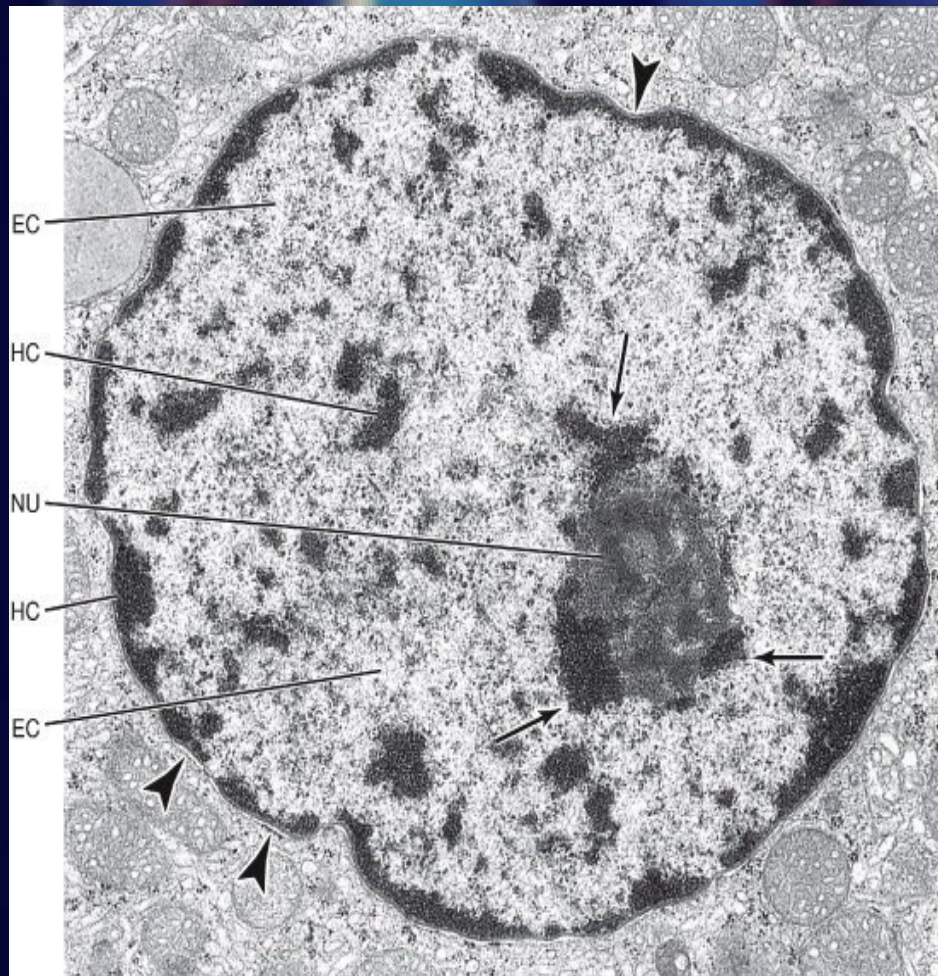


2.4. Nucleolus

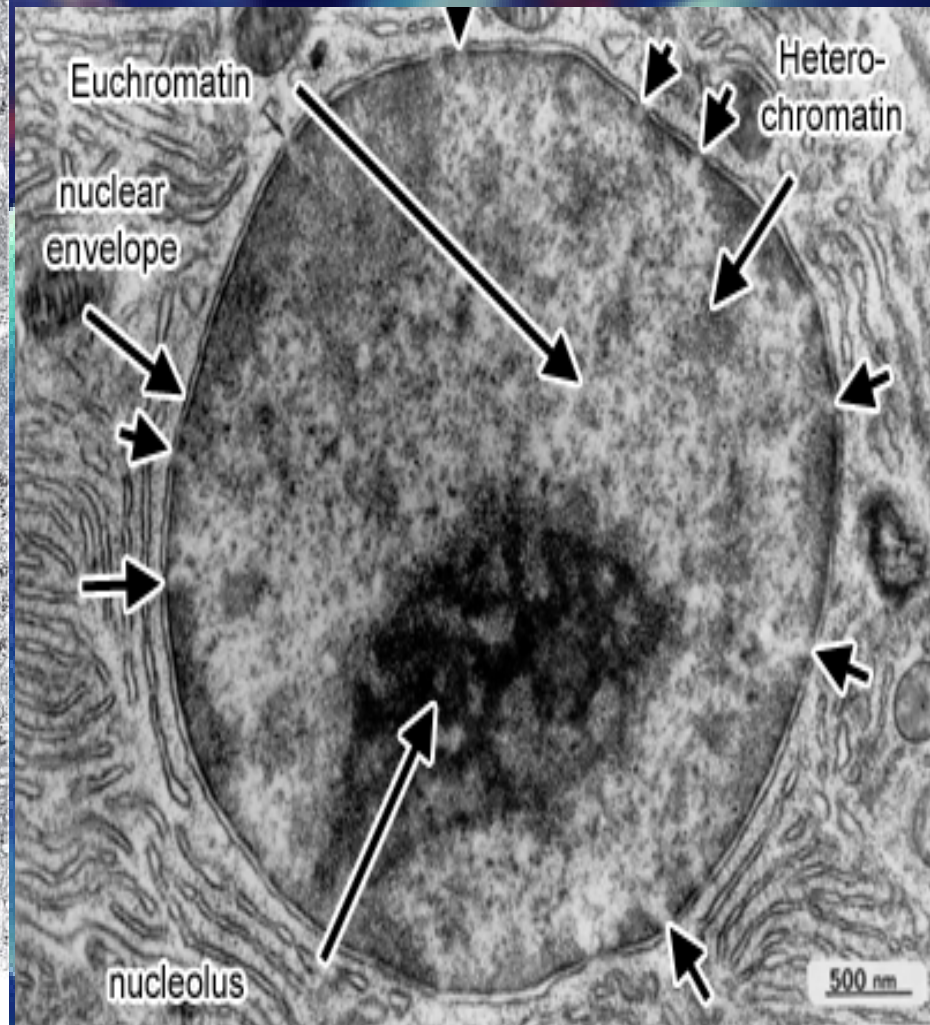
Nucleus



2.4. Nucleolus



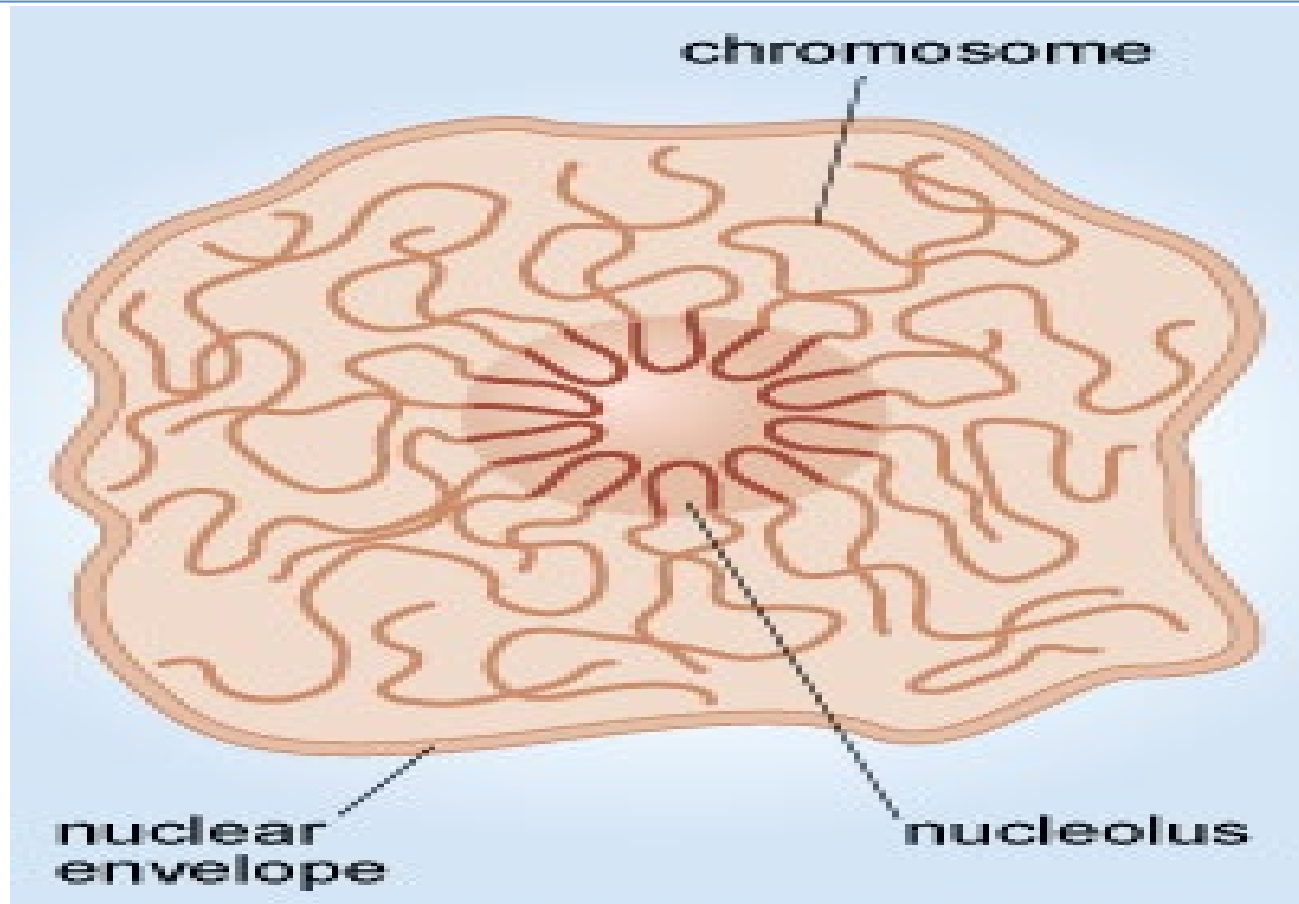
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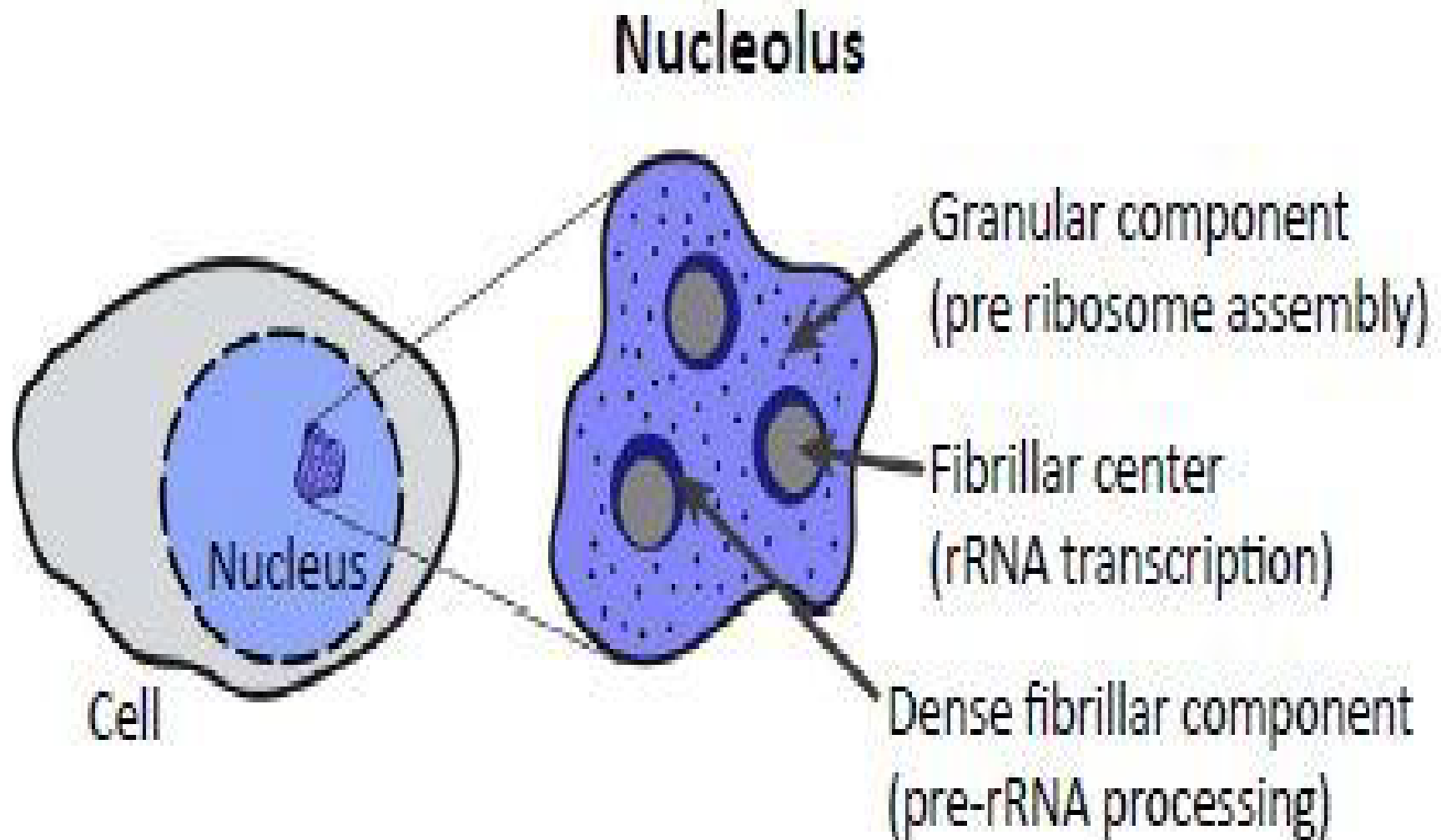
2.4. Nucleolus

Nucleolus components are:

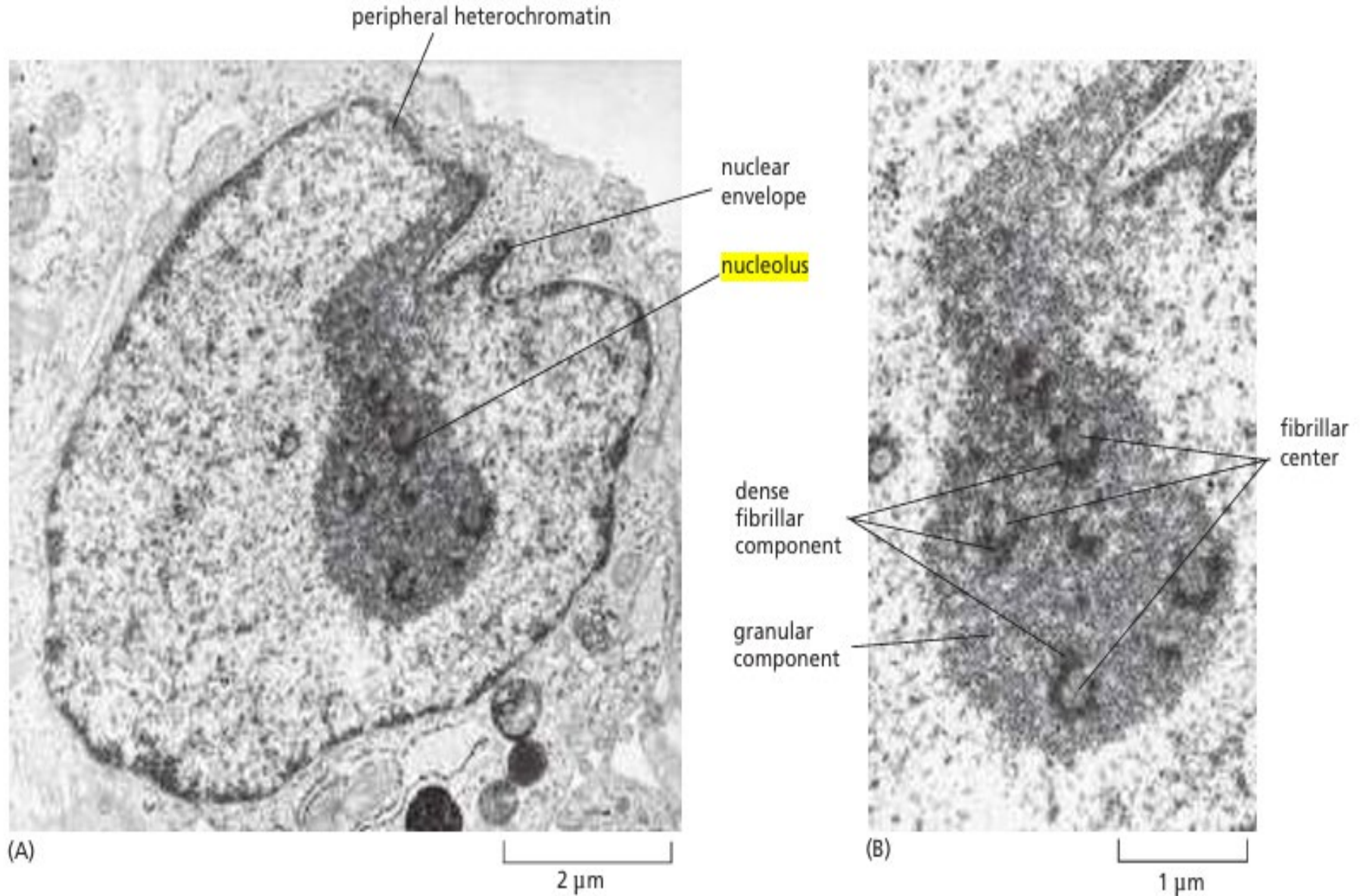
- rDNA (10 chromosomes: 13, 14, 15, 21 and 22)
- Proteins (Ribosomal proteins and RNA polymerase I)
- rRNA: 47S (precursor), 28S, 5.8S, 18S (mature products) and 5S (which imported from the nucleoplasm)



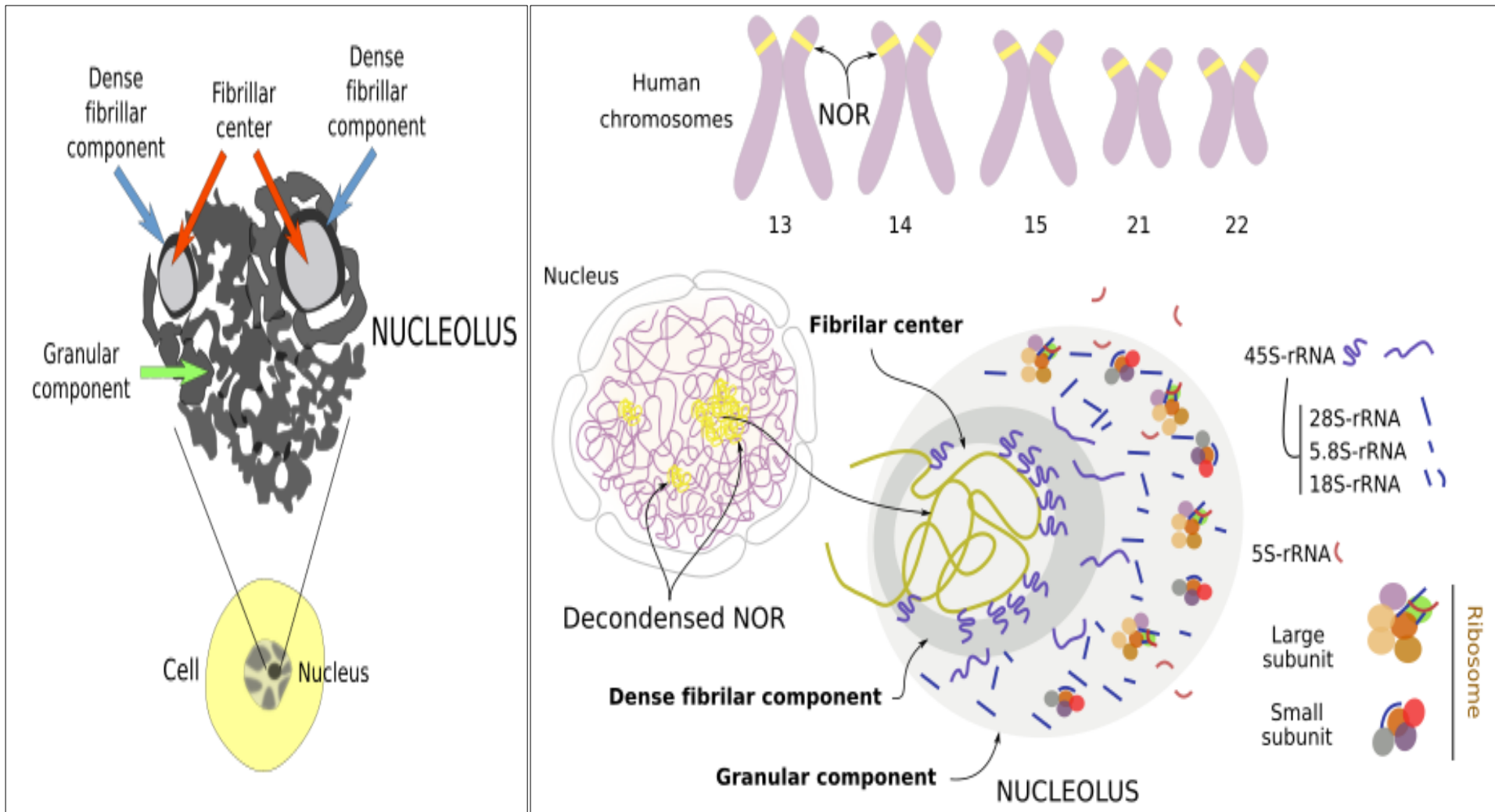
A. The Three Compartments of the Nucleolus



A. The Three Compartments of the Nucleolus

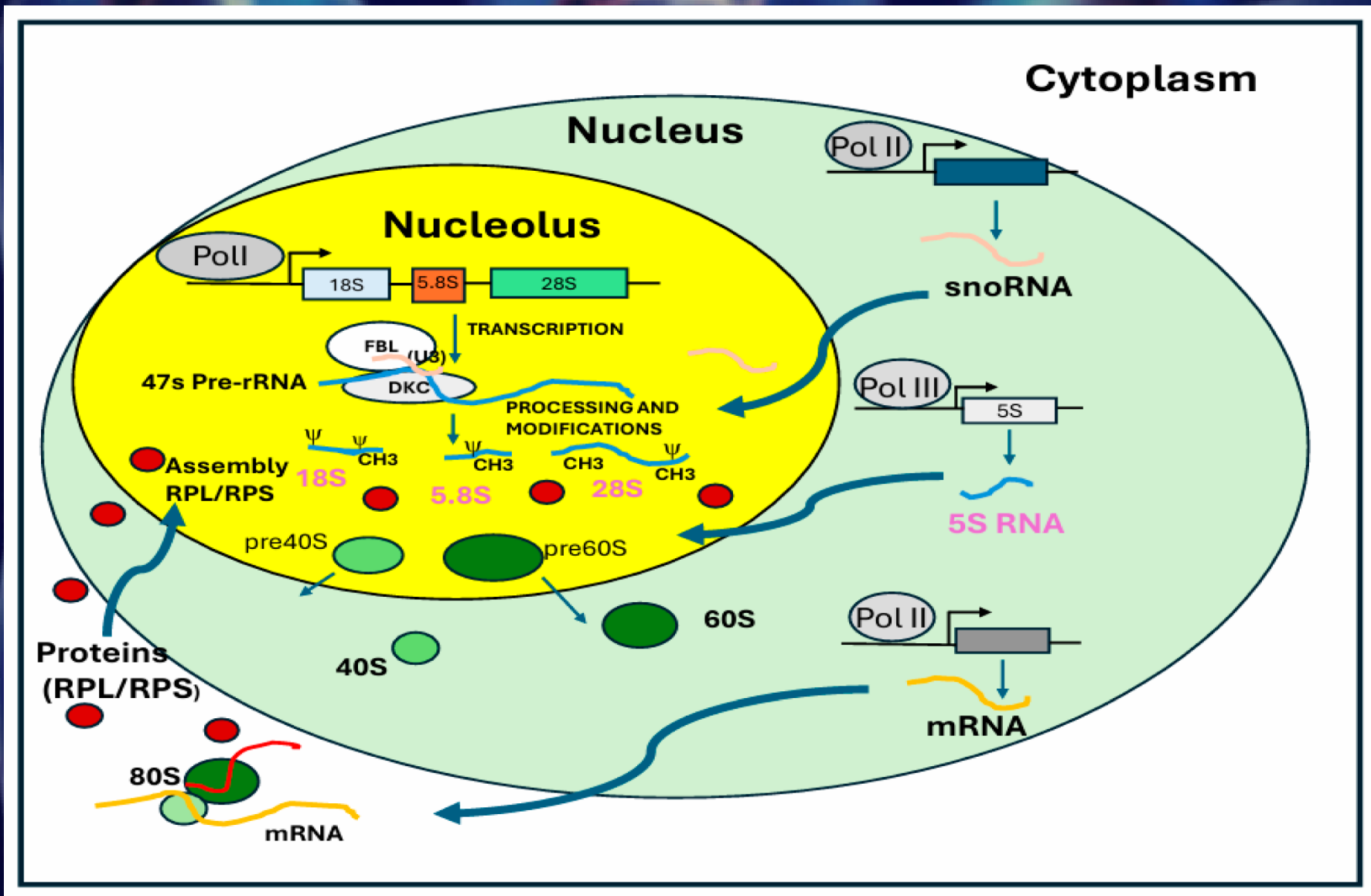


B. Ribosome Biogenesis: From Chromosomes to Assembly

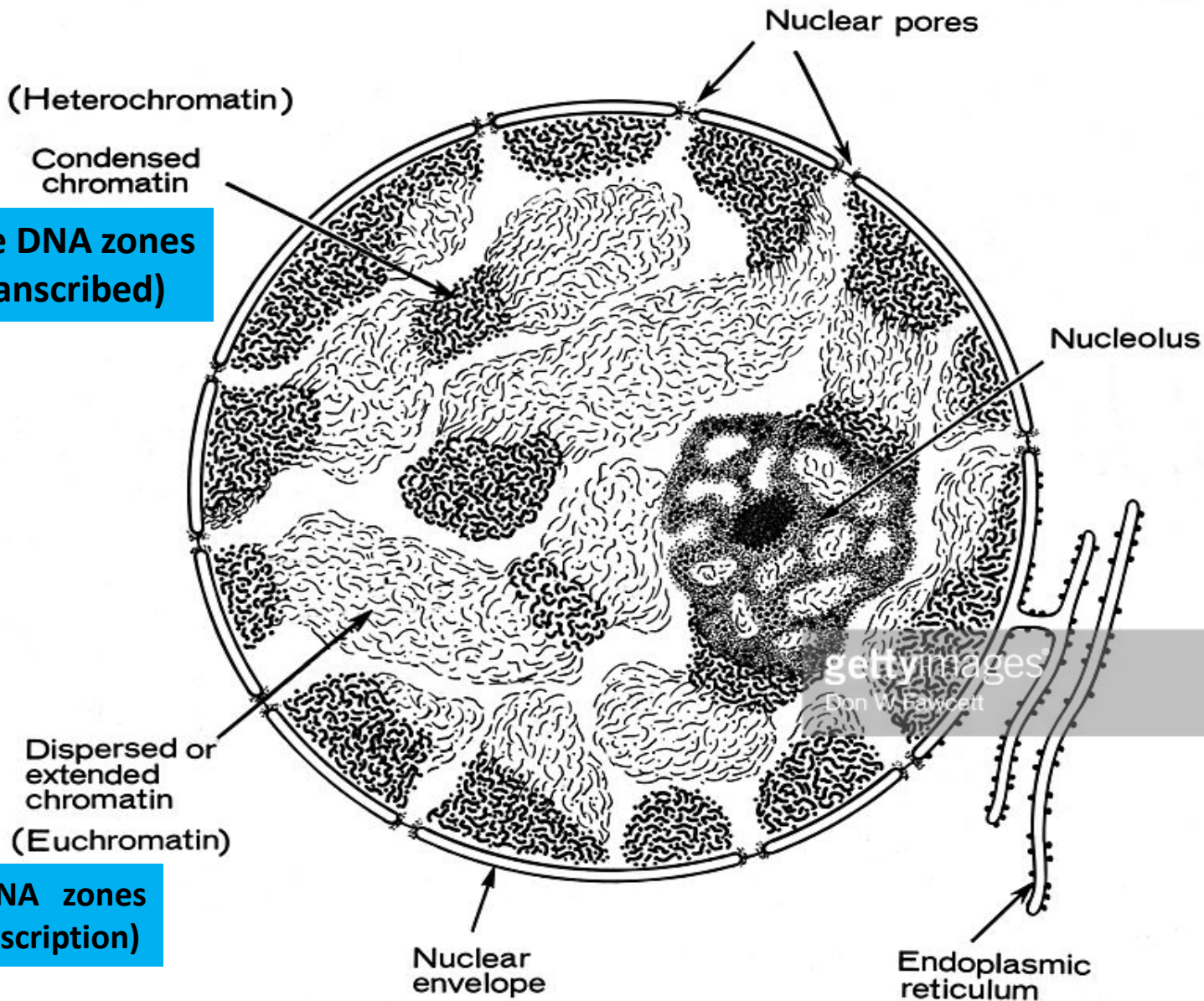


Fibrillar center: is the region where rDNA is transcribed to RNA 47S, **Dense fibrillar component**: region where pre rRNA 47S is located and cleaved. **Granular component**: region where rRNAs and ribosomal proteins are assembled to Large and Small subunits

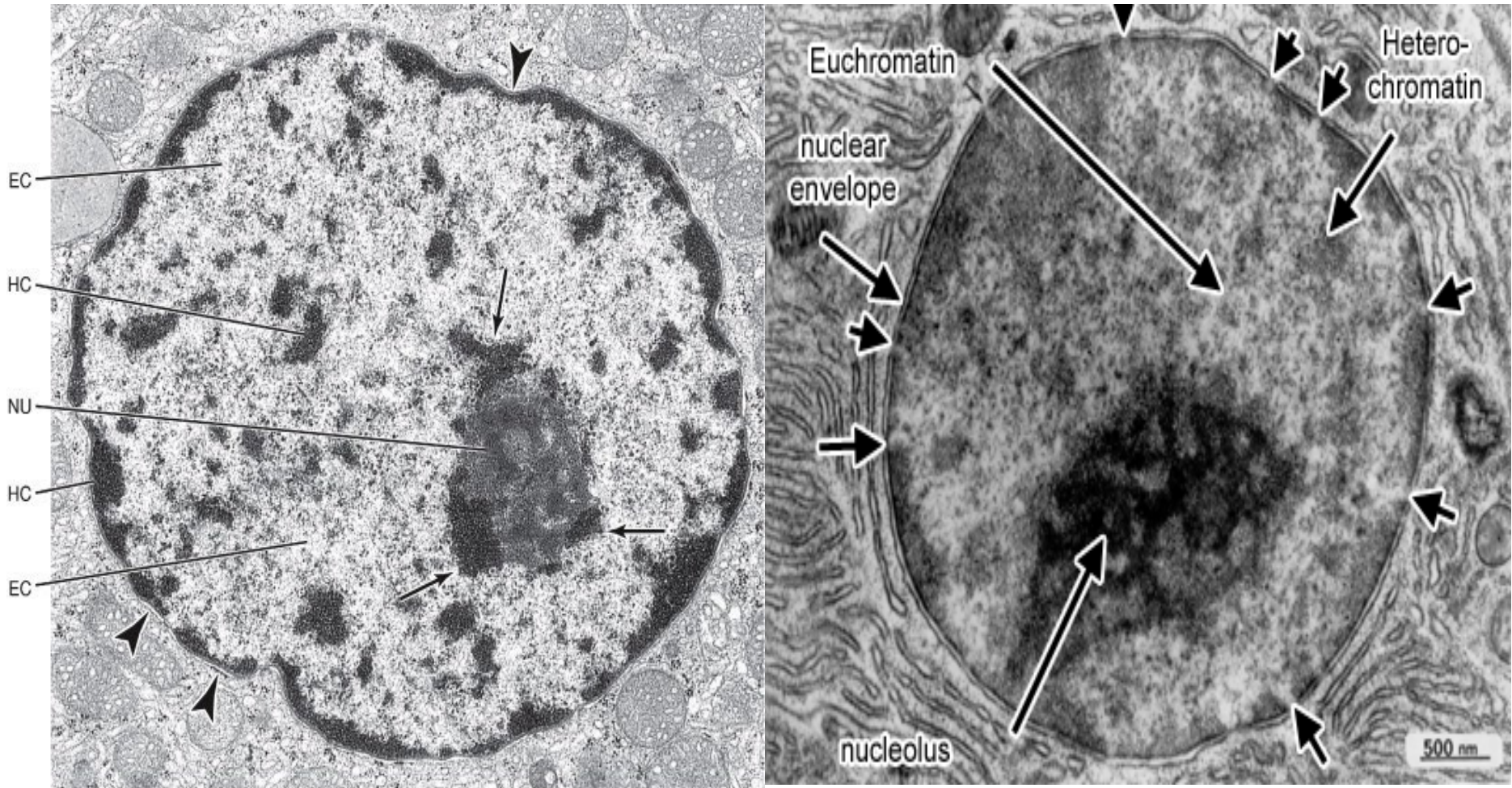
B. Ribosome Biogenesis: From Chromosomes to Assembly



2.5. Chromatin

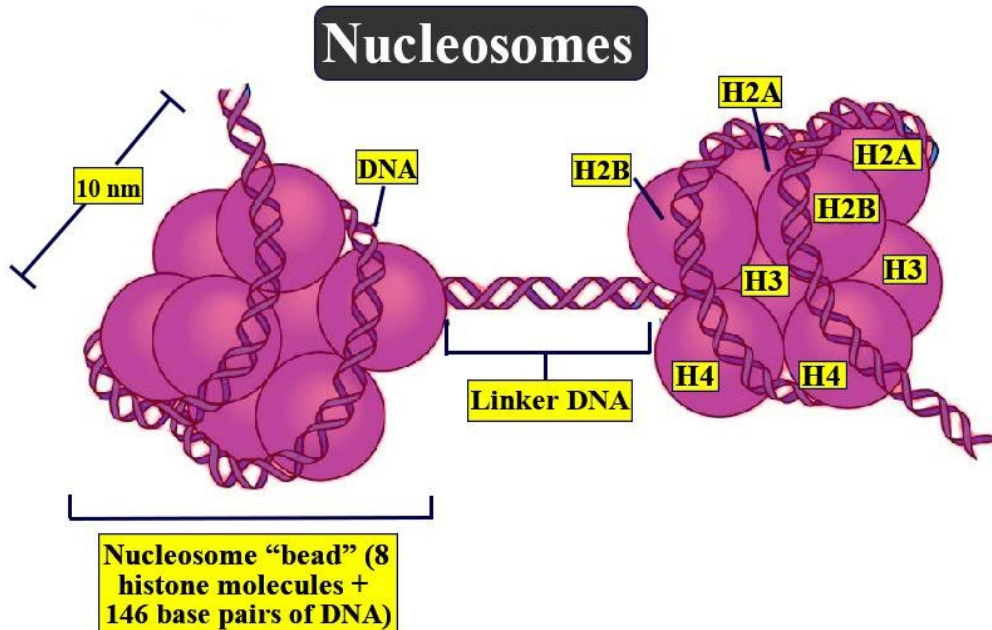
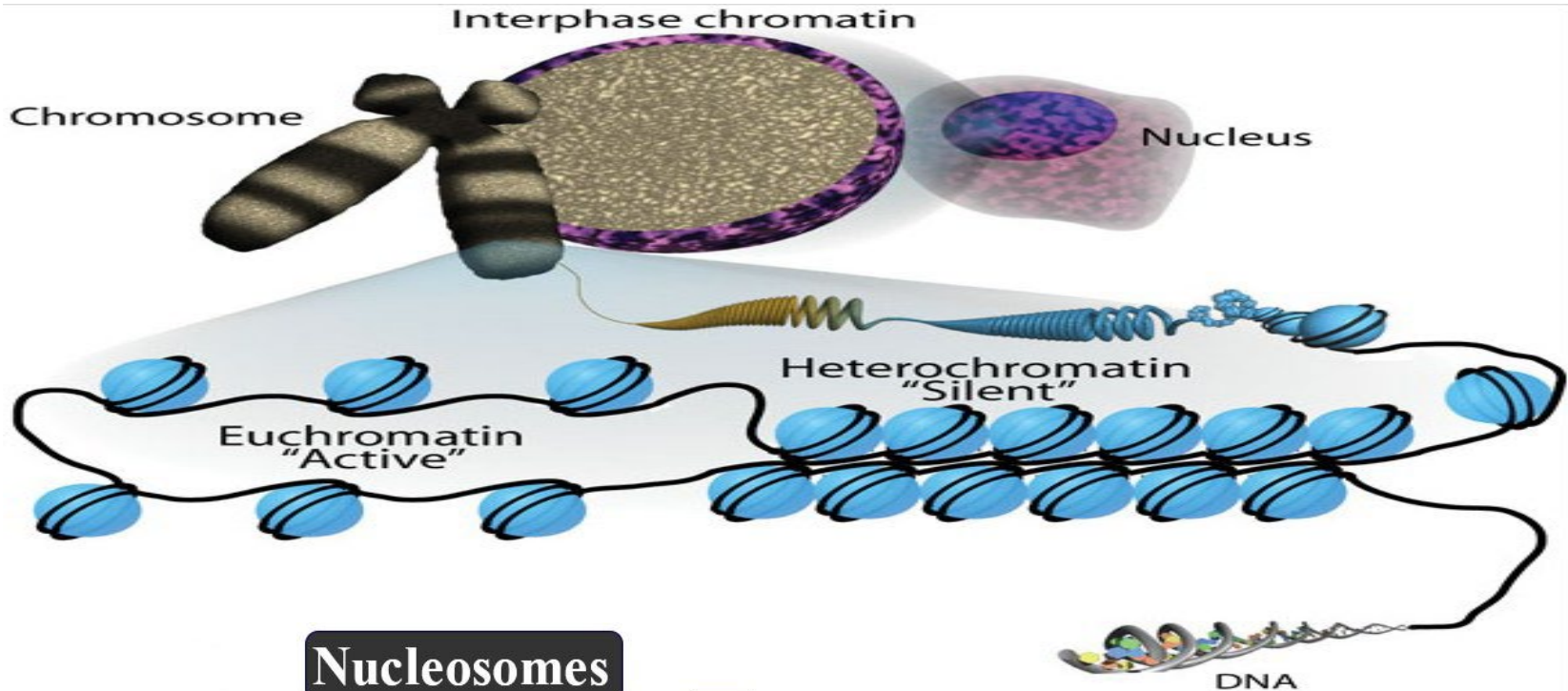


2.5. Chromatin



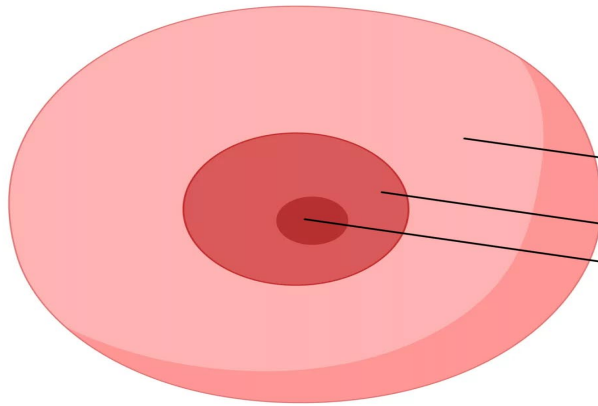
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Structure of chromatin



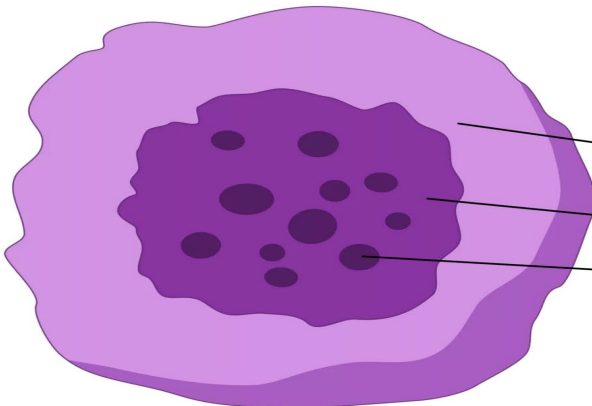
Octamere: H2A, H2B, H3, H4.

Morphological differences between normal and cancer cells



Normal cell

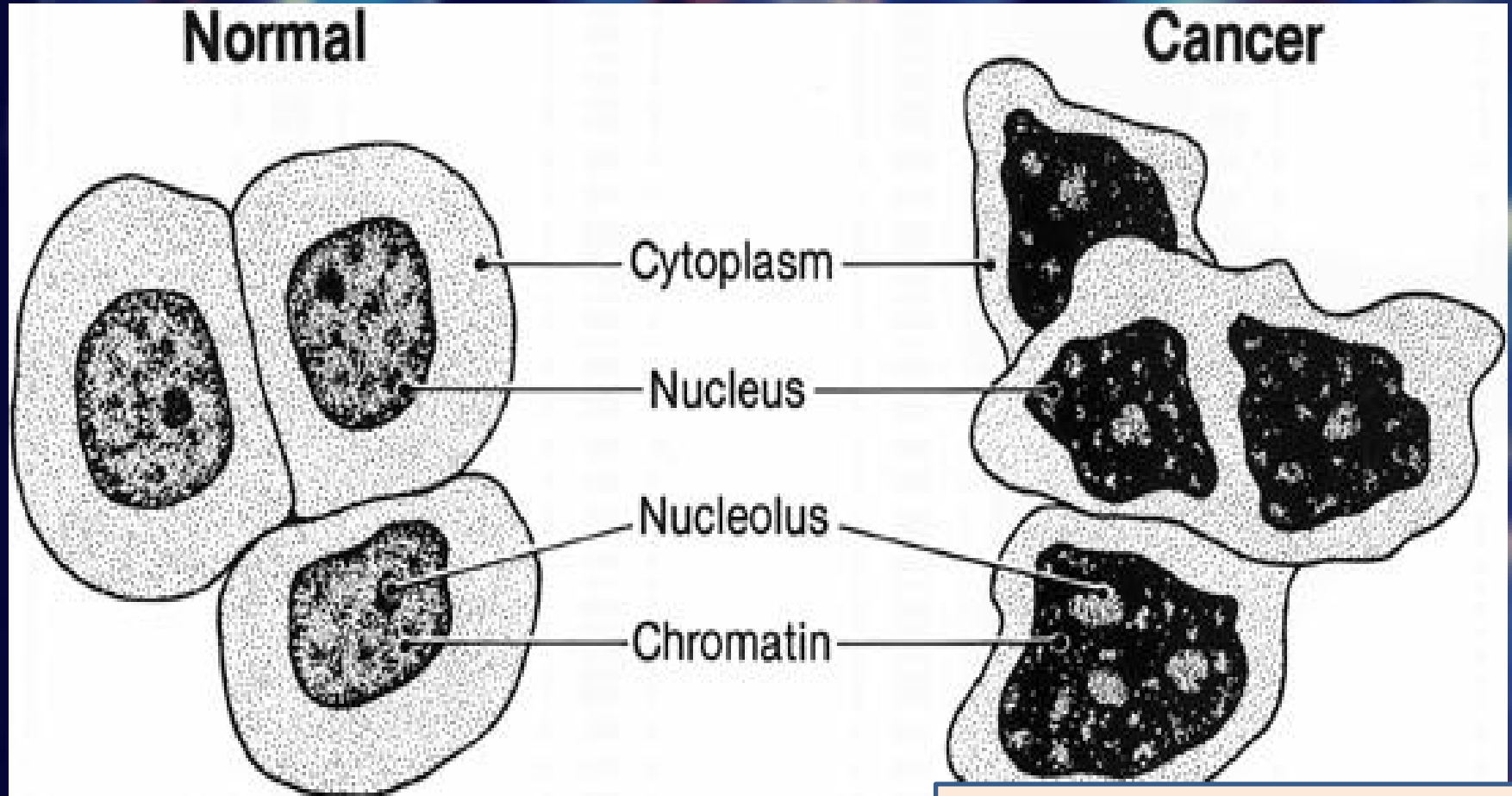
- Large cytoplasm
- Spherical shape of nucleus
- Single nucleolus



Cancer cell

- Small cytoplasm
- Irregular shape of nucleus
- Multiple nucleolus

Morphological differences between normal and cancer cells

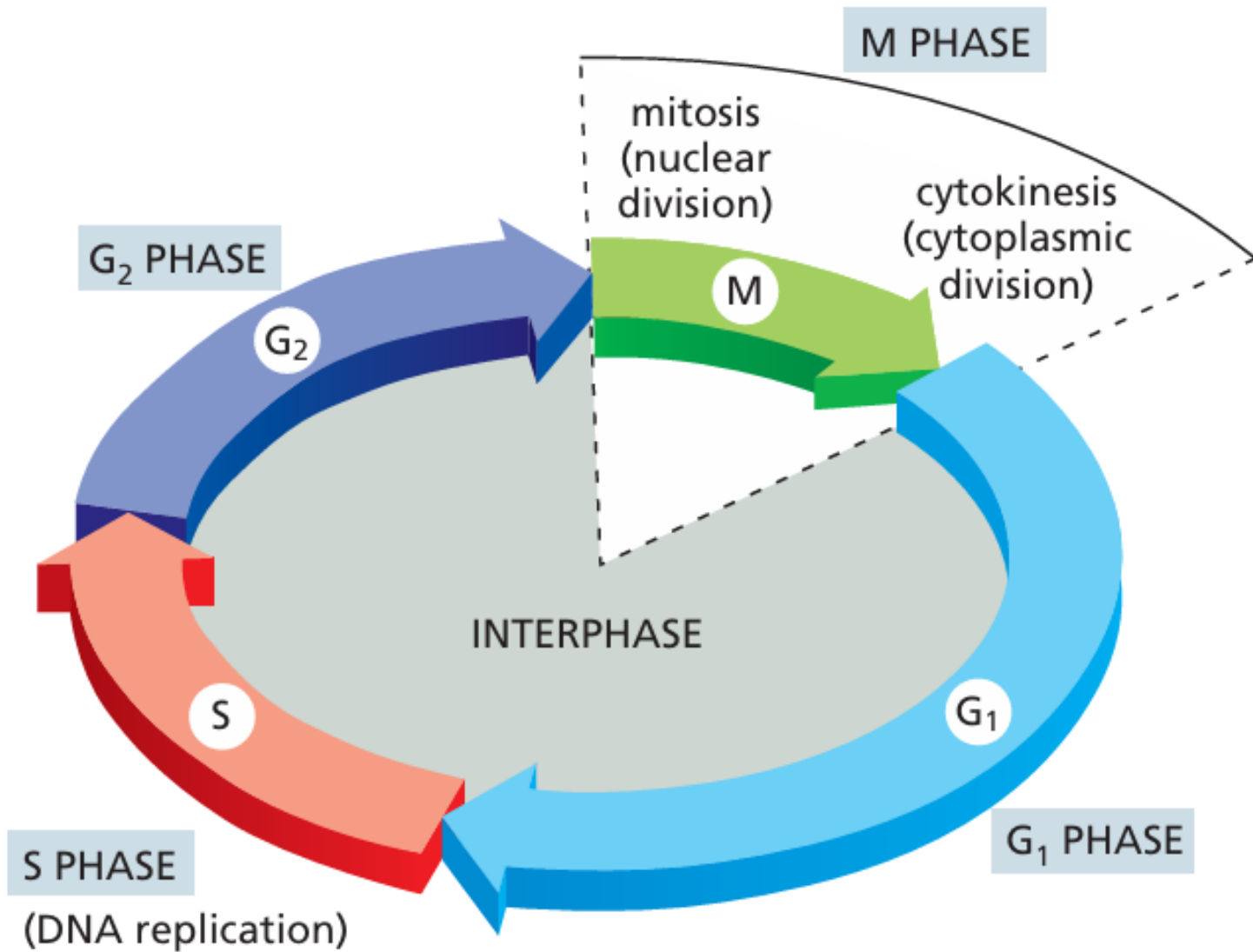


- Large cytoplasm
- Single nucleus
- Single nucleolus
- Fine chromatin

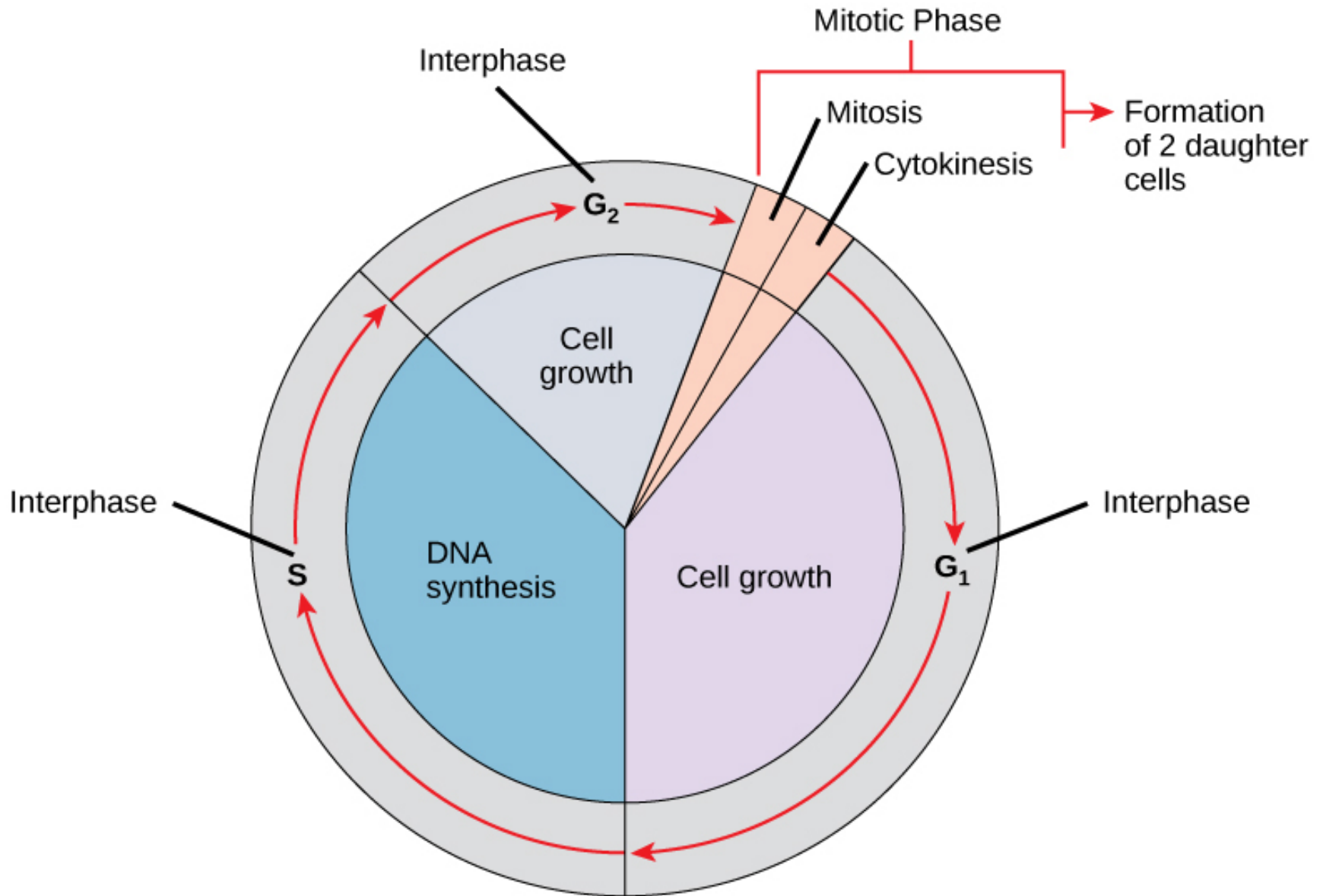
- Small cytoplasm
- Multiple nuclei
- Multiple and large nucleoli
- Coarse chromatin

II. The Cell Cycle

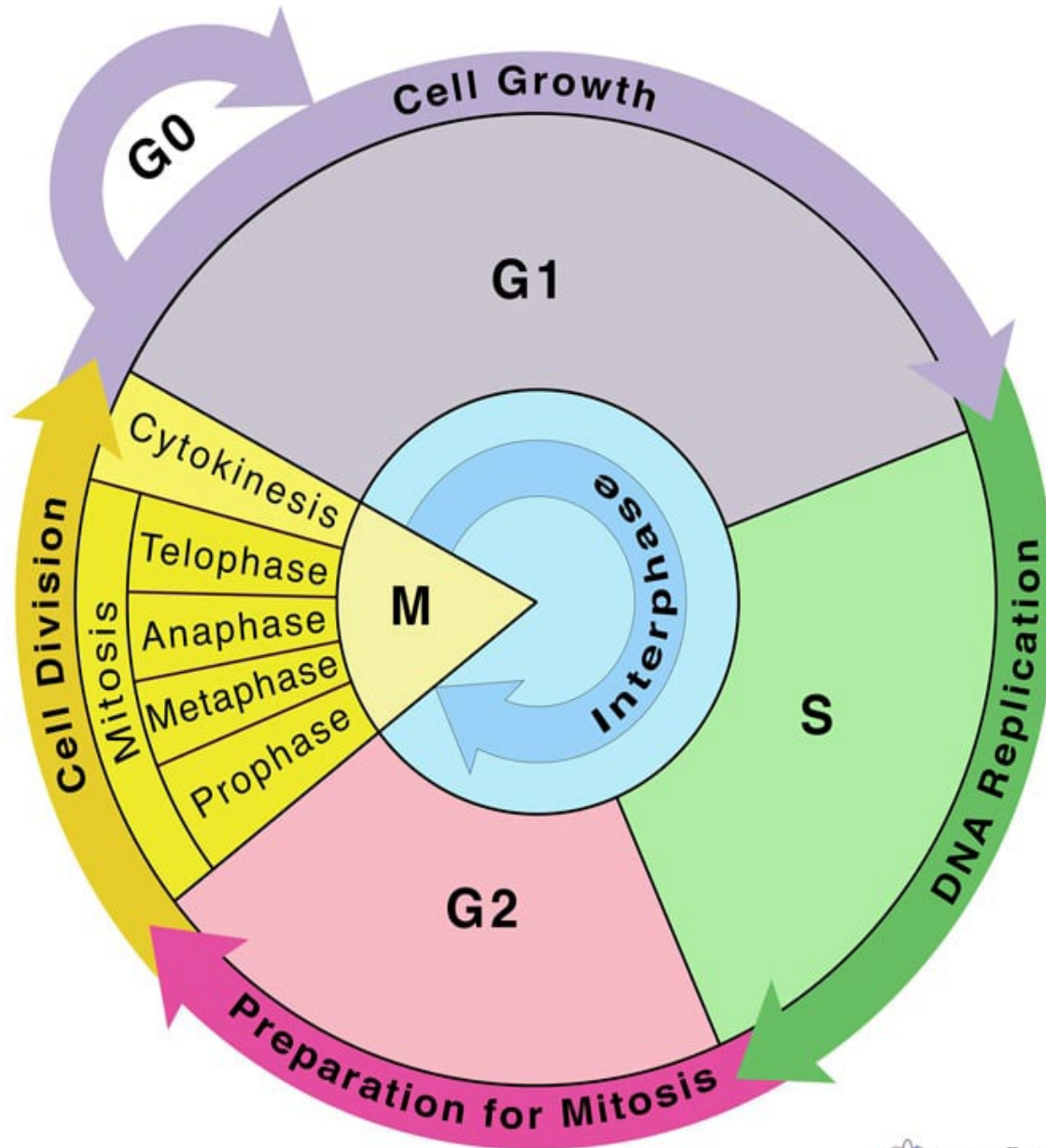
1. Overview of the Cell Cycle
2. Mitosis and Cytokinesis
3. The Cell-Cycle Control System



Cell cycle phases



Cell cycle phases



G1 - Growth

S - DNA synthesis

G2 - Growth and
preparation for
mitosis

M - Mitosis
(cell division)

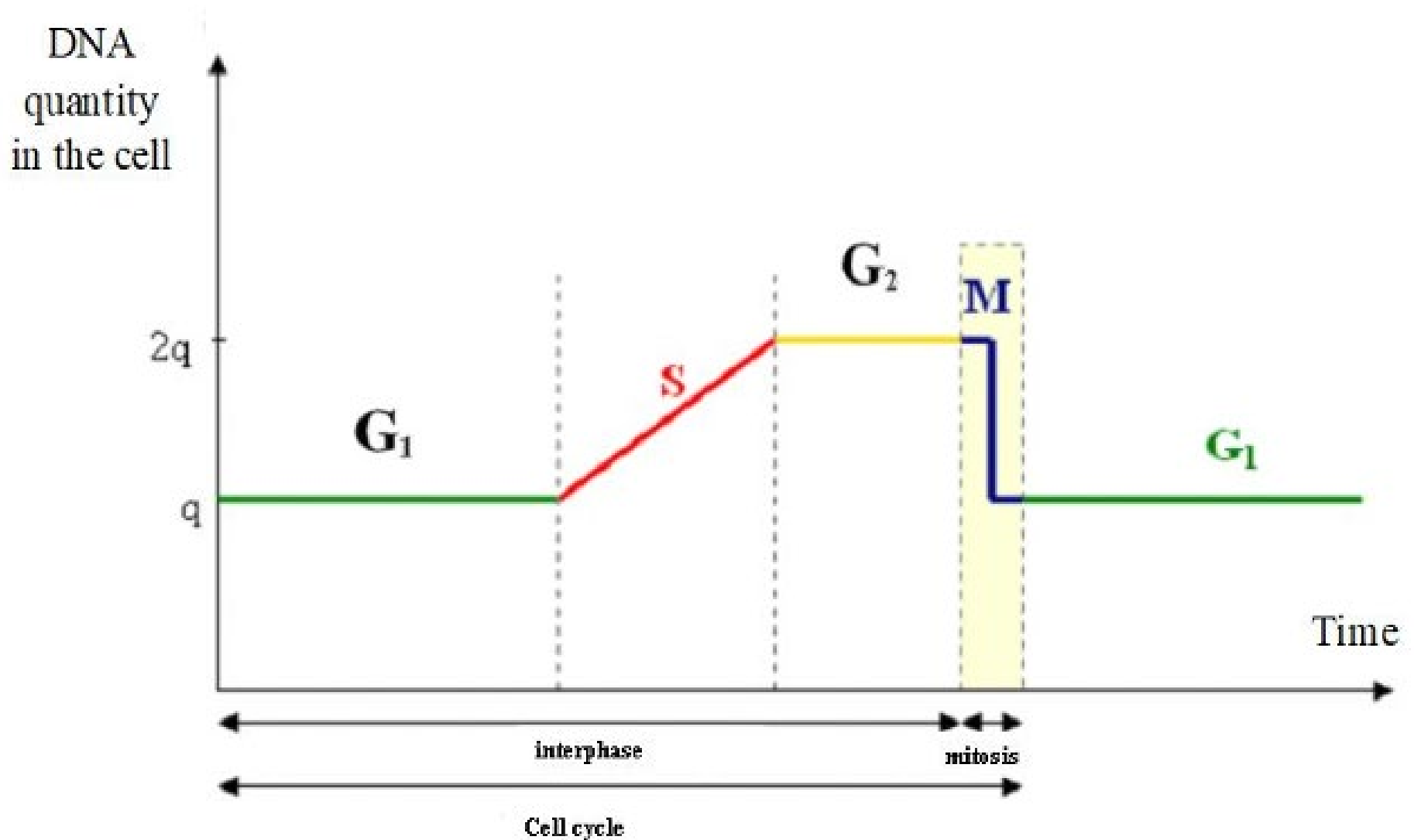
Summary of cell cycle events

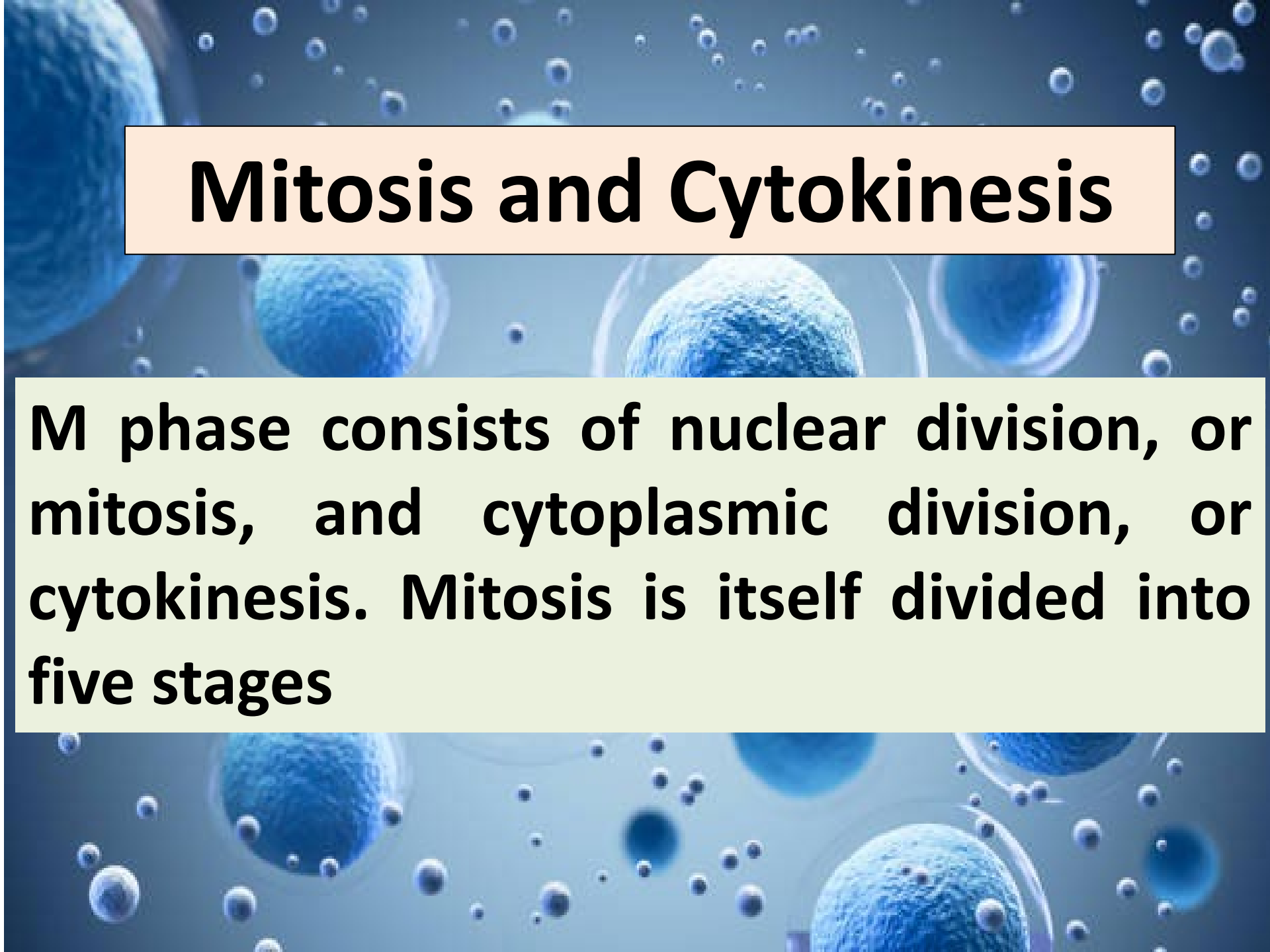
Phase	Core Process	Key Molecular Action
G1	Growth	The cell is in its most active "working" state. Protein synthesis & organelle duplication.
S	DNA Synthesis	DNA replication and centrosome duplication.
G2	Preparation	Final growth & DNA error checking.
M	Division	Mitosis (Nuclear) & Cytokinesis (Cytoplasmic).
G0	Quiescence	Exit from the active cycle.

Cell Cycle Duration by Human Cell Type

Human Cell Type	Total Cycle	G1 Phase	S Phase	G2 Phase	M Phase
Intestinal cell	~12–24 hours	~5–10 hours	~7 hours	~3 hours	~1 hour
Bone Marrow	~18 hours	~5 hours	~8 hours	~4 hours	~1 hour
Cancer Cell	~18–24 hours	~5–10 hours	~8 hours	~3 hours	~1 hour
Hair Follicle	~24–72 hours	~10–50 hours	~8 hours	~4 hours	~1 hour
Skin	~20 days	~19 days	~8 hours	~4 hours	~1 hour
Lung	~15–30 days	~2–4 weeks	~8 hours	~4 hours	~1 hour
Liver	6–12 months	Months	~8 hours	~4 hours	~1 hour
Nerve Cell	N/A (G0 Phase)	Permanent	None	None	None
Heart Muscle	N/A (G0 Phase)	Permanent	None	None	None

Evolution of DNA Content (q) During the Cell Cycle

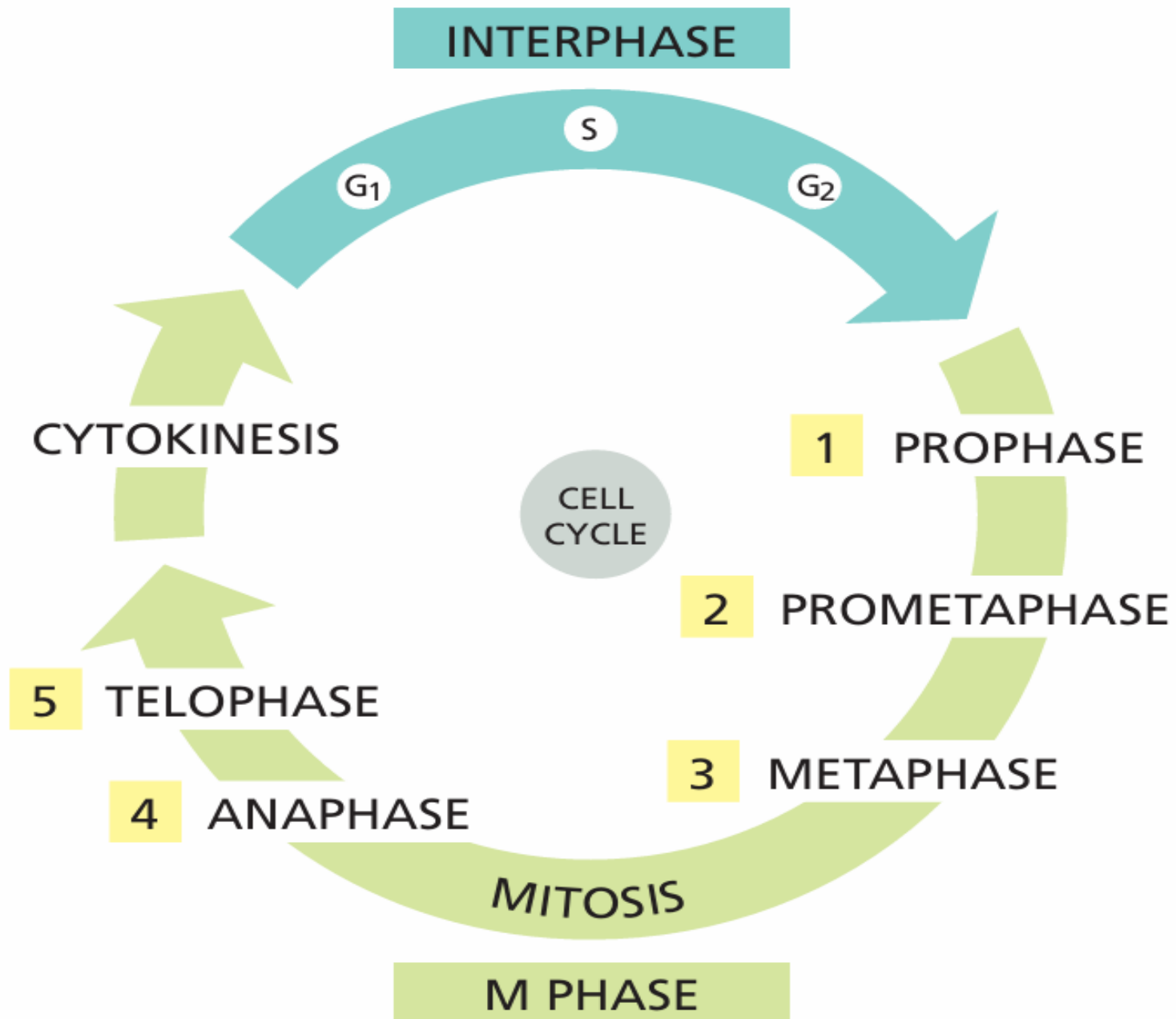


The background of the slide is a microscopic image showing various blue-colored cells. Some cells are large and spherical with a textured surface, while others are smaller and more rounded. The overall color palette is shades of blue, creating a scientific and cellular atmosphere.

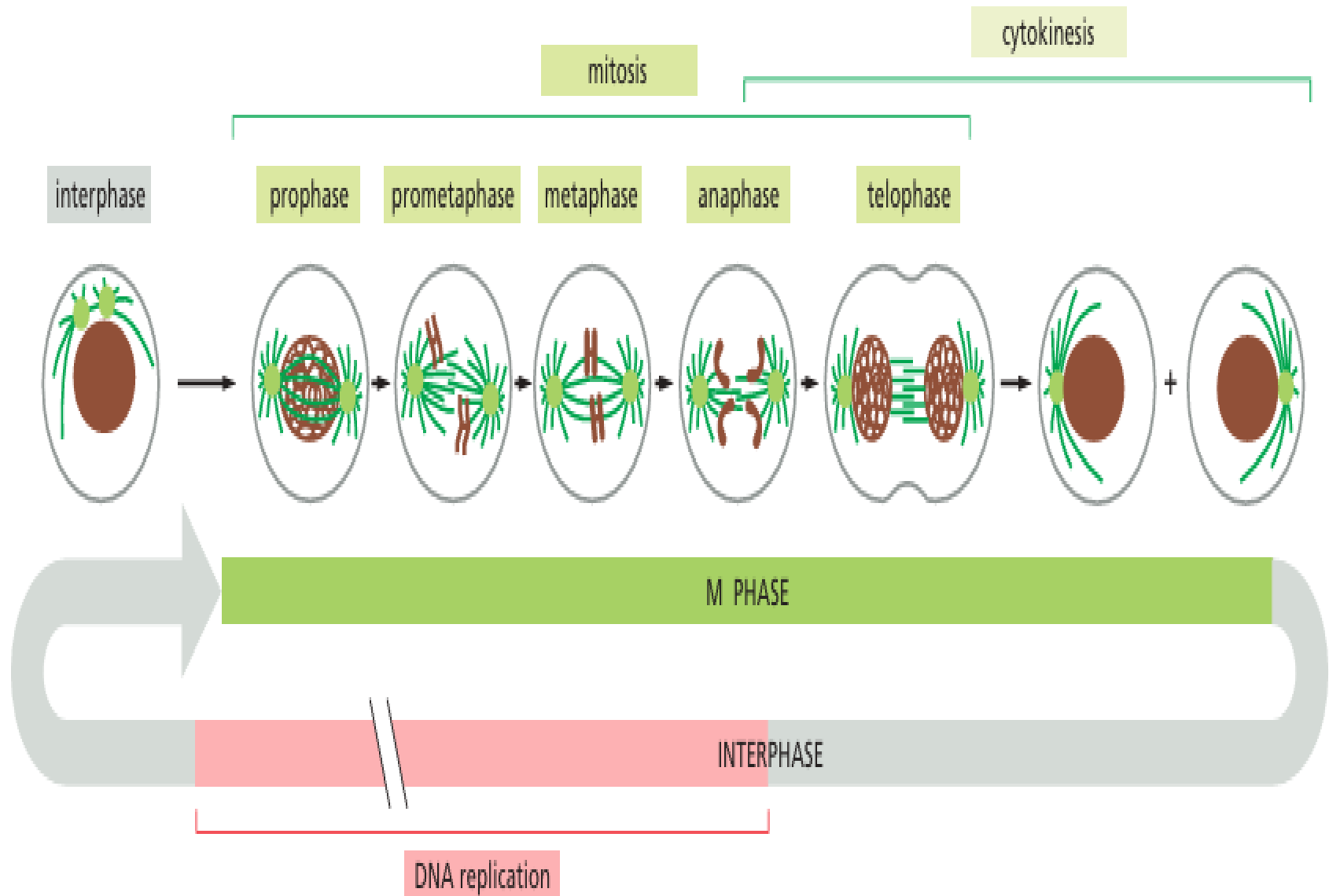
Mitosis and Cytokinesis

M phase consists of nuclear division, or mitosis, and cytoplasmic division, or cytokinesis. Mitosis is itself divided into five stages

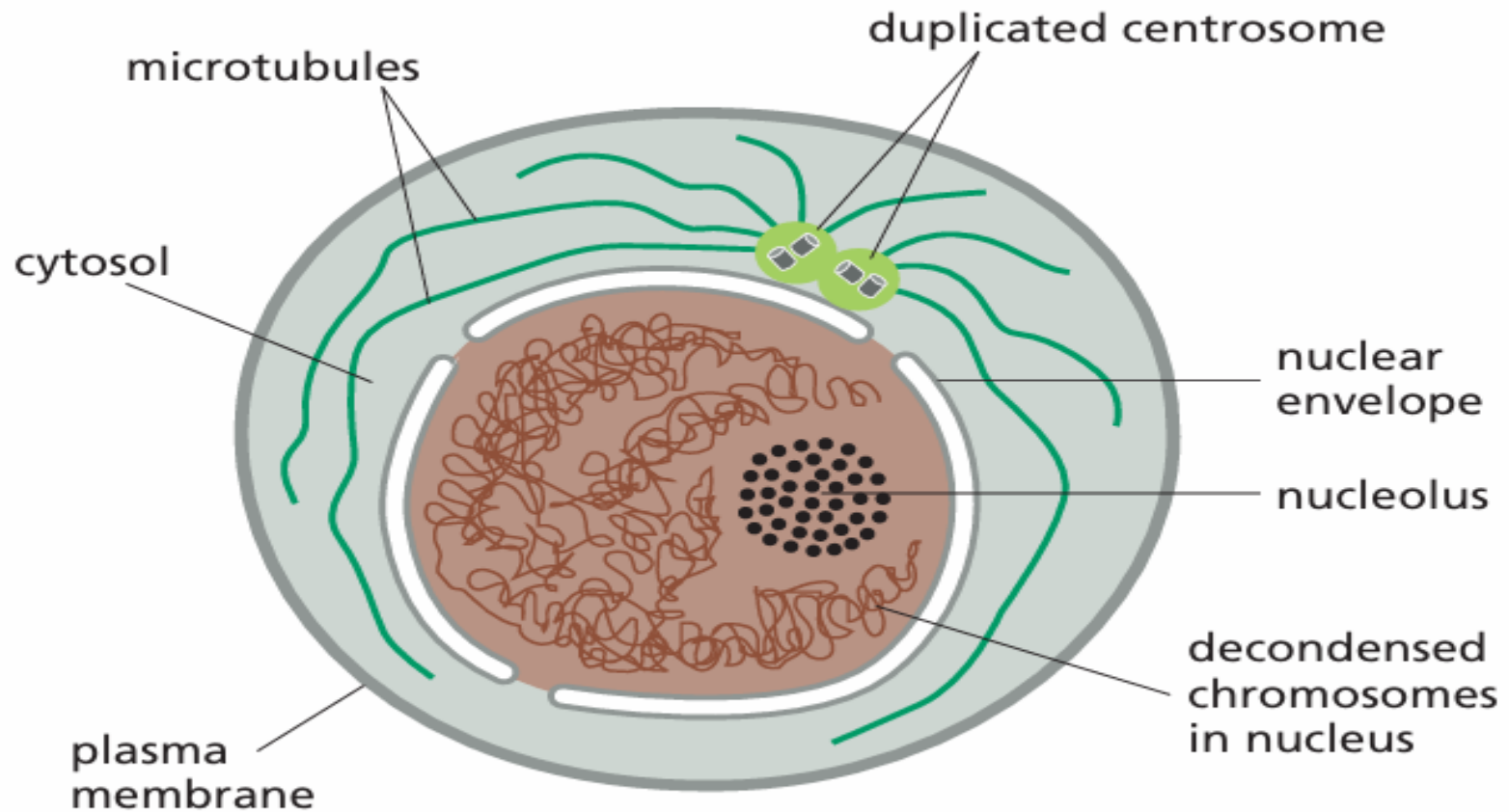
Mitosis and Cytokinesis



The events of human cell division as seen under a microscope

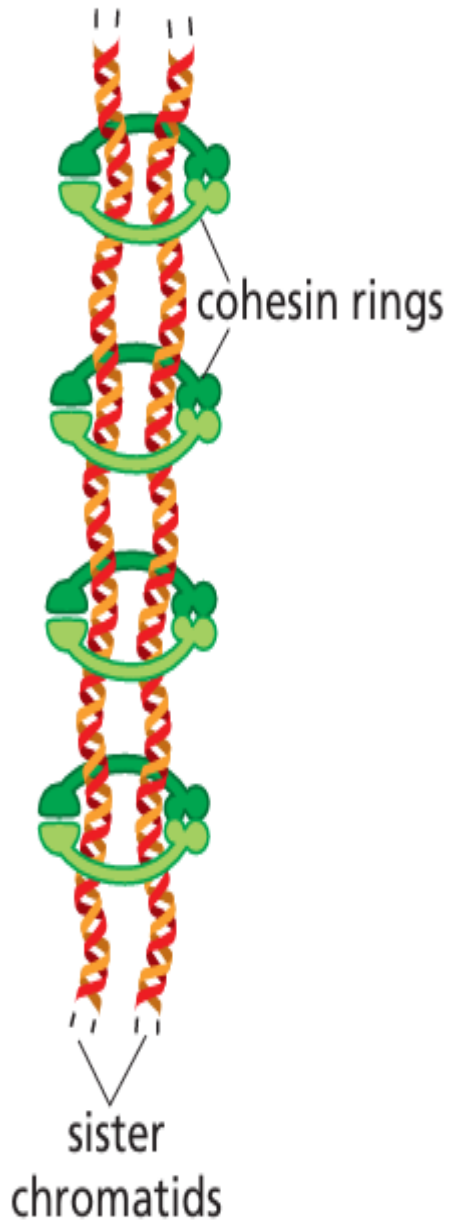


INTERPHASE



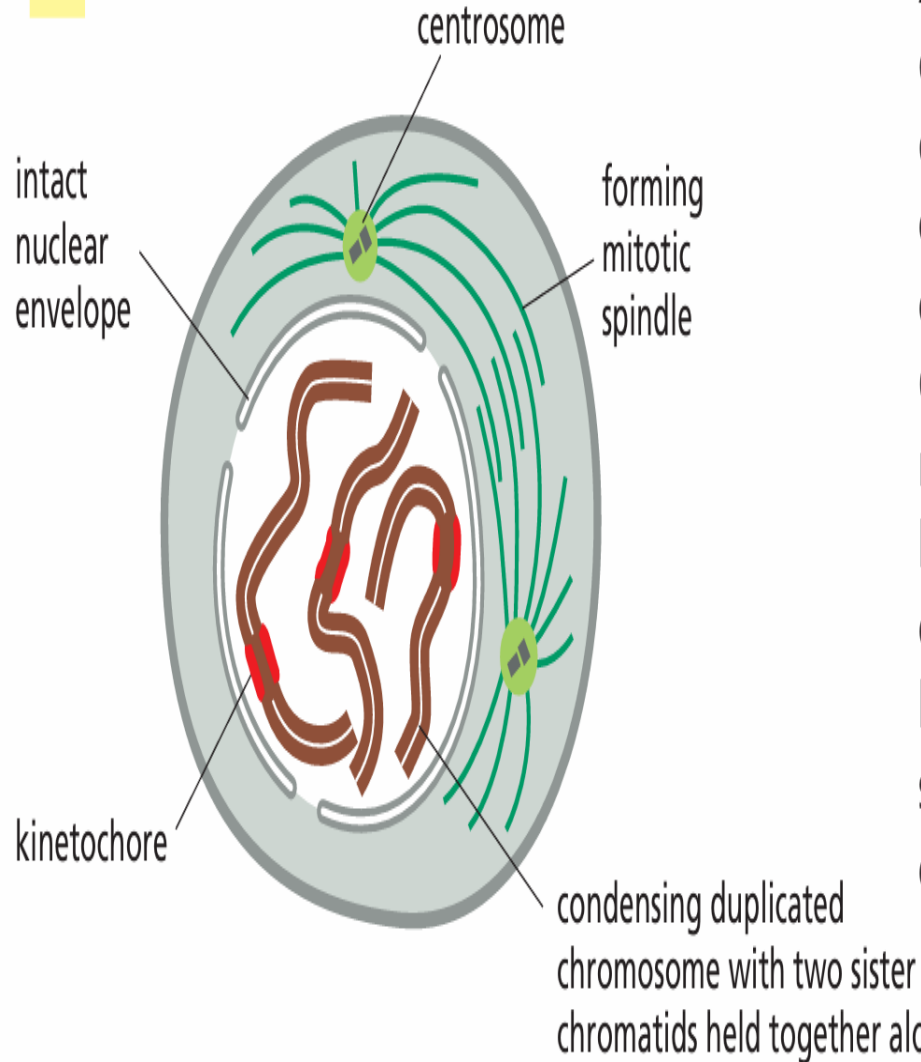
During interphase, the cell increases in size. The DNA of the chromosomes is replicated, and the centrosome is duplicated.

THE PRINCIPAL STAGES OF M PHASE IN A HUMAN CELL



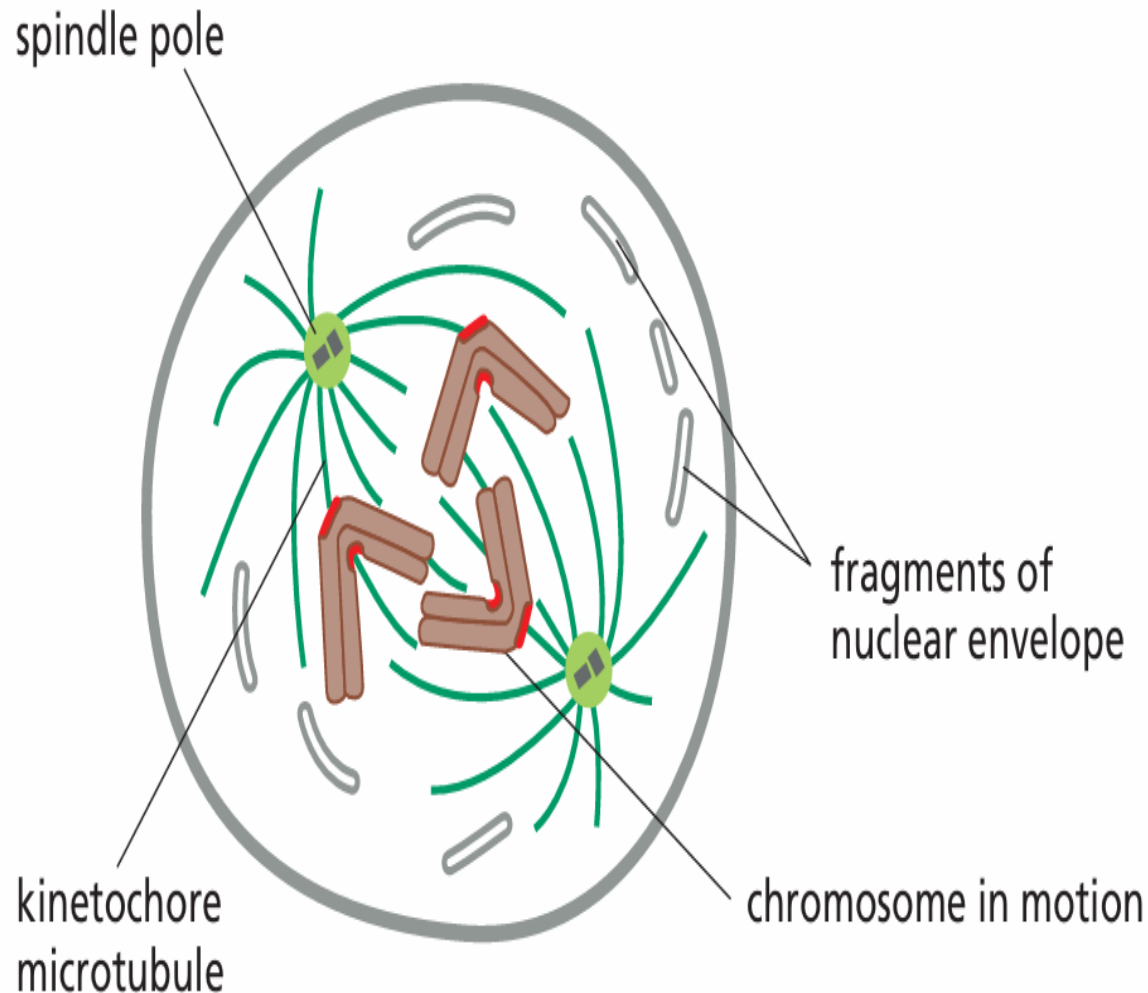
MITOSIS

1 PROPHASE



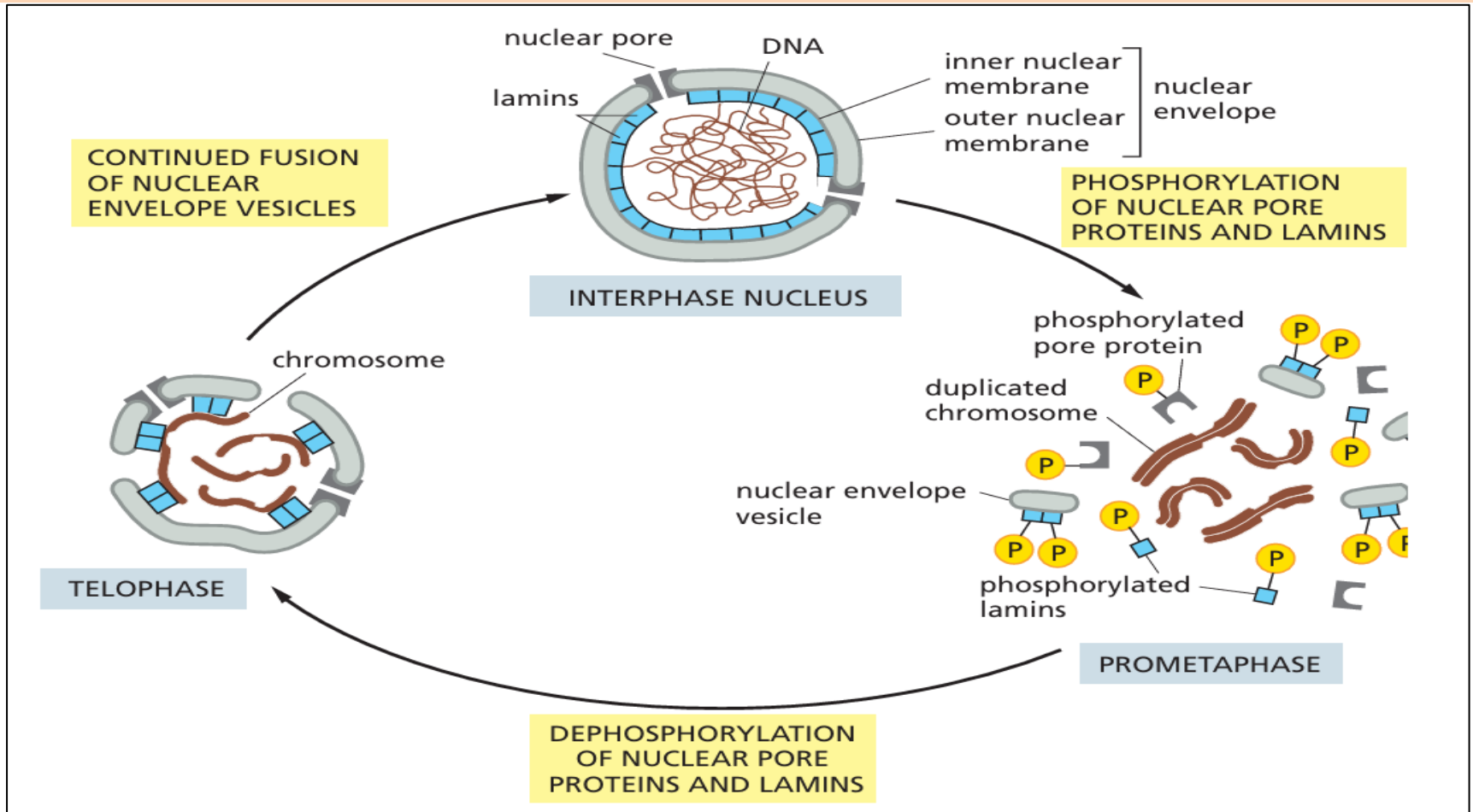
At **prophase**, the duplicated chromosomes, each consisting of two closely associated sister chromatids, condense. Outside the nucleus, the mitotic spindle assembles between the two centrosomes, which have begun to move apart. For simplicity, only three chromosomes are drawn.

2 PROMETAPHASE



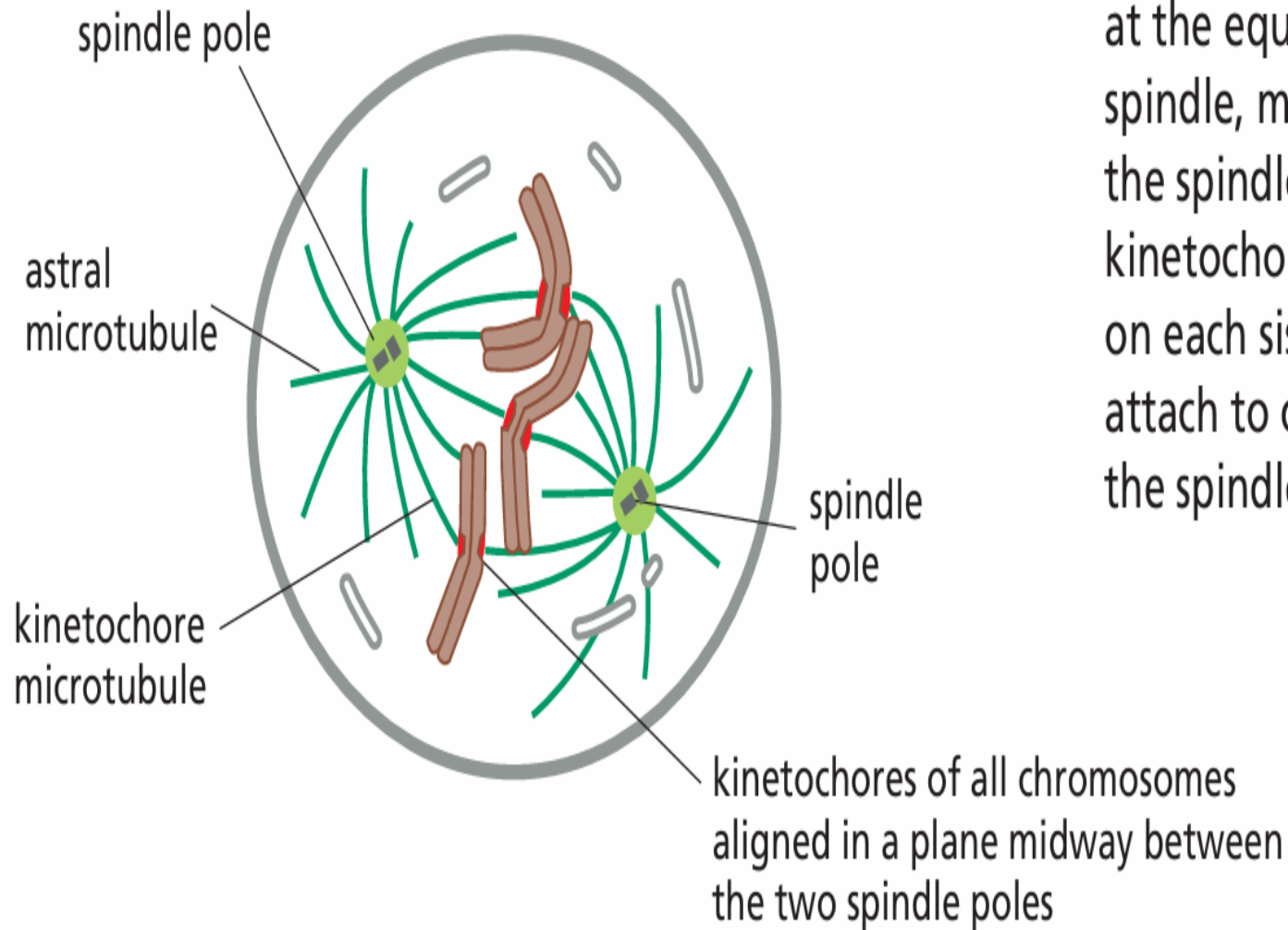
Prometaphase starts abruptly with the breakdown of the nuclear envelope. Chromosomes can now attach to spindle microtubules via their kinetochores and undergo active movement.

Dynamics of Nuclear Envelope Breakdown and Reassembly during Mitosis

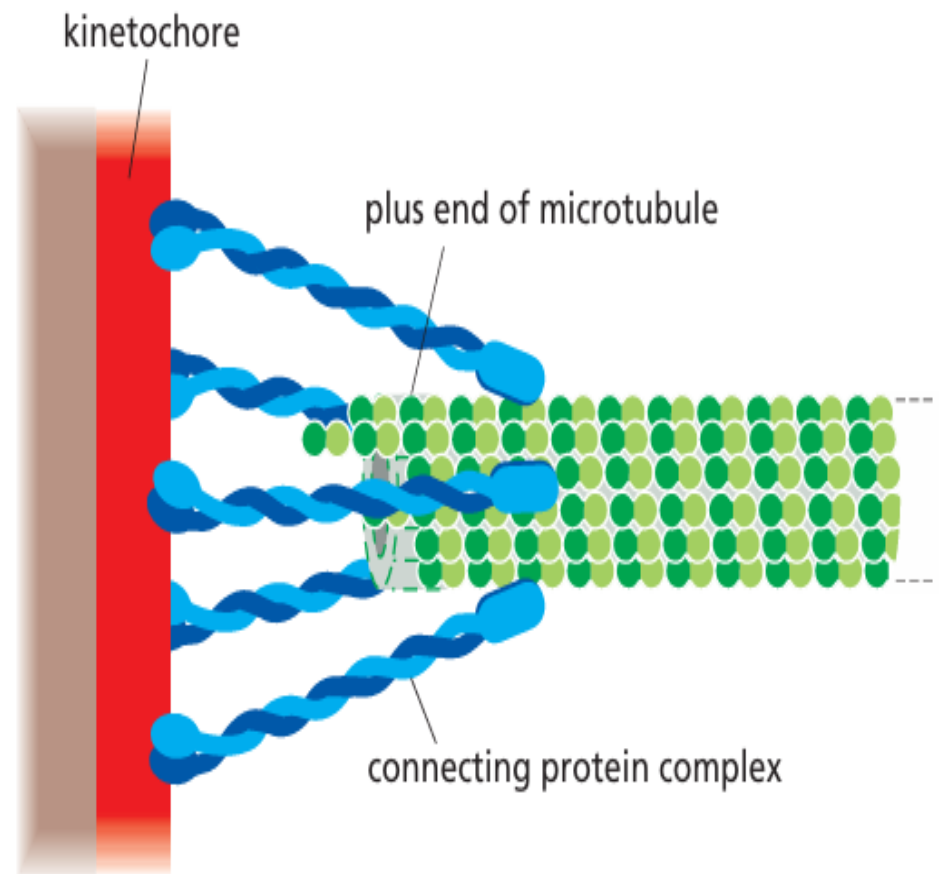
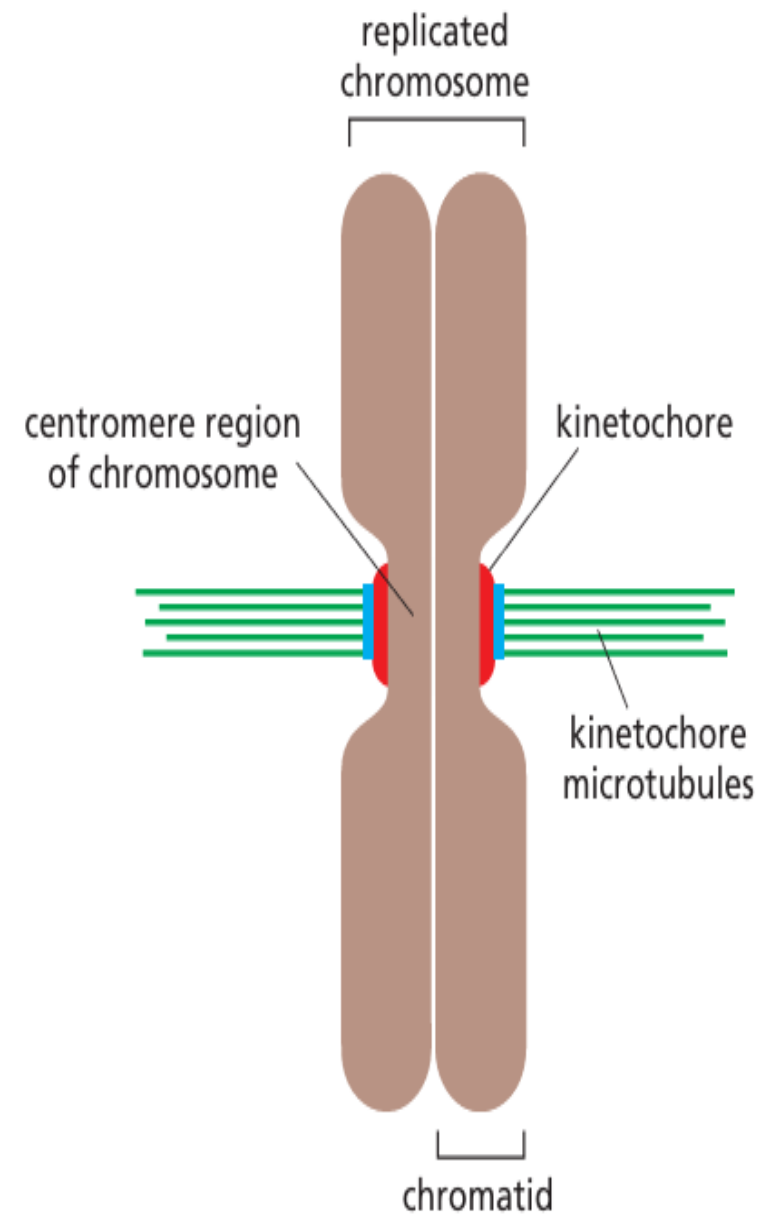


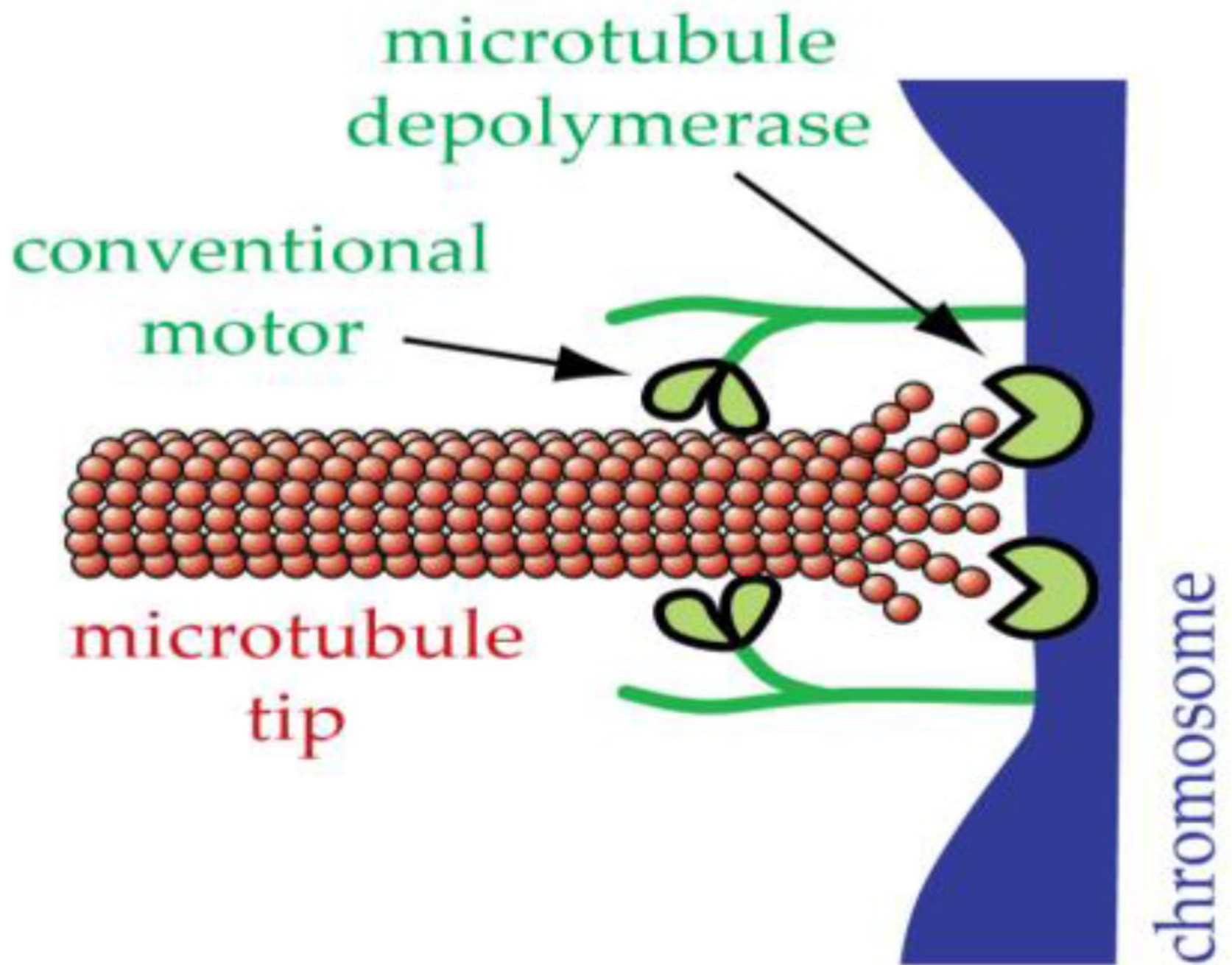
The phosphorylation of nuclear pore proteins and lamins helps trigger the disassembly of the nuclear envelope at prometaphase. Dephosphorylation of these proteins at telophase helps reverse the process.

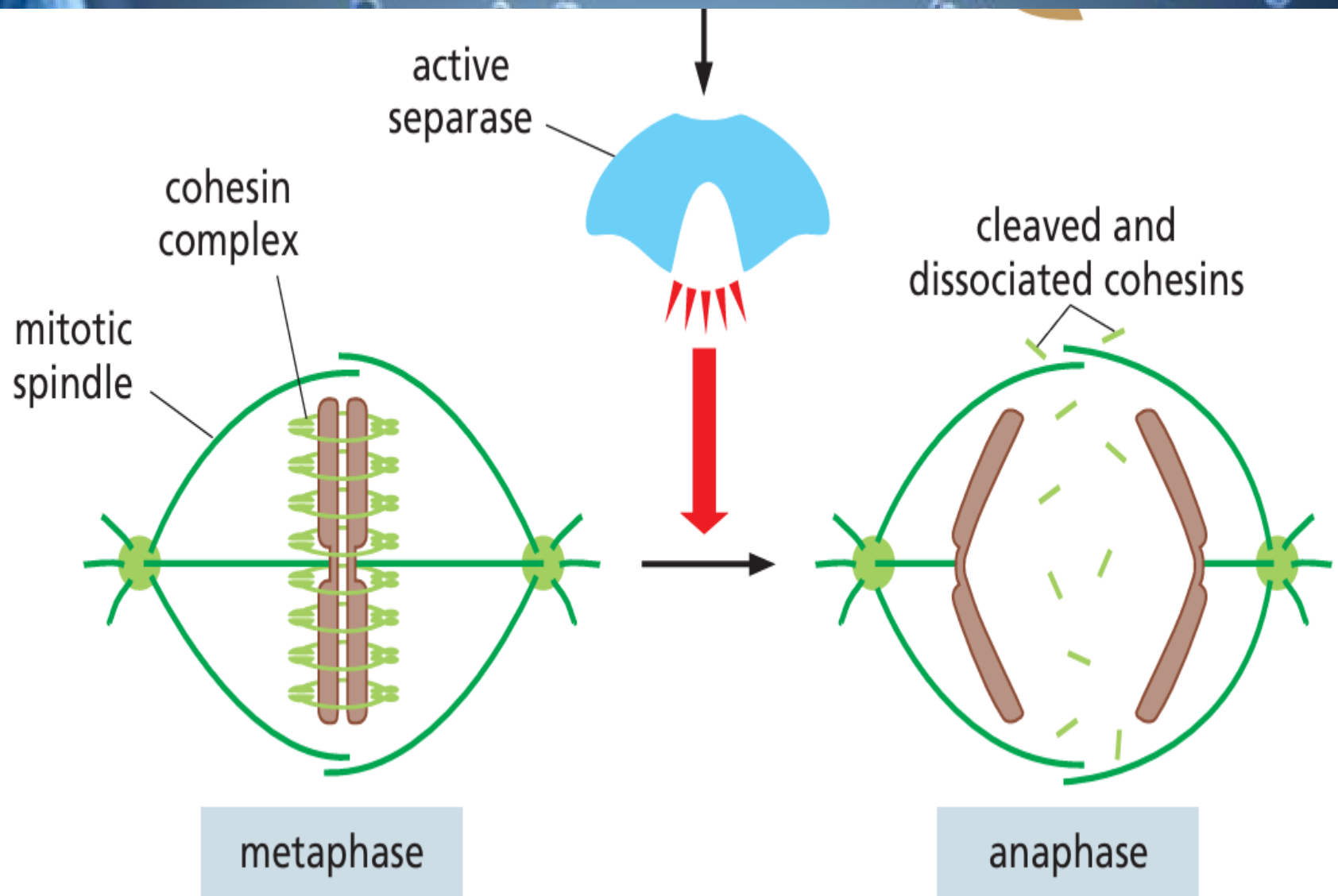
3 METAPHASE



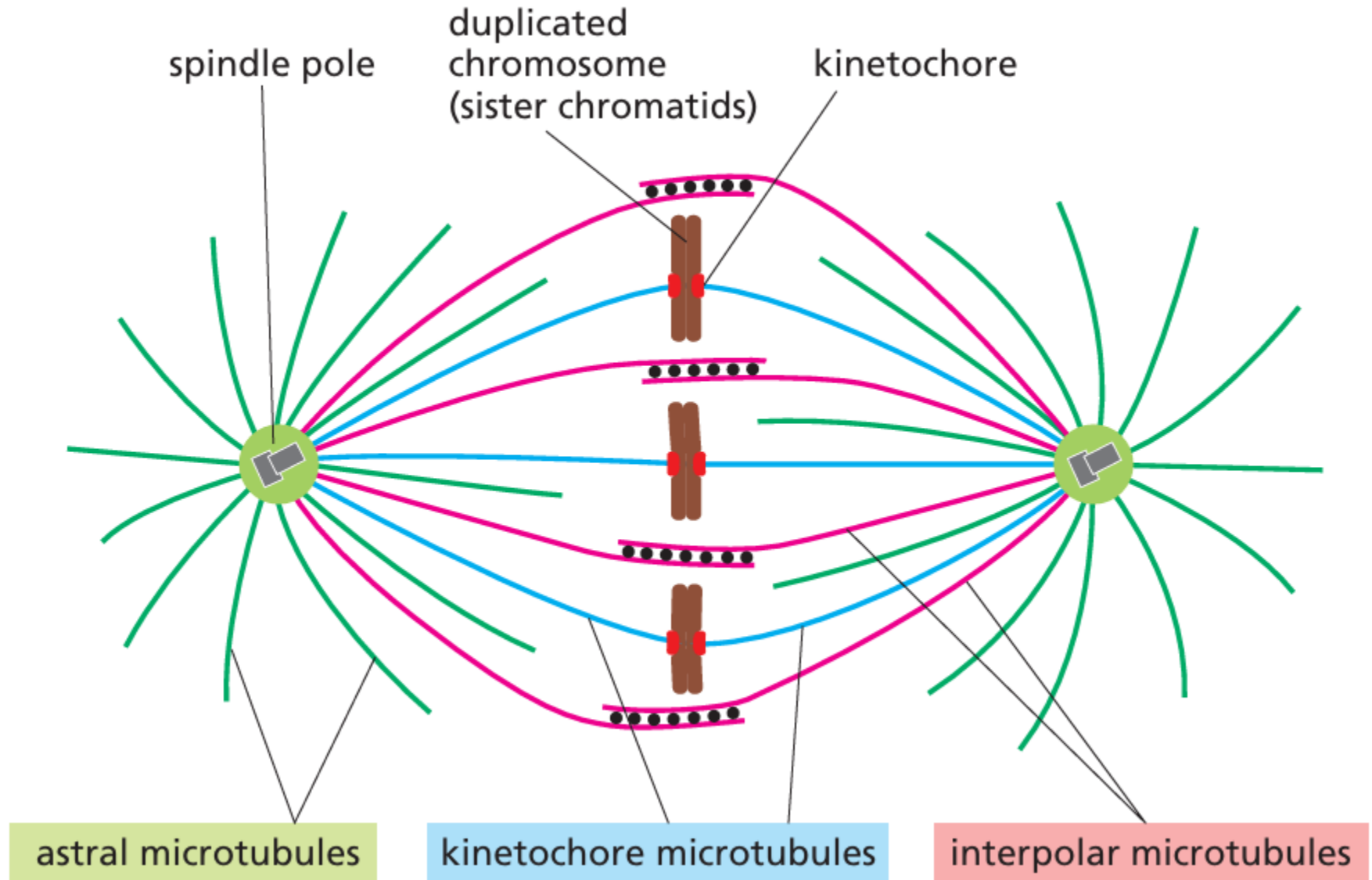
At **metaphase**, the chromosomes are aligned at the equator of the spindle, midway between the spindle poles. The kinetochore microtubules on each sister chromatid attach to opposite poles of the spindle.



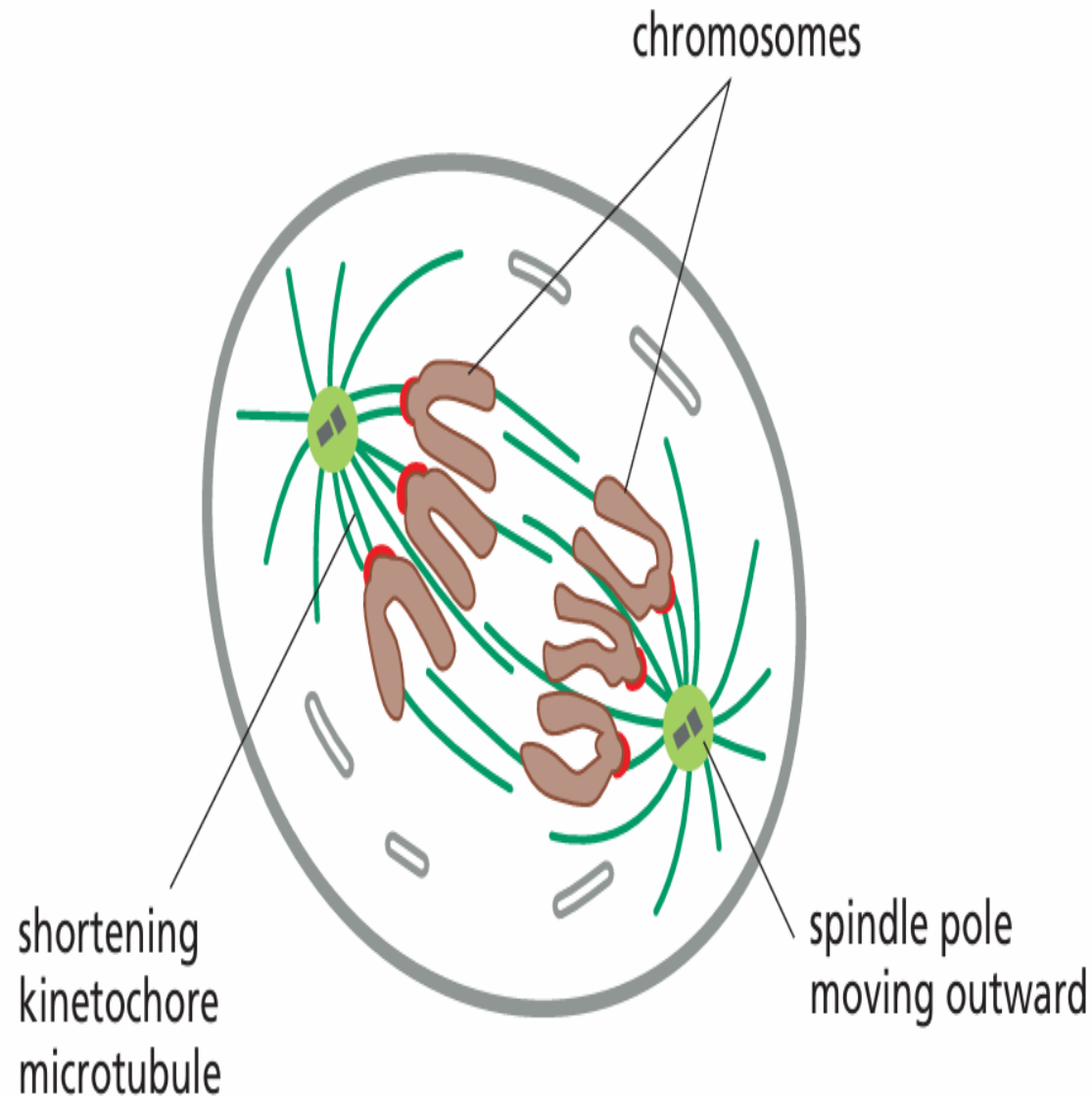




The Three Types of Microtubules in a Mitotic Cell

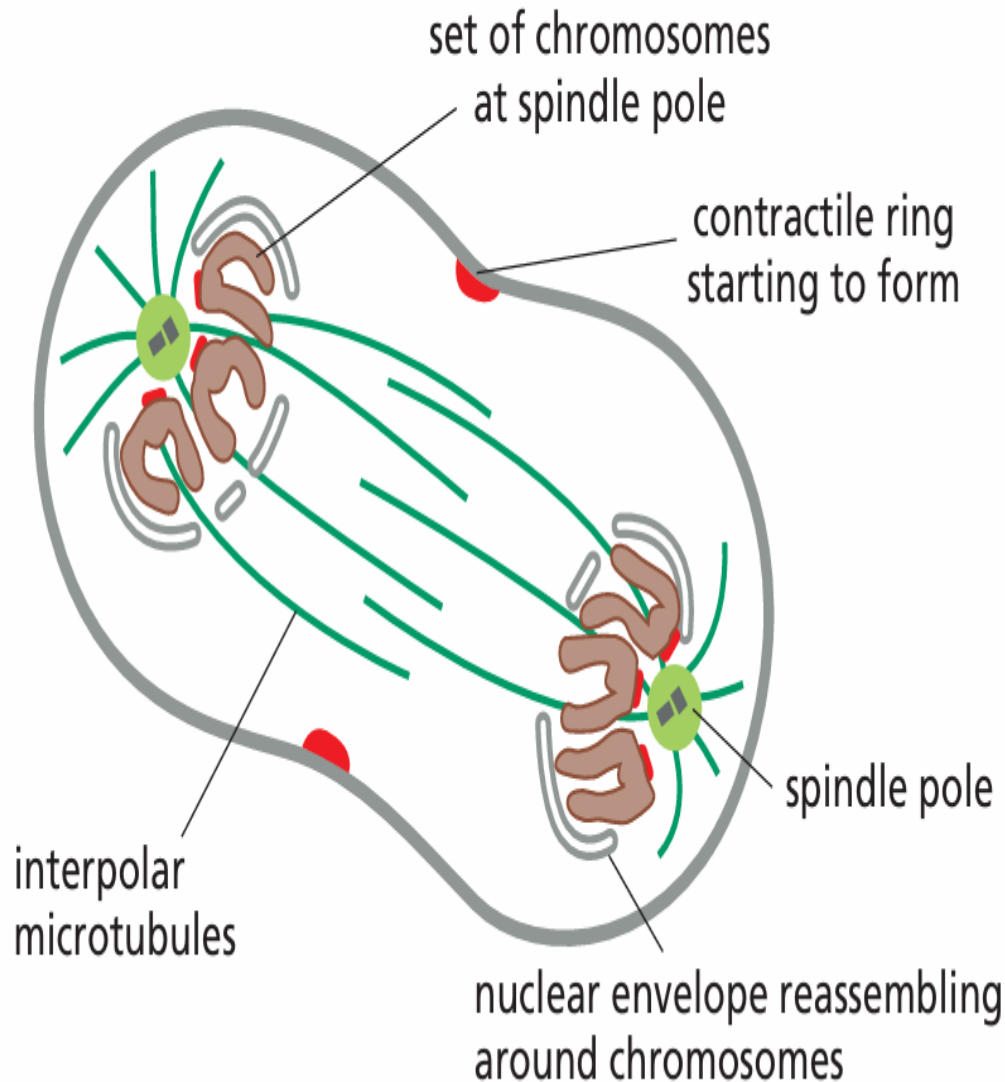


4 ANAPHASE



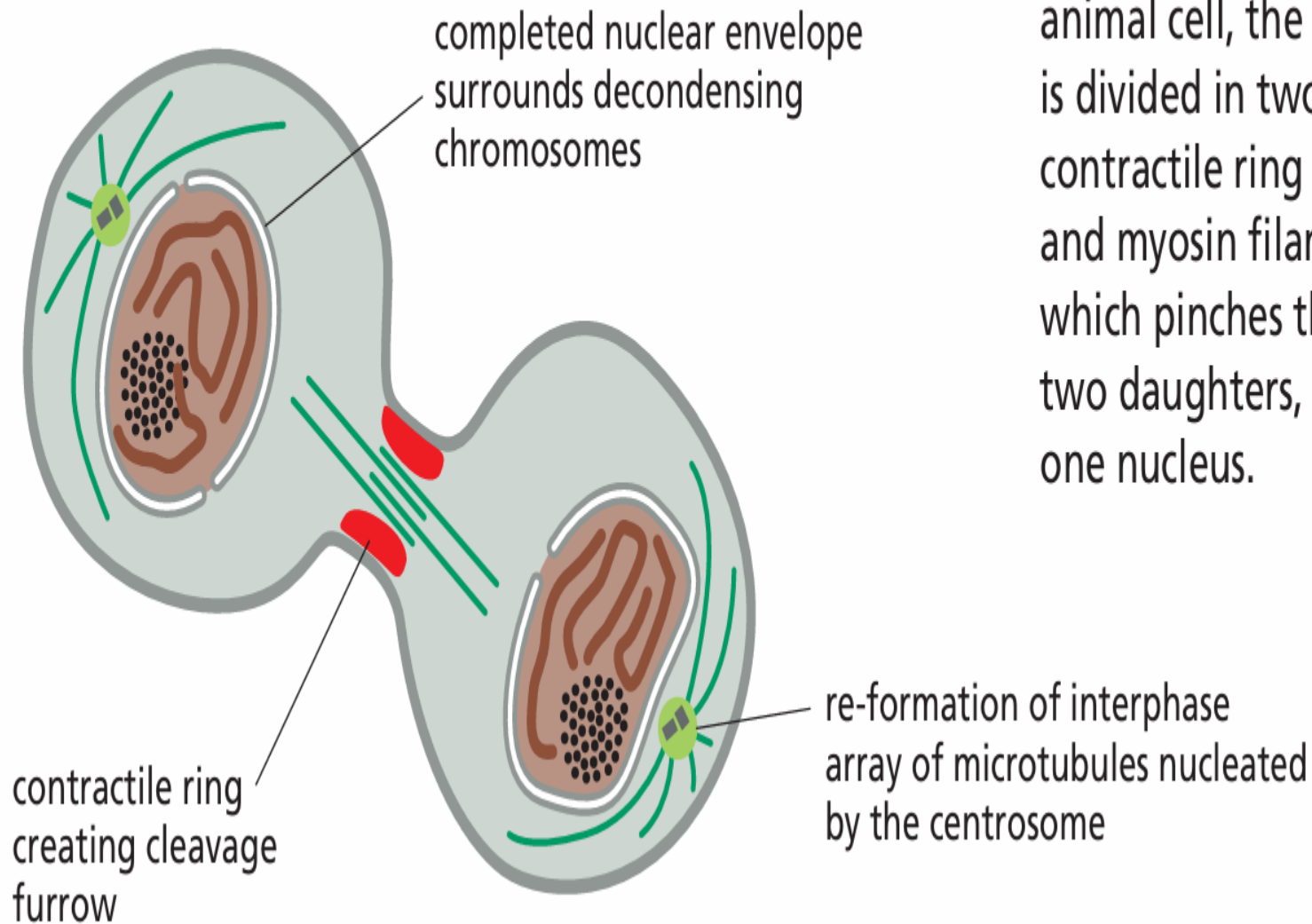
At **anaphase**, the sister chromatids synchronously separate and are pulled slowly toward the spindle pole to which they are attached. The kinetochore microtubules get shorter, and the spindle poles also move apart, both contributing to chromosome segregation.

5 TELOPHASE

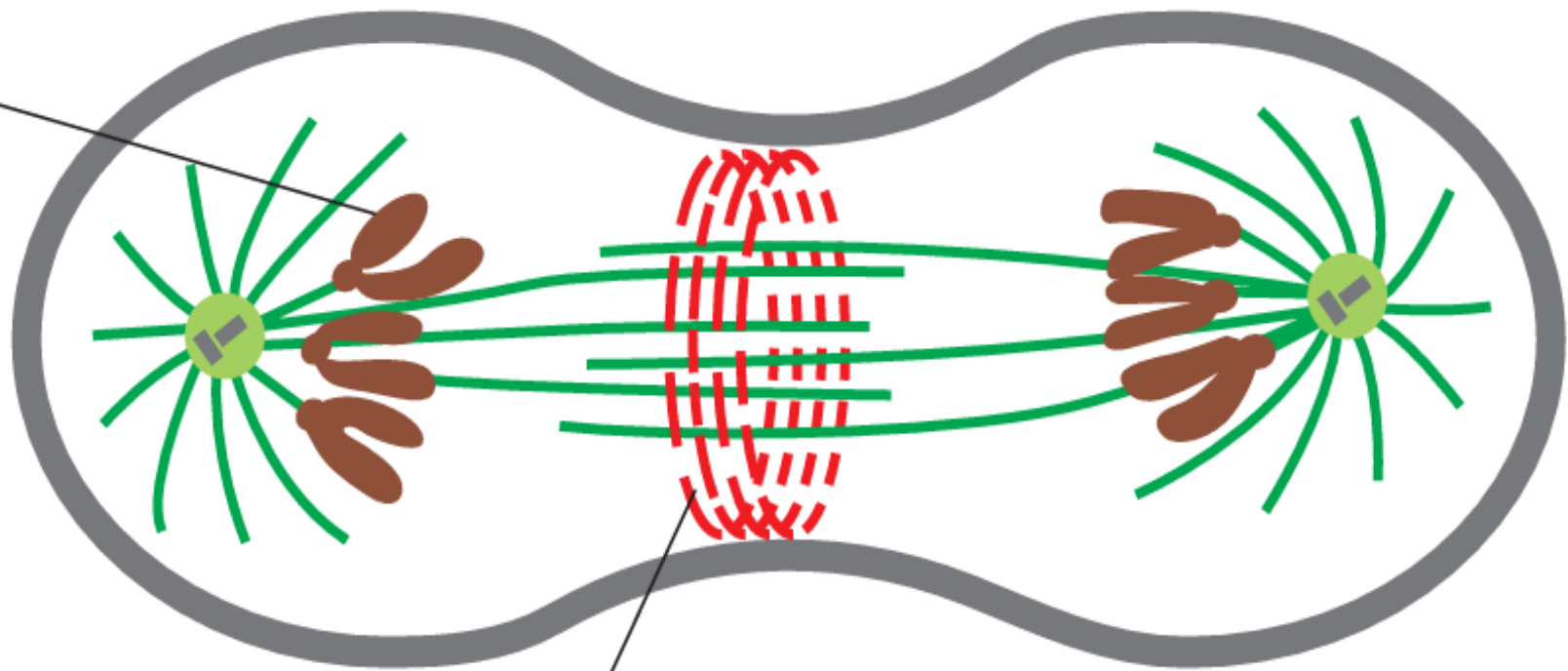


During **telophase**, the two sets of chromosomes arrive at the poles of the spindle. A new nuclear envelope reassembles around each set, completing the formation of two nuclei and marking the end of mitosis. The division of the cytoplasm begins with the assembly of the contractile ring.

CYTOKINESIS



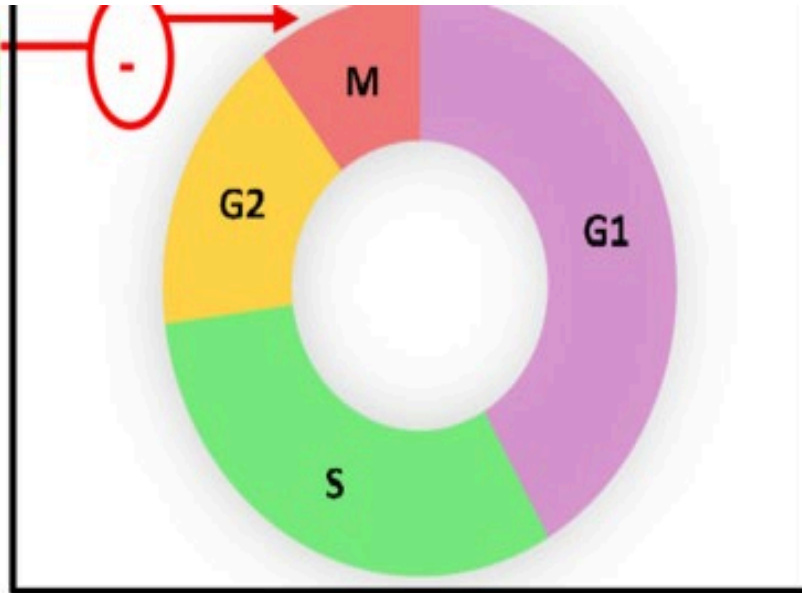
During **cytokinesis** of an animal cell, the cytoplasm is divided in two by a contractile ring of actin and myosin filaments, which pinches the cell into two daughters, each with one nucleus.



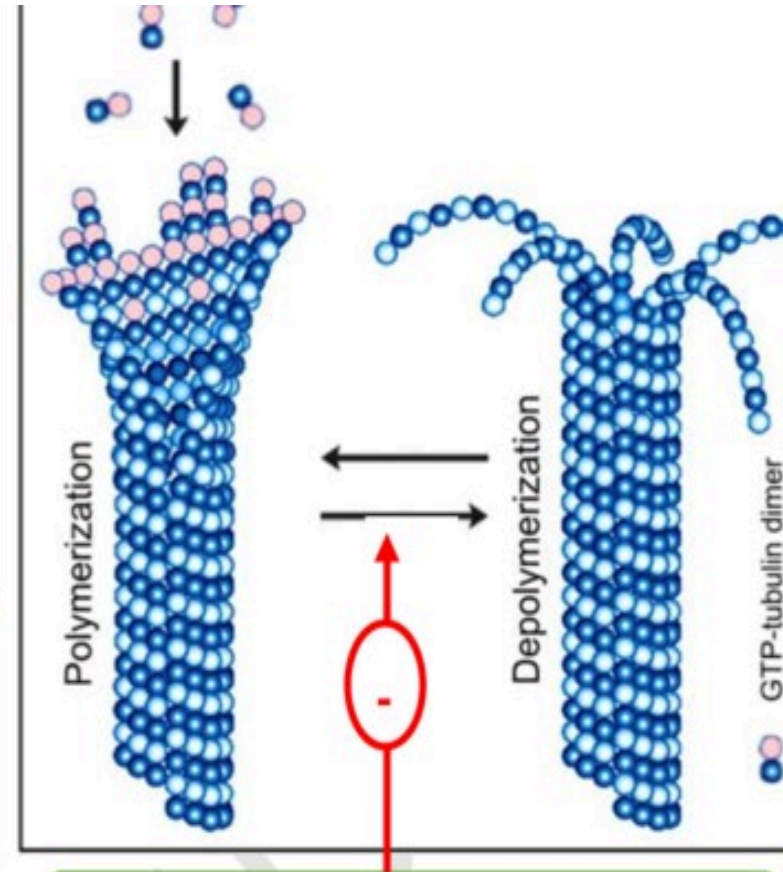
actin and myosin filaments of the
contractile ring

Mitosis Inhibitors (Spindle Poisons)

Taxanes



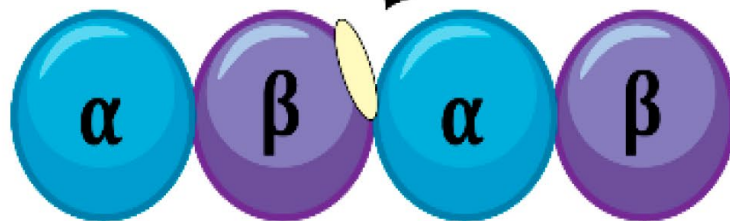
G1: Growth S: DNA synthesis
G2: Growth and preparation for mitosis
M: Mitosis



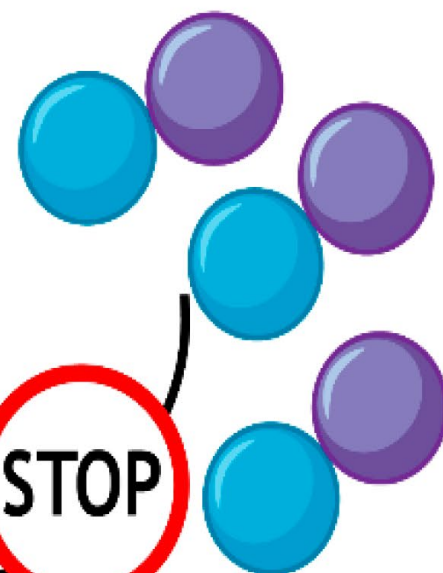
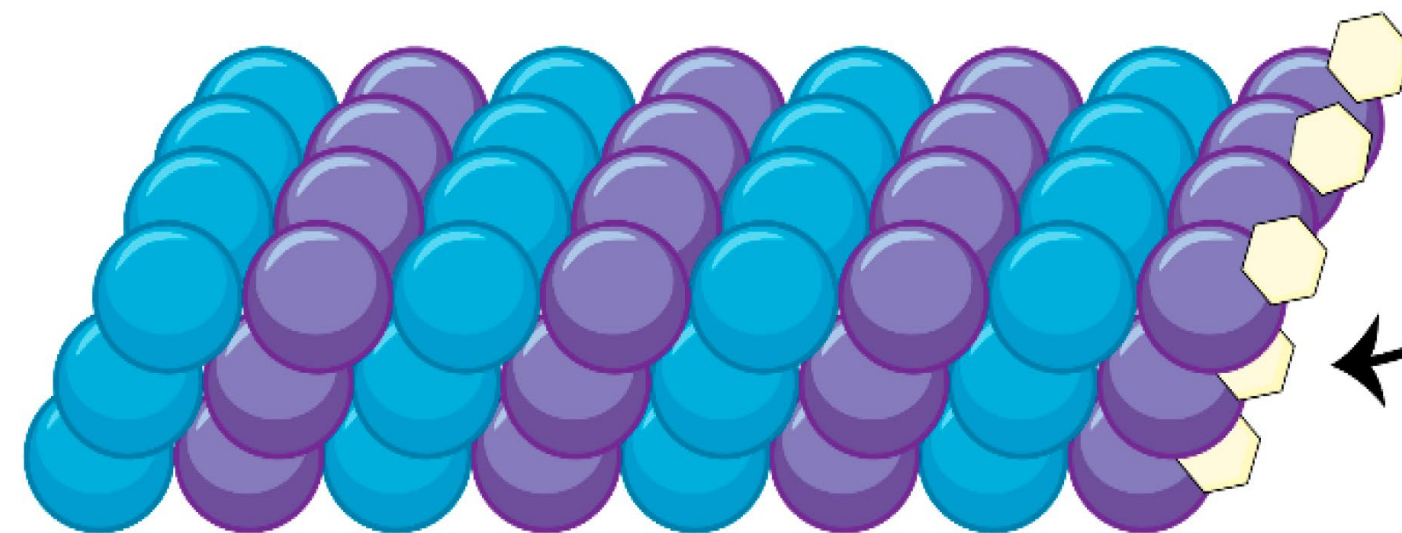
Taxanes (microtubule stabilizer)

Binding site

Vinca alkaloids



Tubulin dimers



STOP

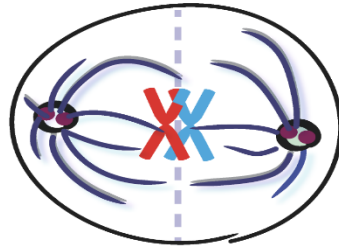
(-)

Microtubule

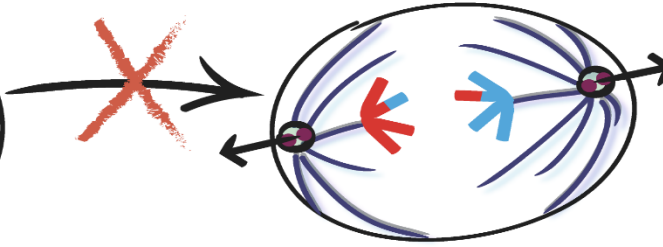
(+)

Microtubule Inhibitors

METAPHASE

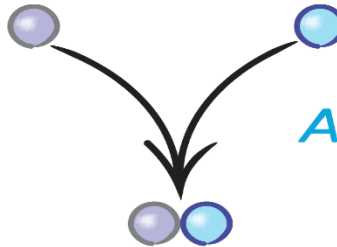


ANAPHASE

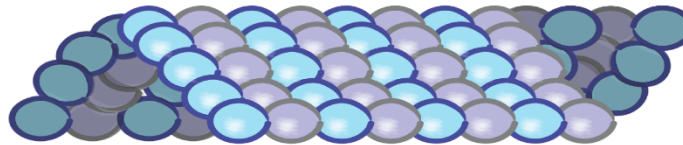


VINCA
ALKALOIDS
Vincristine
Vinblastine

Polymerization

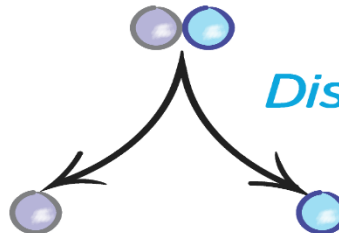


Assembly



TAXANES
Paclitaxel

Depolymerization



Disassembly

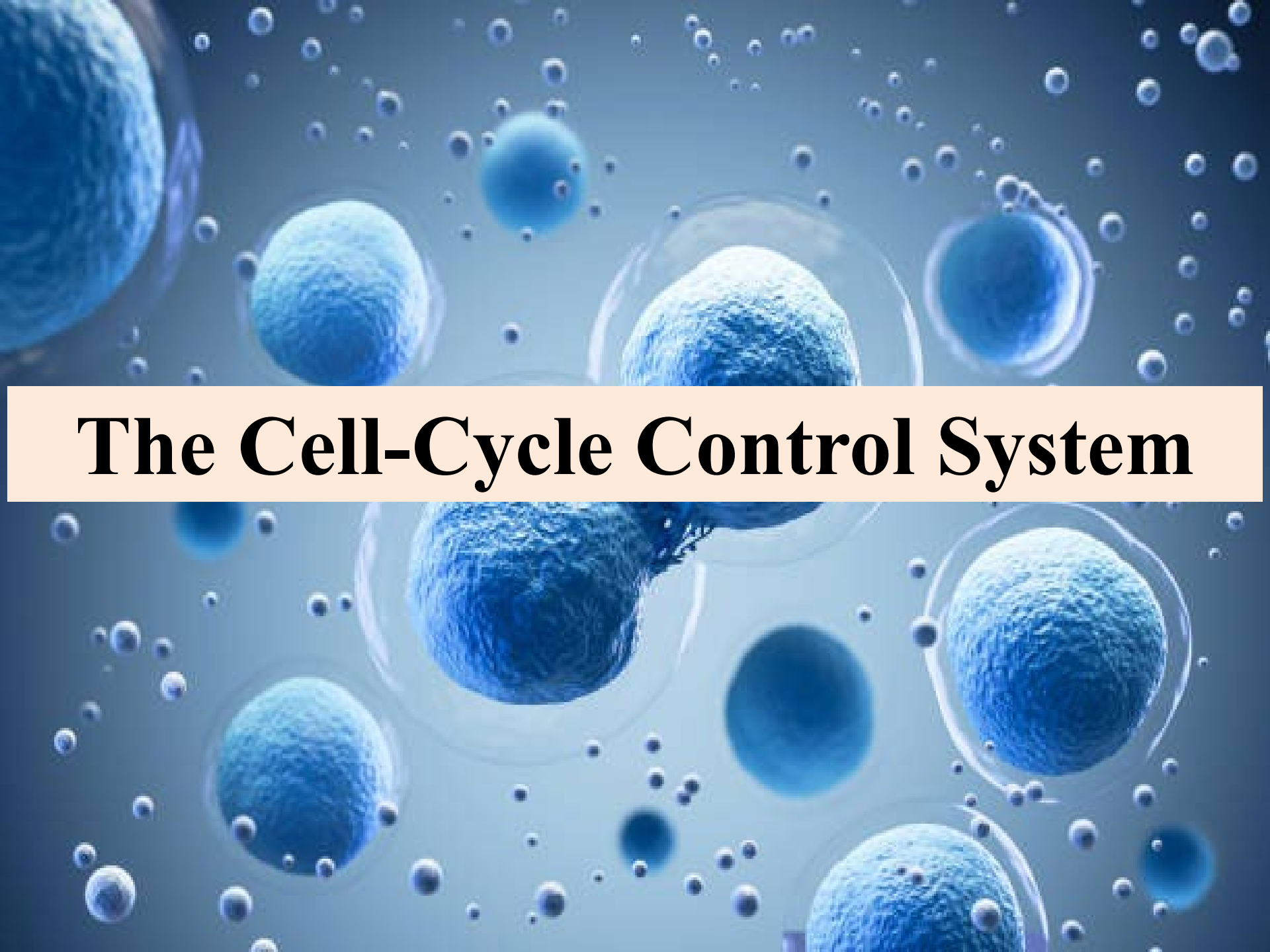
ALPHA (α)
TUBULIN

BETA (β)
TUBULIN

$\alpha\beta$ -TUBULIN
HETERODIMER

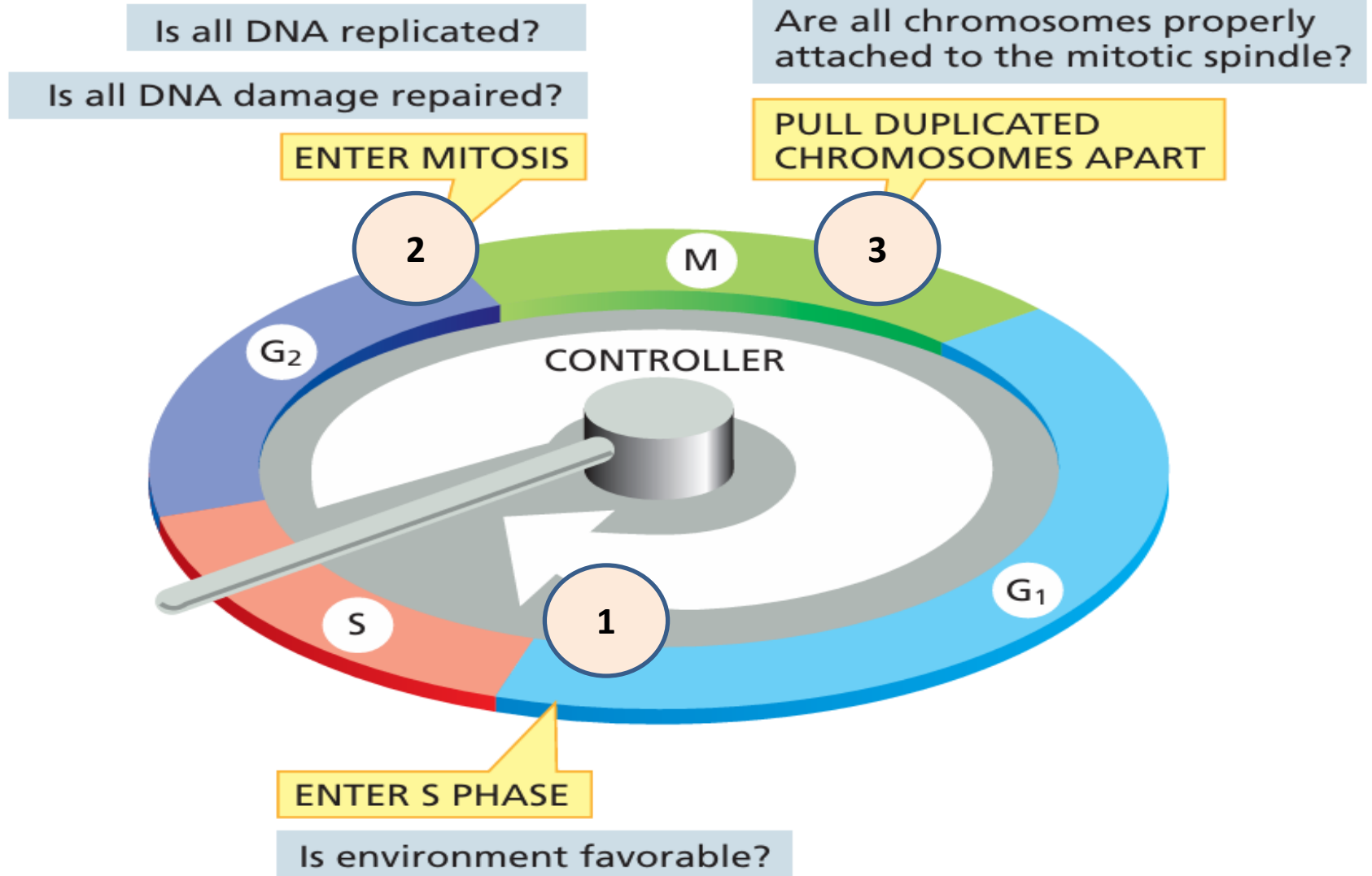
Key Events in Cell Division

Phase	Key Event	Brief Description
Prophase	Condensation	DNA coils into visible chromosomes; spindle fibers begin to form.
Prometaphase	Attachment	The nuclear envelope breaks down (via CDK1/Cyclin B enzymes); microtubules attach to kinetochores.
Metaphase	Alignment	Chromosomes line up at the center; kinetochores are under tension from spindle fibers.
Anaphase	Separation	Sister chromatids are pulled apart toward opposite poles.
Telophase	Reconstruction	New nuclear envelopes reform around the DNA; chromosomes decondense.
Cytokinesis	Division	The cytoplasm splits, resulting in two distinct daughter cells

A microscopic view of various cells, including several large, spherical cells with textured surfaces and many smaller, round cells, all rendered in shades of blue against a dark background. The cells are scattered across the frame, with some appearing more prominent than others.

The Cell-Cycle Control System

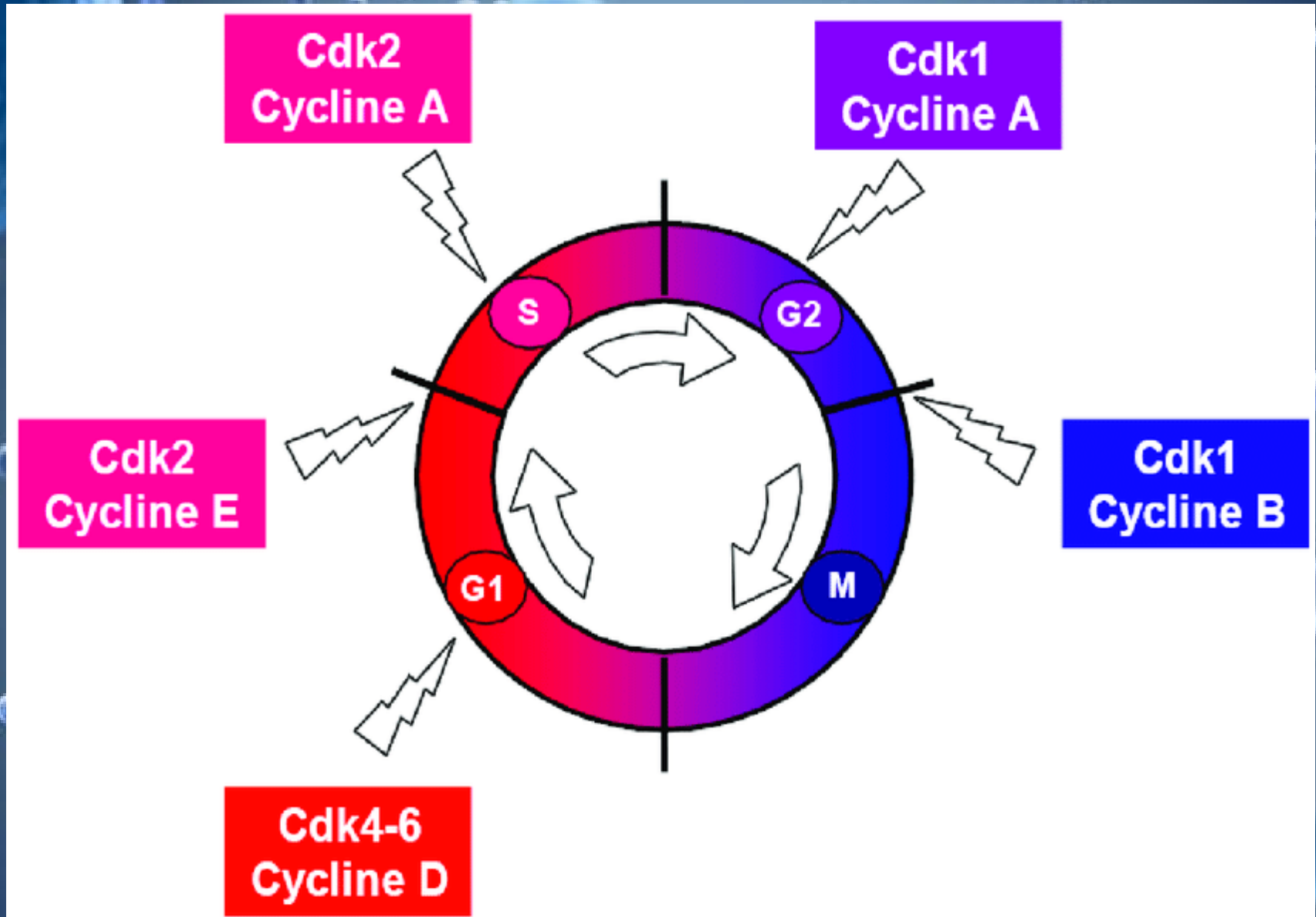
The Three Key Checkpoints

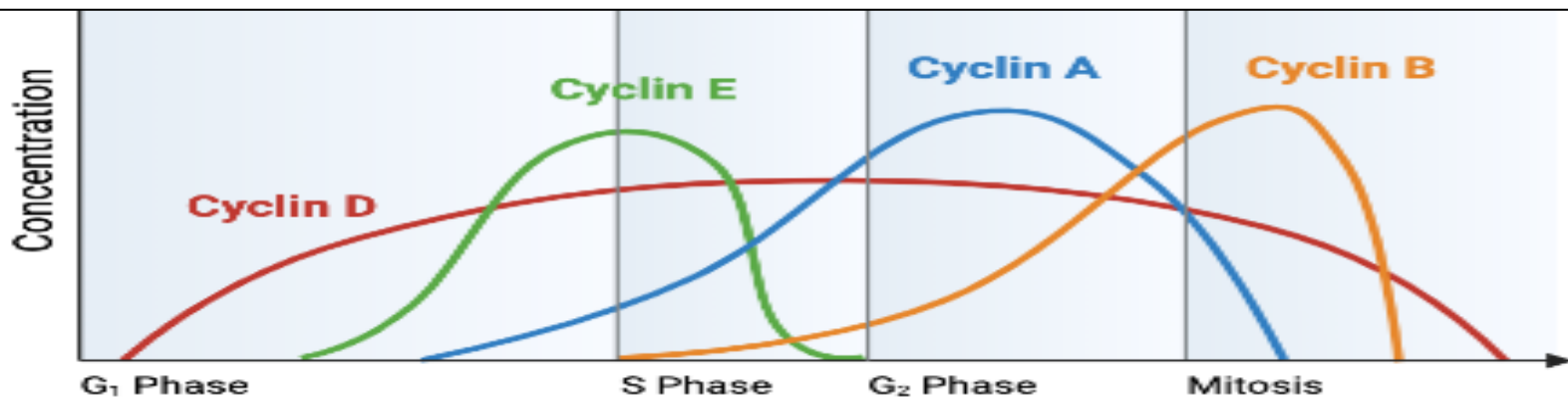
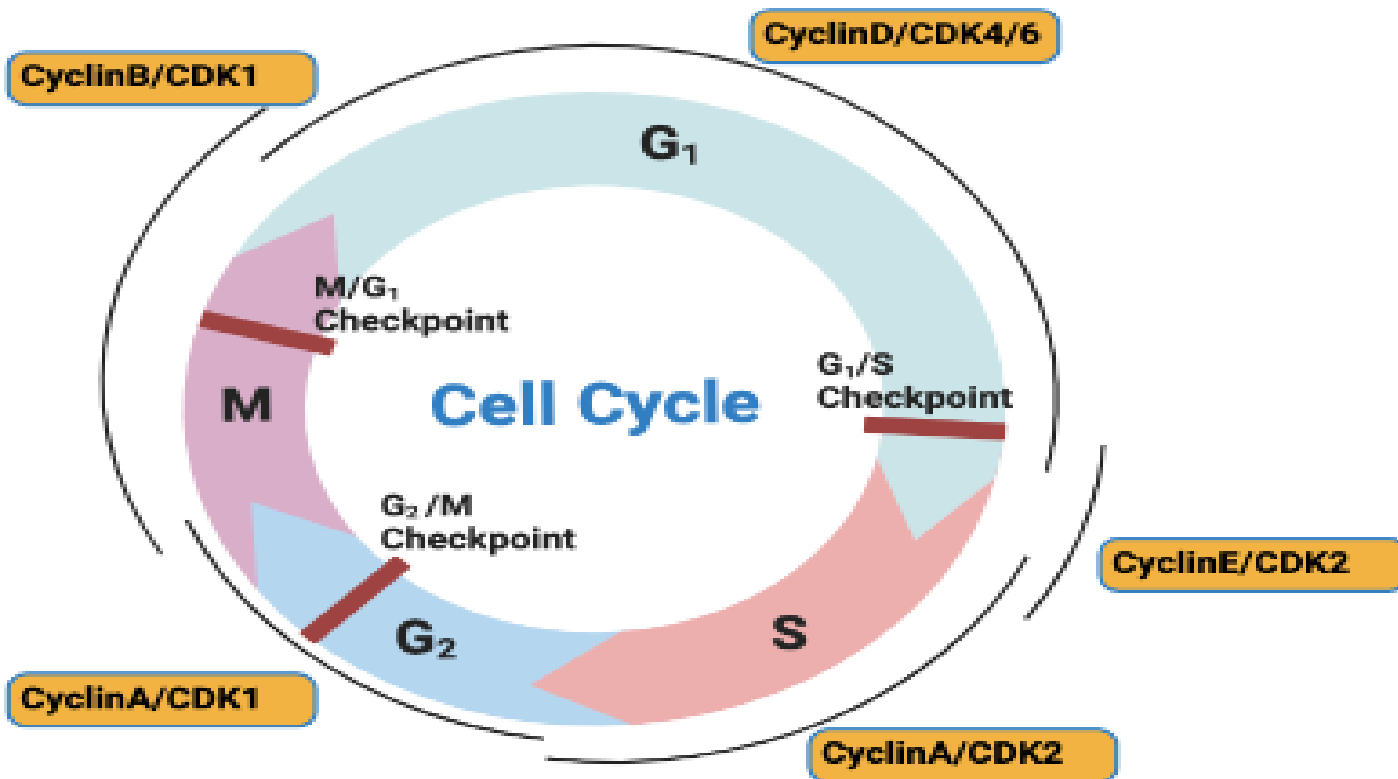


The cell-cycle control system regulates progression through the cell cycle at three main transition points

Checkpoint	What it Checks	Result (If Okay)
G1 Checkpoint	Environmental favorability (nutrients, growth factors) and DNA integrity.	Cell enters S Phase (DNA replication).
G2 Checkpoint	Completion of DNA replication and repair of all DNA damage.	Cell enters Mitosis.
M Checkpoint	Proper attachment of all chromosomes to the mitotic spindle.	Cell pulls duplicated chromosomes apart (Anaphase).

Major Cyclins and Cdks in Cell Cycle Regulation





Cyclin–Cdk Complexes

Cell Cycle Control

Requirements:

Cyclin:Cdk Complex

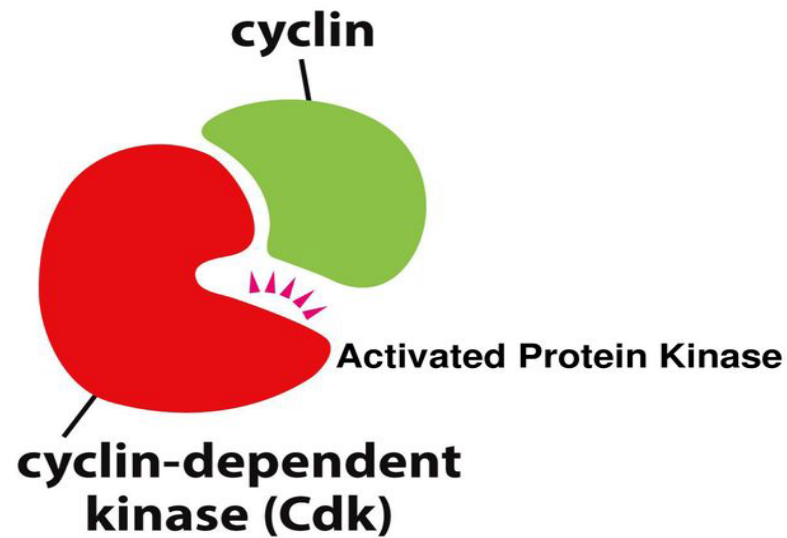
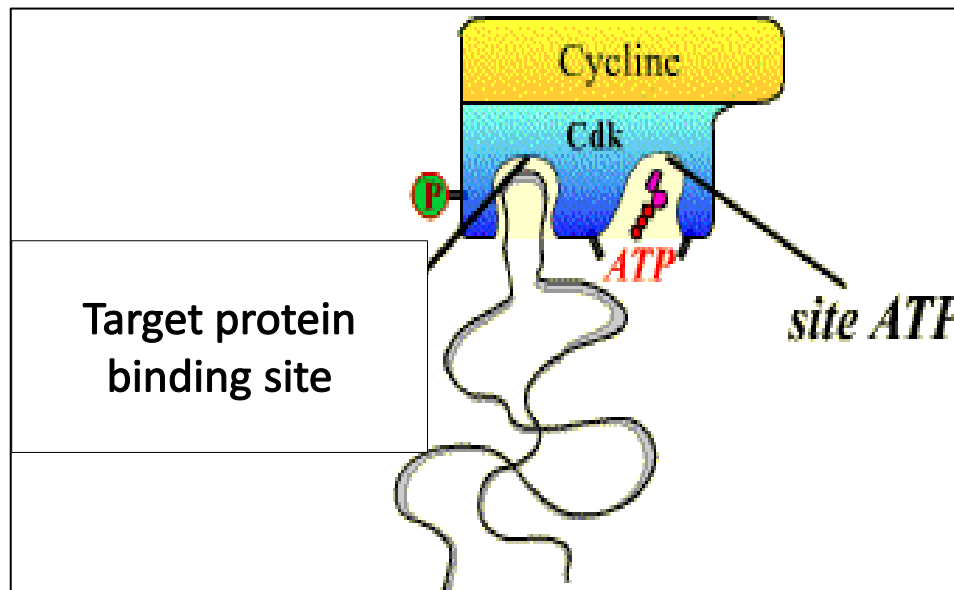
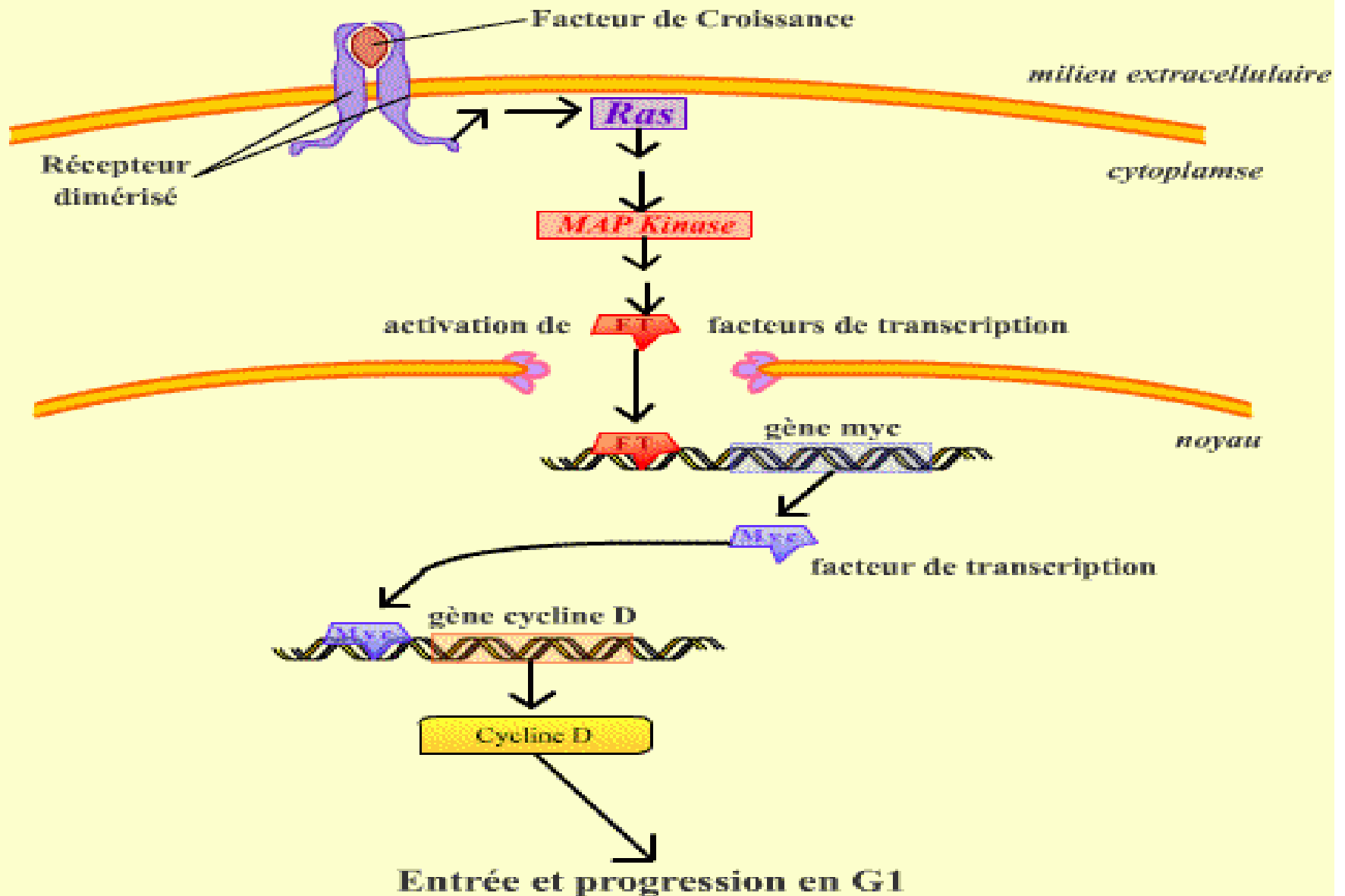


Figure 17-10 Molecular Biology of the Cell 6e (© Garland Science 2015)

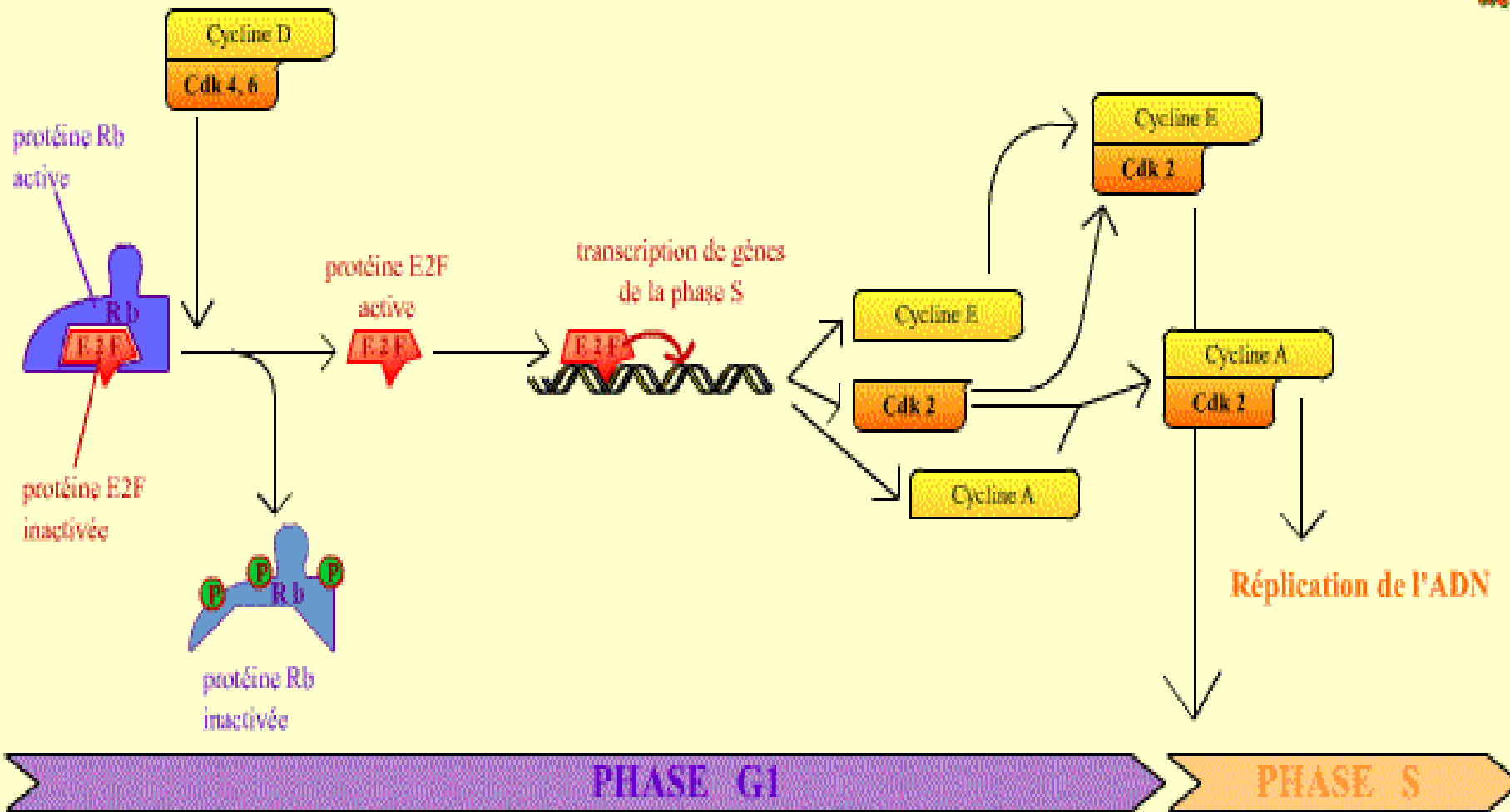


Entrée de la cellule dans un cycle cellulaire



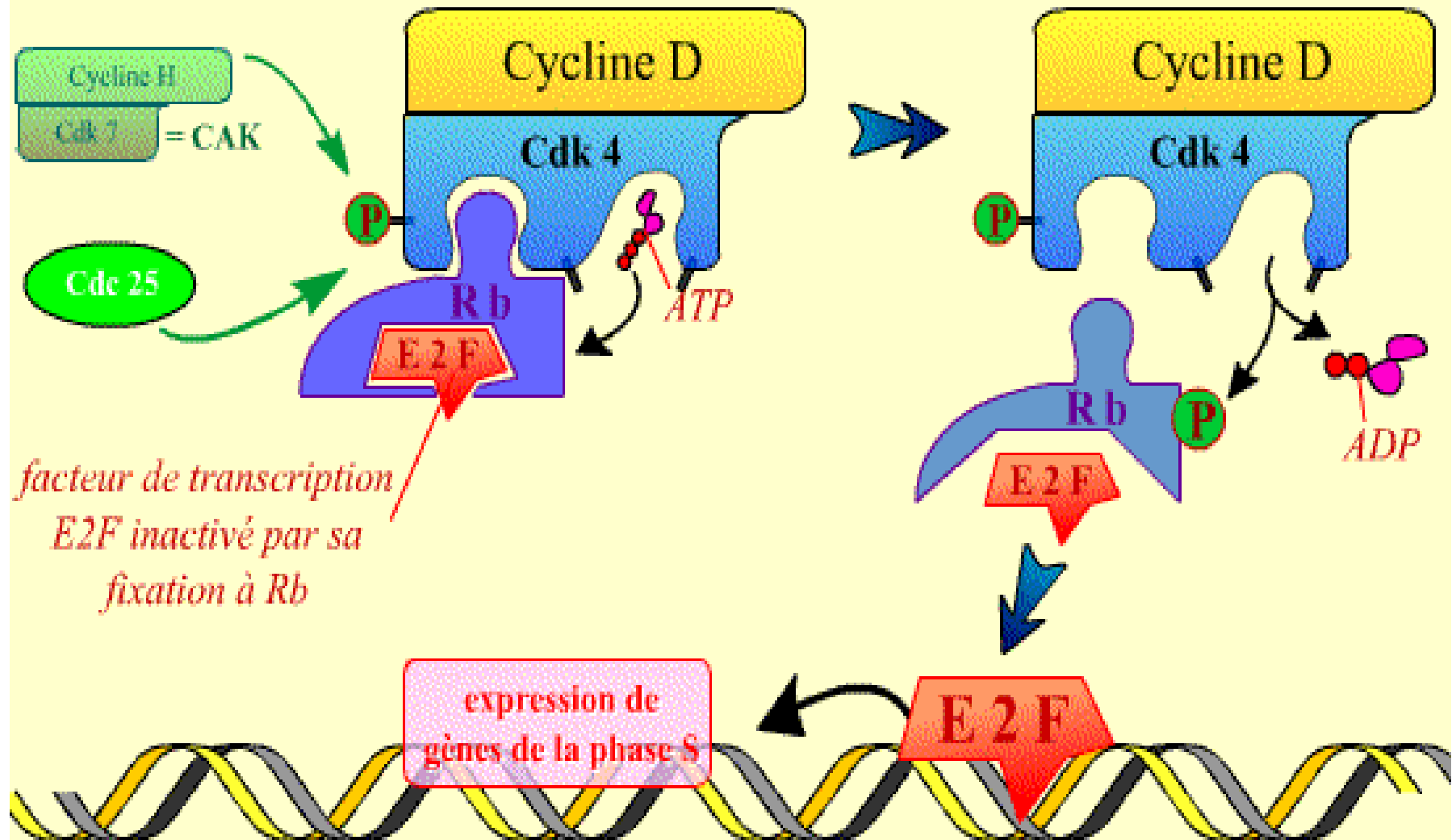
Inactivation de la protéine Rb et libération de E2F

La phase G1 prépare la phase S



Inactivation de la protéine Rb et libération de E2F

Action des CDK 4 / Cycline D sur pRb



Inhibition des Cdk

Régulation de l'activité des Cdk : inhibitions

